

Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type of $n \times m$ Kokotsakis Polyhedra” — Helper

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Tested on: Mathematica 14.0

Section 1

Results of Section 1 are used in the proof of Lemmas 1-2

In[3502]:=

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(* =====*)
(*PART 1:SETUP*)
(* =====*)
ClearAll[α, β, γ, δ, σ, a, b, c, d, r, s, f, M, assumptions, isTrueQ,
  idL, idR, okId, statusChip, smallChip, s1colTest, s1colHdr];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α];
b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ];
d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi, 0 < δ < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
idL[1] = HoldForm[Sin[α] Sin[δ] / (Sin[β] Sin[γ])];
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idR[1] = HoldForm[r (r - M) / (M (r - 1))];
idL[2] = HoldForm[Sin[σ - α] Sin[σ - δ] / (Sin[σ - β] Sin[σ - γ])];
idR[2] = HoldForm[(r - M) / (r (r - 1))];
idL[3] = HoldForm[Sin[γ] Sin[δ] / (Sin[α] Sin[β])];
idR[3] = HoldForm[s (s - M) / (M (s - 1))];
idL[4] = HoldForm[Sin[σ - γ] Sin[σ - δ] / (Sin[σ - α] Sin[σ - β])];
idR[4] = HoldForm[(s - M) / (s (s - 1))];
idL[5] = HoldForm[Sin[α] Sin[γ] / (Sin[β] Sin[δ])];
idR[5] = HoldForm[f (f - M) / (M (f - 1))];
idL[6] = HoldForm[Sin[σ - α] Sin[σ - γ] / (Sin[σ - β] Sin[σ - δ])];
idR[6] = HoldForm[(f - M) / (f (f - 1))];
okId[n_] := okId[n] = isTrueQ[ReleaseHold@idL[n], ReleaseHold@idR[n]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
s1colTest = RGBColor[0.10, 0.35, 0.75];
s1colHdr = RGBColor[0.15, 0.35, 0.45];
(* =====*)
(*PART 2:GIVEN PANEL AND VERIFICATION PANEL*)
(* =====*)
Module[{okAll, starLine, chkLine}, okAll = And@@Table[okId[n], {n, 6}];
  starLine[n1_, n2_] := Style[
    Row[{Style[TraditionalForm[HoldForm[#1 == #2] &[idL[n1], idR[n1]]], Bold],
      ", ", Style[TraditionalForm[HoldForm[#1 == #2] &[idL[n2], idR[n2]]],
        Bold]}], 14, s1colTest];
  chkLine[n1_, n2_] :=
    Style[Row[{TraditionalForm[HoldForm[#1 == #2] &[idL[n1], idR[n1]]], " ",
      smallChip[okId[n1]], " ", TraditionalForm[HoldForm[#1 == #2] &[
        idL[n2], idR[n2]]], " ", smallChip[okId[n2]]}], 13];
  Print[Panel[
    Column[{Style["Given hypotheses:", Bold, 14], Style[Row[{TraditionalForm[
      HoldForm[α], " ", TraditionalForm[HoldForm[β]], " ",
      TraditionalForm[HoldForm[γ]], " ", TraditionalForm[HoldForm[δ]],
      " ∈ (0, π), ", TraditionalForm[HoldForm[σ == (α + β + γ + δ) / 2]],
      " , ", TraditionalForm[HoldForm[σ - α]], " ", TraditionalForm[
        HoldForm[σ - β]], " ", TraditionalForm[HoldForm[σ - γ]],
        " ", TraditionalForm[HoldForm[σ - δ]], " ∉ πℤ"}], 13],
      Style[Row[{TraditionalForm[HoldForm[a == Sin[α] / Sin[σ - α]]],
        " ", TraditionalForm[HoldForm[b == Sin[β] / Sin[σ - β]]], " ",
        TraditionalForm[HoldForm[c == Sin[γ] / Sin[σ - γ]]], " ",
        TraditionalForm[HoldForm[d == Sin[δ] / Sin[σ - δ]]], " ",
        TraditionalForm[HoldForm[r == a d]], " ", TraditionalForm[

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HoldForm[s == c d]], " ", TraditionalForm[HoldForm[f == a c]],
", ", TraditionalForm[HoldForm[M == a b c d]]}], 13],
Style[Row[{TraditionalForm[HoldForm[r]], " ", TraditionalForm[
HoldForm[s]], " ", TraditionalForm[HoldForm[f]], " ",
TraditionalForm[HoldForm[M]], " ", TraditionalForm[HoldForm[M / r]],
", ", TraditionalForm[HoldForm[M / s]], " ", TraditionalForm[
HoldForm[M / f]], " ≠ 0, 1 ; hence both sides of the identities ",
Style["★", 13, s1colTest], " below are well-defined"}], 13],
"", Style[Row[{"Each ", smallChip[True],
" below certifies that the statement displayed beside it has been
verified by a symbolic computation at that point; a ",
smallChip[False], " would mean that this verification failed."}],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test the following identities ", Bold, 14],
Style["★", 14, s1colTest], Style[" :", Bold, 14]}]],
starLine[1, 2], starLine[3, 4], starLine[5, 6]], Spacings → 1],
Background → RGBColor[0.99, 0.96, 0.88],
FrameMargins → 12]]];
Print[
Panel[Column[{Row[{Style["The six ratio identities", Bold, 15, s1colHdr],
Spacer[20], statusChip[okAll]}], Style["Verification:", Bold, 13],
Style[Row[{"The identities ", Style["★", 13, s1colTest],
" hold under the given hypotheses:"}], 13],
chkLine[1, 2], chkLine[3, 4], chkLine[5, 6]], Spacings → 1],
Background → Lighter[RGBColor[0.15, 0.35, 0.45], 0.93],
FrameMargins → 12]]];];

```

Given hypotheses:

$$\alpha, \beta, \gamma, \delta \in (0, \pi), \quad \sigma = \frac{1}{2}(\alpha + \beta + \gamma + \delta), \quad \sigma - \alpha, \sigma - \beta, \sigma - \gamma, \sigma - \delta \notin \pi\mathbb{Z}$$

$$a = \frac{\sin(\alpha)}{\sin(\sigma - \alpha)}, \quad b = \frac{\sin(\beta)}{\sin(\sigma - \beta)}, \quad c = \frac{\sin(\gamma)}{\sin(\sigma - \gamma)}, \quad d = \frac{\sin(\delta)}{\sin(\sigma - \delta)}, \quad r = a d, \quad s = c d, \quad f = a c, \quad M = a b c d$$

$$r, s, f, M, \frac{M}{r}, \frac{M}{s}, \frac{M}{f} \neq 0, 1; \text{ hence both sides of the identities } \star \text{ below are well-defined}$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test the following identities ★ :

$$\frac{\sin(\alpha) \sin(\delta)}{\sin(\beta) \sin(\gamma)} = \frac{r(r-M)}{M(r-1)}, \quad \frac{\sin(\sigma - \alpha) \sin(\sigma - \delta)}{\sin(\sigma - \beta) \sin(\sigma - \gamma)} = \frac{r-M}{r(r-1)}$$

$$\frac{\sin(\gamma) \sin(\delta)}{\sin(\alpha) \sin(\beta)} = \frac{s(s-M)}{M(s-1)}, \quad \frac{\sin(\sigma - \gamma) \sin(\sigma - \delta)}{\sin(\sigma - \alpha) \sin(\sigma - \beta)} = \frac{s-M}{s(s-1)}$$

$$\frac{\sin(\alpha) \sin(\gamma)}{\sin(\beta) \sin(\delta)} = \frac{f(f-M)}{M(f-1)}, \quad \frac{\sin(\sigma - \alpha) \sin(\sigma - \gamma)}{\sin(\sigma - \beta) \sin(\sigma - \delta)} = \frac{f-M}{f(f-1)}$$

The six ratio identities

Verified

Verification:

The identities ★ hold under the given hypotheses:

$$\begin{array}{ll} \frac{\sin(\alpha) \sin(\delta)}{\sin(\beta) \sin(\gamma)} = \frac{r(r-M)}{M(r-1)} \quad \checkmark & \frac{\sin(\sigma-\alpha) \sin(\sigma-\delta)}{\sin(\sigma-\beta) \sin(\sigma-\gamma)} = \frac{r-M}{r(r-1)} \quad \checkmark \\ \frac{\sin(\gamma) \sin(\delta)}{\sin(\alpha) \sin(\beta)} = \frac{s(s-M)}{M(s-1)} \quad \checkmark & \frac{\sin(\sigma-\gamma) \sin(\sigma-\delta)}{\sin(\sigma-\alpha) \sin(\sigma-\beta)} = \frac{s-M}{s(s-1)} \quad \checkmark \\ \frac{\sin(\alpha) \sin(\gamma)}{\sin(\beta) \sin(\delta)} = \frac{f(f-M)}{M(f-1)} \quad \checkmark & \frac{\sin(\sigma-\alpha) \sin(\sigma-\gamma)}{\sin(\sigma-\beta) \sin(\sigma-\delta)} = \frac{f-M}{f(f-1)} \quad \checkmark \end{array}$$

Section 2

Results of Section 2 are used in the proof of Lemma 2

In[3549]:=

```
(* =====*)
(*PART 1:SETUP*)
(* =====*)
ClearAll[f, l, m, M, isTrueQ, idL, idR,
  okId, statusChip, smallChip, s2colTest, s2colHdr];
(*The identities below are identities of
  rational functions in Subscript[f,l],Subscript[f,m],M;
  the subscripted symbols remain inert,so no domain assumptions are needed*)
isTrueQ[lhs_, rhs_] := TrueQ[Simplify[lhs == rhs]];
idL[1] =
  HoldForm[Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1)) -
    Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))];
idR[1] = HoldForm[(Subscript[f, l] - Subscript[f, m])
  (Subscript[f, l] Subscript[f, m] - (Subscript[f, l] + Subscript[f, m]) + M) /
  (M (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
idL[2] =
  HoldForm[(Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1))) *
  (Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))) - 1];
idR[2] = HoldForm[(Subscript[f, l] Subscript[f, m] - M)
  (Subscript[f, l] Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
  (M^2 (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
idL[3] = HoldForm[(Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1)) -
  (Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))];
idR[3] = HoldForm[(Subscript[f, m] - Subscript[f, l])
  (Subscript[f, l] Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
  (Subscript[f, l] Subscript[f, m]
  (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
idL[4] = HoldForm[((Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1))) *
  ((Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))) - 1];
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idR[4] = HoldForm[(Subscript[f, l] Subscript[f, m] - M)
  (Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] - M) /
  (Subscript[f, l] Subscript[f, m]
    (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
okId[n_] := okId[n] = isTrueQ[ReleaseHold[idL[n], ReleaseHold[idR[n]]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
s2colTest = RGBColor[0.10, 0.35, 0.75];
s2colHdr = RGBColor[0.15, 0.35, 0.45];
(* =====*)
(*PART 2:GIVEN PANEL AND VERIFICATION PANEL*)
(* =====*)
Module[{okAll, starLine, chkLine}, okAll = And@@Table[okId[n], {n, 4}];
starLine[n_] :=
  Style[Style[TraditionalForm[HoldForm[#1 == #2] &[idL[n], idR[n]]], Bold],
    14, s2colTest];
chkLine[n_] :=
  Style[Row[{TraditionalForm[HoldForm[#1 == #2] &[idL[n], idR[n]]],
    " ", smallChip[okId[n]]}], 13];
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
  Style[Row[{"For ", TraditionalForm[HoldForm[l]], ", ",
    TraditionalForm[HoldForm[m]], " ∈ {1, ..., 4}, ",
    TraditionalForm[HoldForm[l ≠ m]], " :"}], Bold, 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[f, l]]], ", ",
    TraditionalForm[HoldForm[Subscript[f, m]]], ", ", TraditionalForm[
      HoldForm[M]], ", ", TraditionalForm[HoldForm[M / Subscript[f, l]]],
    ", ", TraditionalForm[HoldForm[M / Subscript[f, m]]],
    " ≠ 0, 1 ; hence both sides of the identities ",
    Style["*"], 13, s2colTest], " below are well-defined"}], 13],
  "", Style[Row[{"Each ", smallChip[True],
    " below certifies that the statement displayed beside it has been
      verified by a symbolic computation at that point; a ",
    smallChip[False], " would mean that this verification failed."}],
    12, GrayLevel[0.35]], "", Style[
  Row[{Style["We want to test the following identities ", Bold, 14],
    Style["*"], 14, s2colTest], Style[" :", Bold, 14]}]],
  starLine[1], starLine[2], starLine[3], starLine[4]], Spacings → 1],
  Background → RGBColor[0.99, 0.96, 0.88],
  FrameMargins → 12]];
Print[Panel[Column[
  {Row[{Style["The four factorization identities", Bold, 15, s2colHdr],

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Spacer[20], statusChip[okAll]]], Style["Verification:", Bold, 13],
Style[Row[{"The identities  ", Style["★", 13, s2colTest],
" hold under the given hypotheses:"}], 13],
chkLine[1], chkLine[2], chkLine[3], chkLine[4]], Spacings → 1],
Background → Lighter[RGBColor[0.15, 0.35, 0.45], 0.93],
FrameMargins → 12]]];

```

Given hypotheses:

For $l, m \in \{1, \dots, 4\}$, $l \neq m$:

$f_l, f_m, M, \frac{M}{f_l}, \frac{M}{f_m} \neq 0, 1$; hence both sides of the identities ★ below are well-defined

Each ✓ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✗ would mean that this verification failed.

We want to test the following identities ★ :

$$\frac{f_l(f_l-M)}{M(f_l-1)} - \frac{f_m(f_m-M)}{M(f_m-1)} = \frac{(f_l-f_m)(f_l f_m - (f_l+f_m)+M)}{M(f_l-1)(f_m-1)}$$

$$\frac{(f_l(f_l-M))(f_m(f_m-M))}{(M(f_l-1))(M(f_m-1))} - 1 = \frac{(f_l f_m - M)(f_l f_m - M(f_l+f_m-1))}{M^2(f_l-1)(f_m-1)}$$

$$\frac{f_l-M}{f_l(f_l-1)} - \frac{f_m-M}{f_m(f_m-1)} = \frac{(f_m-f_l)(f_l f_m - M(f_l+f_m-1))}{f_l f_m(f_l-1)(f_m-1)}$$

$$\frac{(f_l-M)(f_m-M)}{(f_l(f_l-1))(f_m(f_m-1))} - 1 = \frac{(f_l f_m - M)(f_l+f_m-f_l f_m-M)}{f_l f_m(f_l-1)(f_m-1)}$$

The four factorization identities

Verified

Verification:

The identities ★ hold under the given hypotheses:

$$\frac{f_l(f_l-M)}{M(f_l-1)} - \frac{f_m(f_m-M)}{M(f_m-1)} = \frac{(f_l-f_m)(f_l f_m - (f_l+f_m)+M)}{M(f_l-1)(f_m-1)} \quad \checkmark$$

$$\frac{(f_l(f_l-M))(f_m(f_m-M))}{(M(f_l-1))(M(f_m-1))} - 1 = \frac{(f_l f_m - M)(f_l f_m - M(f_l+f_m-1))}{M^2(f_l-1)(f_m-1)} \quad \checkmark$$

$$\frac{f_l-M}{f_l(f_l-1)} - \frac{f_m-M}{f_m(f_m-1)} = \frac{(f_m-f_l)(f_l f_m - M(f_l+f_m-1))}{f_l f_m(f_l-1)(f_m-1)} \quad \checkmark$$

$$\frac{(f_l-M)(f_m-M)}{(f_l(f_l-1))(f_m(f_m-1))} - 1 = \frac{(f_l f_m - M)(f_l+f_m-f_l f_m-M)}{f_l f_m(f_l-1)(f_m-1)} \quad \checkmark$$

Section 3

Results of Section 3 are used in the proof of Lemma 4

In[3389]:=

```

(* =====*)
(*PART 1:SETUP*)
(* =====*)
ClearAll[α, β, γ, δ, σ, ε, a, b, c, d, r, s, f, M, rad, sq,

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assumptions, okSq, okEps, okRadLow, okRadHigh, okRoot, okMerge, s3chk,
cellChk, clSym, intvStr, intIdx, snVal, cnVal, lblCell, statusChip,
smallChip, s3colGiven, s3colTest, s3colHdr, dn, t, p, q, k];

$$\sigma = (\alpha + \beta + \gamma + \delta) / 2;$$

a = Sin[ $\alpha$ ] / Sin[ $\sigma - \alpha$ ];
b = Sin[ $\beta$ ] / Sin[ $\sigma - \beta$ ];
c = Sin[ $\gamma$ ] / Sin[ $\sigma - \gamma$ ];
d = Sin[ $\delta$ ] / Sin[ $\sigma - \delta$ ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
these hold under the given hypotheses*)
assumptions = {0 <  $\alpha$  < Pi, 0 <  $\beta$  < Pi, 0 <  $\gamma$  < Pi,
0 <  $\delta$  < Pi, 0 <  $\sigma - \alpha$  < Pi, 0 <  $\sigma - \beta$  < Pi, 0 <  $\sigma - \gamma$  < Pi, 0 <  $\sigma - \delta$  < Pi};
rad = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq = (Sin[ $\sigma$ ] / Sin[ $\sigma - \delta$ ]) ^2;
(*Two-stage verification of the sigma-
identity:STAGE 1 (okSq) identifies rad with the perfect square
sq.STAGE 2 (okEps) uses Stage 1:for t=Sin[sigma]/Sin[sigma-delta],
the eps-split assumptions give t the sign eps (u/v checks),
so Sqrt[rad]=Sqrt[t^2]=eps*t (abstract root law),
and eps*Sqrt[rad]*Sin[sigma-delta]=
eps^2*Sin[sigma]=Sin[sigma] (cancellation).No Sqrt
over trigonometric arguments reaches the kernel.*)
okSq = TrueQ[FullSimplify[rad == sq, Assumptions -> assumptions]];
okEps =
Module[{tt, u, v}, okSq && TrueQ[Assuming[u > 0 && v > 0, Simplify[u / v > 0]]] &&
TrueQ[Assuming[u < 0 && v > 0, Simplify[u / v < 0]]] &&
TrueQ[Assuming[tt > 0, Simplify[Sqrt[tt^2] == tt]]] &&
TrueQ[Assuming[tt < 0, Simplify[Sqrt[tt^2] == -tt]]] &&
TrueQ[Simplify[(Sin[ $\sigma$ ] / Sin[ $\sigma - \delta$ ]) Sin[ $\sigma - \delta$ ] == Sin[ $\sigma$ ]]];
s3chk[e1_, e2_] := Together[e1 - e2] === 0 || Simplify[e1 - e2] === 0;
(*Radicand identities per regime,rational in fresh symbols*)
okRadLow =
Module[{rv, sv, fv, mv}, s3chk[fv (fv - mv) (rv - 1) (sv - 1) / (fv (1 - mv)),
(fv - mv) (rv - 1) (sv - 1) / (1 - mv)]];
okRadHigh = Module[{rv, sv, fv, mv}, s3chk[(fv - mv) (rv - 1) (sv - 1) / (fv (1 - mv)),
(1 / fv - 1 / mv) (rv - 1) (sv - 1) / (1 - 1 / mv)]];
(*Abstract root laws:k-extraction and the dn-fold*)
okRoot = Module[{x, kk, dd},
TrueQ[Assuming[kk > 0 && x >= 0, Simplify[Sqrt[x / kk^2] == Sqrt[x] / kk]]] &&
TrueQ[Assuming[dd > 0 && x >= 0, Simplify[dd Sqrt[x] == Sqrt[dd^2 x]]]];
(*Class-wise radical merges,as displayed:atoms m,w,v,
A>0 parametrize|p q|,Sqrt[dn^2 t-1],Sqrt[dn^2 t-k'^2],Sqrt[k'^2-dn^2 t];
in class 2,dn^2 t-1=-(A^2+k^2) and dn^2 t-k'^2=-A^2*)
okMerge[1] = Module[{m, w, v},
TrueQ[Assuming[m > 0 && w > 0, Simplify[Sqrt[w^2 m^2] == m Sqrt[w^2]]]] &&

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TrueQ[Assuming[m > 0 && v > 0, Simplify[Sqrt[v^2 m^2] == m Sqrt[v^2]]]];
okMerge[2] = Module[{m, A, kk},
  TrueQ[Assuming[m > 0 && A > 0 && kk > 0, Simplify[Sqrt[(- (A^2 + kk^2)) (-m^2)] ==
    -I (I m) Sqrt[A^2 + kk^2]]]] && TrueQ[Assuming[m > 0 && A > 0,
    Simplify[Sqrt[(-A^2) (-m^2)] == -I (I m) Sqrt[A^2]]]]];
okMerge[3] = Module[{m, w, v},
  TrueQ[Assuming[m > 0 && w > 0, Simplify[Sqrt[w^2 m^2] == -(-m) Sqrt[w^2]]]] &&
  TrueQ[Assuming[m > 0 && v > 0, Simplify[Sqrt[v^2 m^2] == -(-m) Sqrt[v^2]]]]];
(*Cell checks: class cl of p q (1:pos real, 2:pos imaginary, 3:neg real) and eps=
  sgn(sin sigma);
same atoms;
radicals merged per the given sign identity;
returns the pair of verdicts {sn agrees, cn agrees}*)
cellChk[cl_, eps_] :=
Module[{kk, m, w, A, pq, W, V, snS, cnS, snT, cnT}, Which[cl == 1, pq = m;
  W = w m;
  V = m Sqrt[w^2 + kk^2];
  snT = If[eps == 1, I w / kk, -I w / kk];
  cnT = If[eps == 1, Sqrt[w^2 + kk^2] / kk, -Sqrt[w^2 + kk^2] / kk],
  cl == 2, pq = I m;
  W = m Sqrt[kk^2 + A^2];
  V = A m;
  snT = If[eps == 1, Sqrt[kk^2 + A^2] / kk, -Sqrt[kk^2 + A^2] / kk];
  cnT = If[eps == 1, -I A / kk, I A / kk], cl == 3, pq = -m; W = w m;
  V = m Sqrt[w^2 + kk^2];
  snT = If[eps == 1, -I w / kk, I w / kk];
  cnT = If[eps == 1, -Sqrt[w^2 + kk^2] / kk, Sqrt[w^2 + kk^2] / kk]];
snS = I eps W / (kk pq); cnS = eps V / (kk pq);
{s3chk[snS, snT], s3chk[cnS, cnT]}}];
clSym[1] = Subscript["R", ">0"];
clSym[2] = Row[{"i ", Subscript["R", ">0"]]];
clSym[3] = Subscript["R", "<0"];
intvStr[1, 1] = "(0, iK)";
intvStr[1, -1] = "(2K, 2K + iK)";
intvStr[2, 1] = "(K, K + iK)";
intvStr[2, -1] = "(3K, 3K + iK)";
intvStr[3, 1] = "(2K, 2K + iK)";
intvStr[3, -1] = "(0, iK)";
intIdx[1, 1] = 1; intIdx[1, -1] = 3; intIdx[2, 1] = 2;
intIdx[2, -1] = 4; intIdx[3, 1] = 3; intIdx[3, -1] = 1;
snVal[1] = TraditionalForm[HoldForm[i Sqrt[Superscript[dn, 2] t - 1] / k]];
snVal[2] = TraditionalForm[HoldForm[Sqrt[1 - Superscript[dn, 2] t] / k]];
snVal[3] = TraditionalForm[HoldForm[-i Sqrt[Superscript[dn, 2] t - 1] / k]];
snVal[4] = TraditionalForm[HoldForm[-Sqrt[1 - Superscript[dn, 2] t] / k]];
cnVal[1] = TraditionalForm[HoldForm[Sqrt[Superscript[dn, 2] t - 1 + k^2] / k]];
cnVal[2] = TraditionalForm[HoldForm[-i Sqrt[1 - k^2 - Superscript[dn, 2] t] / k]];
cnVal[3] = TraditionalForm[HoldForm[-Sqrt[Superscript[dn, 2] t - 1 + k^2] / k]];

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cnVal[4] = TraditionalForm[HoldForm[ $\pm \sqrt{1 - k^2 - \text{Superscript}[dn, 2] t} / k$ ]];
lblCell[cl_, eps_] := Row[{TraditionalForm[HoldForm[p q]],
  " ∈ ", clSym[cl], ", ", TraditionalForm[HoldForm[σ]],
  If[eps == 1, " < π", " > π"], " → ", TraditionalForm[HoldForm[ε]],
  If[eps == 1, " = 1", " = -1"], ", ", t ∈ ", intvStr[cl, eps]}}];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
s3colGiven = RGBColor[0.85, 0.45, 0.00];
s3colTest = RGBColor[0.10, 0.35, 0.75];
s3colHdr = RGBColor[0.15, 0.35, 0.45];
(* =====*)
(*PART 2:GIVEN PANEL AND VERIFICATION PANEL*)
(* =====*)
Module[{cells, gridPairs, gridSn, gridCn, okAll, hdrCell, tab1, tab2, mergeRow},
  cells = {{1, 1}, {1, -1}, {2, 1}, {2, -1}, {3, 1}, {3, -1}};
  gridPairs = cellChk@@@cells;
  gridSn = gridPairs[[All, 1]]; gridCn = gridPairs[[All, 2]];
  okAll = okSq && okEps && okRadLow && okRadHigh && okRoot && okMerge[1] &&
    okMerge[2] && okMerge[3] && (And@@gridSn) && (And@@gridCn);
  hdrCell[cl_] := Row[{TraditionalForm[HoldForm[p q]], " ∈ ", clSym[cl]}}];
  tab1 = Grid[{{"", hdrCell[1], hdrCell[2], hdrCell[3]},
    {Row[{TraditionalForm[HoldForm[σ]], " < π"}], intvStr[1, 1], intvStr[2, 1],
    intvStr[3, 1]}, {Row[{TraditionalForm[HoldForm[σ]], " > π"}],
    intvStr[1, -1], intvStr[2, -1], intvStr[3, -1]}},
    Frame → All, FrameStyle → GrayLevel[0.75], Spacings → {2, 0.8}];
  tab2 = Grid[{{Style["t", Italic], intvStr[1, 1], intvStr[2, 1], intvStr[3, 1],
    intvStr[2, -1]}, {Row[{"sn ", Style["t", Italic]}], snVal[1],
    snVal[2], snVal[3], snVal[4]}, {Row[{"cn ", Style["t", Italic]}],
    cnVal[1], cnVal[2], cnVal[3], cnVal[4]}},
    Frame → All, FrameStyle → GrayLevel[0.75], Spacings → {2, 0.8}];
  mergeRow[1] = Row[{TraditionalForm[HoldForm[p q]], " ∈ ", clSym[1], " : ",
    TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] t - 1) p^2 q^2] ==
    p q Sqrt[Superscript[dn, 2] t - 1]]], ", ",
    TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] t - k'^2) p^2 q^2] ==
    p q Sqrt[Superscript[dn, 2] t - k'^2]]],
    " ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]],
    " ", smallChip[okMerge[1]]}];
  mergeRow[2] = Row[{TraditionalForm[HoldForm[p q]], " ∈ ", clSym[2], " : ",
    TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] t - 1) p^2 q^2] ==
    - $\pm$  p q Sqrt[1 - Superscript[dn, 2] t]]], ", ",
    TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] t - k'^2) p^2 q^2] ==

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- $\frac{1}{2} p q \sqrt{k'^2 - \text{Superscript}[dn, 2] t}]$ ,
"      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]],
"      ", smallChip[okMerge[2]]];
mergeRow[3] = Row[{TraditionalForm[HoldForm[p q]], " ∈ ", c1Sym[3], " : ",
TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] t - 1) p^2 q^2] =
-p q Sqrt[Superscript[dn, 2] t - 1]]], " ",
TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] t - k'^2) p^2 q^2] =
-p q Sqrt[Superscript[dn, 2] t - k'^2]]],
"      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]],
"      ", smallChip[okMerge[3]]];
Print[Panel[
Column[{Style["Given hypotheses:", Bold, 14], Style[Row[{TraditionalForm[
HoldForm[α]], " ", TraditionalForm[HoldForm[β]], " ",
TraditionalForm[HoldForm[γ]], " ", TraditionalForm[HoldForm[δ]],
" ∈ (0, π), ", TraditionalForm[HoldForm[σ = (α + β + γ + δ) / 2]],
" , ", TraditionalForm[HoldForm[σ - α]], " ", TraditionalForm[
HoldForm[σ - β]], " ", TraditionalForm[HoldForm[σ - γ]],
" ", TraditionalForm[HoldForm[σ - δ]], " ∈ (0, π), ",
TraditionalForm[HoldForm[Sin[σ] ≠ 0]], " , ",
TraditionalForm[HoldForm[ε = Sign[Sin[σ]]], " ∈ {±1}"}], 13],
Style[Row[{TraditionalForm[HoldForm[a = Sin[α] / Sin[σ - α]],
" , ", TraditionalForm[HoldForm[b = Sin[β] / Sin[σ - β]], " , ",
TraditionalForm[HoldForm[c = Sin[γ] / Sin[σ - γ]], " , ",
TraditionalForm[HoldForm[d = Sin[δ] / Sin[σ - δ]], " , ",
TraditionalForm[HoldForm[r = a d]], " , ", TraditionalForm[
HoldForm[s = c d]], " , ", TraditionalForm[HoldForm[f = a c]],
" , ", TraditionalForm[HoldForm[M = a b c d]]}], 13],
Style[Row[{TraditionalForm[HoldForm[r]], " , ", TraditionalForm[
HoldForm[s]], " , ", TraditionalForm[HoldForm[f]], " , ",
TraditionalForm[HoldForm[M]], " , ", TraditionalForm[HoldForm[M / f]],
" ≠ 0, 1, ", TraditionalForm[HoldForm[f > 0]],
" , ", TraditionalForm[HoldForm[M > 0]]}], 13],
Style[Row[{TraditionalForm[HoldForm[k = Sqrt[1 - M]]], " , ",
TraditionalForm[HoldForm[k' = Sqrt[M]]],
" , ", "dn ", Style["t", Italic], " = ",
TraditionalForm[HoldForm[Sqrt[f]]], " if M < 1"}], 13],
Style[Row[{TraditionalForm[HoldForm[k = Sqrt[1 - 1 / M]]],
" , ", TraditionalForm[HoldForm[k' = Sqrt[1 / M]]],
" , ", "dn ", Style["t", Italic], " = ",
TraditionalForm[HoldForm[1 / Sqrt[f]]], " if M > 1"}], 13],
Style[Row[{TraditionalForm[HoldForm[p = Sqrt[r - 1]]], " , ",
TraditionalForm[HoldForm[q = Sqrt[s - 1]]], " ∈ ",
Subscript["ℝ", ">0"], " ∪ ", "i ", Subscript["ℝ", ">0"]}], 13],
Style[Row[{TraditionalForm[HoldForm[k]], " , ",
TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 13],
Style[Row[{TraditionalForm[HoldForm[Sign[Superscript[dn, 2] t - 1] =
Sign[Superscript[dn, 2] t - k'^2] = Sign[p^2 q^2]]], " ≠ 0"}],

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13], "", Style["Given formulas:", Bold, 14], Style[Row[
  {Style[TraditionalForm[HoldForm[cn t == dn t Sin[σ] / (p q Sin[σ - δ])]],
    Bold], "  if  M < 1 ;      ",
    Style[TraditionalForm[HoldForm[cn t == Sin[σ] / (p q Sin[σ - δ])]],
      Bold], "  if  M > 1"}], 14, s3colGiven], "",
Style[Row[{"Table 1 (the interval of  ", Style["t", Italic], " ):"}],
  Bold, 14], tab1, "",
Style[Row[{"Table 2 (the values of  sn ", Style["t", Italic],
  "  and  cn ", Style["t", Italic], "  on each interval):"}], Bold, 14],
tab2, "", Style[Row[{"Each  ", smallChip[True],
  "  below certifies that the statement displayed beside it has been
  verified by a symbolic computation at that point; a  ",
  smallChip[False], "  would mean that this verification failed."}],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test the following formulas  ", Bold, 14],
  Style["*", 14, s3colTest], Style[" : ", Bold, 14]}]],
Style[Row[{Style[TraditionalForm[HoldForm[sn t ==
  i ∈ Sqrt[(Superscript[dn, 2] t - 1) p^2 q^2] / (k p q)]], Bold],
  ",      ", Style[TraditionalForm[HoldForm[cn t ==
  ε Sqrt[(Superscript[dn, 2] t - k'^2) p^2 q^2] / (k p q)]], Bold]}],
14, s3colTest]], Spacings → 1], Background → RGBColor[
0.99, 0.96, 0.88], FrameMargins → 12]];
Print[
Panel[Column[{Row[{Style["The derivation:  the formulas for  ", Bold, 15,
  s3colHdr], Style[Row[{"sn ", Style["t", Italic]}], Bold,
  15, s3colHdr], Style["  and  ", Bold, 15, s3colHdr],
  Style[Row[{"cn ", Style["t", Italic]}], Bold, 15, s3colHdr],
  Spacer[20], statusChip[okAll]}], Style["Verification:", Bold, 13],
Style["The following identities hold:", 13],
Style[Row[{TraditionalForm[HoldForm[(f - M) (r - 1) (s - 1) / (f (1 - M)) ==
  (Sin[σ] / Sin[σ - δ])^2]], "  ", smallChip[okSq]}],
13], Style[Row[{TraditionalForm[HoldForm[Sin[σ] ==
  ε Sqrt[(f - M) (r - 1) (s - 1) / (f (1 - M))] Sin[σ - δ]], "      ",
  Style["●", 14, RGBColor[0.55, 0.33, 0.10]], "  ", smallChip[okEps]}],
13], Style[Row[{"Hence, substituting the identity  ",
  Style["●", 13, RGBColor[0.55, 0.33, 0.10]], "  into ",
  Style["the given formulas for  cn t", Italic, 13, s3colGiven],
  " ,  and using  ", TraditionalForm[
    HoldForm[p^2 q^2 == (r - 1) (s - 1)]], "  and the values of  ",
    TraditionalForm[HoldForm[k]], " , ", TraditionalForm[HoldForm[k']],
    " ,  dn ", Style["t", Italic], "  above, we get"}], 13],
Style[Row[{TraditionalForm[HoldForm[cn t ==
  ε Sqrt[(Superscript[dn, 2] t - k'^2) p^2 q^2 / k^2] / (p q)],
  "      ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]],
  "  ", Style["( M < 1 : ", 12, GrayLevel[0.3]],
  "  ", smallChip[okSq && okRadLow && okRoot],
  Style[" ;  M > 1 : ", 12, GrayLevel[0.3]], "  ", smallChip[

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okSq&&okRadHigh&&okRoot], Style[" )", 12, GrayLevel[0.3]]], 13],
Style[Row[{"and, because ", TraditionalForm[HoldForm[k > 0]],
" , the identity ", Style["●", 13, RGBColor[0.00, 0.45, 0.25]],
" gives the formula ", Style["*", 13, s3colTest],
" for cn ", Style["t", Italic], " ."}], 13], "", Style[Row[
{"By the given sign identity, the radicals merge on each class of ",
TraditionalForm[HoldForm[p q]], " as follows:"}], 13],
mergeRow[1], mergeRow[2], mergeRow[3], "",
Style[Row[{"Finally, on each cell below, the range of ",
TraditionalForm[HoldForm[σ]], " determines ",
TraditionalForm[HoldForm[ε = Sign[Sin[σ]]]],
" , the interval of t is read from Table 1 and the values of sn
", Style["t", Italic], " and cn ", Style["t", Italic],
" from Table 2, and, substituting the identities ",
Style["●", 13, RGBColor[0.40, 0.15, 0.60]], " , the formulas ",
Style["*", 13, s3colTest], " for sn ", Style["t", Italic],
" and cn ", Style["t", Italic], " (for cn ", Style["t", Italic],
" , this is a second, independent verification of the formula
derived above) agree with these values:"}], 13], Grid[
Table[{Style[lblCell@@cells[[ic]], 12, GrayLevel[0.3]], Column[{Style[
Row[{"sn ", Style["t", Italic], " = ", TraditionalForm[HoldForm[
 $\frac{1}{k} \varepsilon \sqrt{(\text{Superscript}[dn, 2] \, t - 1) \, p^2 \, q^2} / (k \, p \, q)}$ ]], " = ",
snVal[intIdx@@cells[[ic]]}], 12, GrayLevel[0.3]], Style[
Row[{"cn ", Style["t", Italic], " = ", TraditionalForm[HoldForm[
 $\varepsilon \sqrt{(\text{Superscript}[dn, 2] \, t - k'^2) \, p^2 \, q^2} / (k \, p \, q)}$ ]],
" = ", cnVal[intIdx@@cells[[ic]]}], 12, GrayLevel[0.3]]}],
smallChip[gridSn[[ic]]&&gridCn[[ic]]], {ic, Length[cells]}],
Alignment → {Left, Center}, Spacings → {2, 1}], "",
Style[Row[{"Together, these give the formulas ",
Style["*", 13, s3colTest], " ."}], 13}], Spacings → 1],
Background → Lighter[RGBColor[0.15, 0.35, 0.45], 0.93],
FrameMargins →
12]]];];

```

Given hypotheses:

$$\begin{aligned} \alpha, \beta, \gamma, \delta &\in (0, \pi), \quad \sigma = \frac{1}{2}(\alpha + \beta + \gamma + \delta), \quad \sigma - \alpha, \\ \sigma - \beta, \sigma - \gamma, \sigma - \delta &\in (0, \pi), \quad \sin(\sigma) \neq 0, \quad \varepsilon = \operatorname{sgn}(\sin(\sigma)) \in \{\pm 1\} \\ a &= \frac{\sin(\alpha)}{\sin(\sigma - \alpha)}, \quad b = \frac{\sin(\beta)}{\sin(\sigma - \beta)}, \quad c = \frac{\sin(\gamma)}{\sin(\sigma - \gamma)}, \quad d = \frac{\sin(\delta)}{\sin(\sigma - \delta)}, \quad r = ad, \quad s = cd, \quad f = ac, \quad M = abcd \\ r, s, f, M, \frac{M}{f} &\neq 0, 1, \quad f > 0, \quad M > 0 \\ k &= \sqrt{1 - M}, \quad k' = \sqrt{M}, \quad \operatorname{dn} t = \sqrt{f} \quad \text{if } M < 1 \\ k &= \sqrt{1 - \frac{1}{M}}, \quad k' = \sqrt{\frac{1}{M}}, \quad \operatorname{dn} t = \frac{1}{\sqrt{f}} \quad \text{if } M > 1 \\ p &= \sqrt{r - 1}, \quad q = \sqrt{s - 1} \in \mathbb{R}_{>0} \cup i \mathbb{R}_{>0} \\ k, k' &\in (0, 1) \\ \operatorname{sgn}(\operatorname{dn}^2 t - 1) &= \operatorname{sgn}(\operatorname{dn}^2 t - (k')^2) = \operatorname{sgn}(p^2 q^2) \neq 0 \end{aligned}$$

Given formulas:

$$\operatorname{cn} t = \frac{\operatorname{dn} t \sin(\sigma)}{p q \sin(\sigma - \delta)} \quad \text{if } M < 1; \quad \operatorname{cn} t = \frac{\sin(\sigma)}{p q \sin(\sigma - \delta)} \quad \text{if } M > 1$$

Table 1 (the interval of t):

	$p q \in \mathbb{R}_{>0}$	$p q \in i \mathbb{R}_{>0}$	$p q \in \mathbb{R}_{<0}$
$\sigma < \pi$	$(0, iK)$	$(K, K + iK)$	$(2K, 2K + iK)$
$\sigma > \pi$	$(2K, 2K + iK)$	$(3K, 3K + iK)$	$(0, iK)$

Table 2 (the values of $\operatorname{sn} t$ and $\operatorname{cn} t$ on each interval):

t	$(0, iK)$	$(K, K + iK)$	$(2K, 2K + iK)$	$(3K, 3K + iK)$
$\operatorname{sn} t$	$\frac{i \sqrt{\operatorname{dn}^2 t - 1}}{k}$	$\frac{\sqrt{1 - \operatorname{dn}^2 t}}{k}$	$-\frac{i \sqrt{\operatorname{dn}^2 t - 1}}{k}$	$-\frac{\sqrt{1 - \operatorname{dn}^2 t}}{k}$
$\operatorname{cn} t$	$\frac{\sqrt{\operatorname{dn}^2 t - 1 + k^2}}{k}$	$-\frac{i \sqrt{1 - k^2 - \operatorname{dn}^2 t}}{k}$	$-\frac{\sqrt{\operatorname{dn}^2 t - 1 + k^2}}{k}$	$\frac{i \sqrt{1 - k^2 - \operatorname{dn}^2 t}}{k}$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test the following formulas :

$$\operatorname{sn} t = \frac{i \varepsilon \sqrt{(\operatorname{dn}^2 t - 1) p^2 q^2}}{k p q}, \quad \operatorname{cn} t = \frac{\varepsilon \sqrt{(\operatorname{dn}^2 t - (k')^2) p^2 q^2}}{k p q}$$

The derivation: the formulas for $\operatorname{sn} t$ and $\operatorname{cn} t$

Verified

Verification:

The following identities hold:

$$\frac{(f-M)(r-1)(s-1)}{f(1-M)} = \left(\frac{\sin(\sigma)}{\sin(\sigma - \delta)} \right)^2 \quad \checkmark$$

$$\sin(\sigma) = \varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{f(1-M)}} \sin(\sigma - \delta) \quad \bullet \quad \checkmark$$

Hence, substituting the identity  into the given formulas for $\operatorname{cn} t$

, and using $p^2 q^2 = (r-1)(s-1)$ and the values of $k, k', \text{dn } t$ above, we get

$$\text{cn } t = \frac{\varepsilon \sqrt{\frac{(\text{dn}^2 t - (k')^2) p^2 q^2}{k^2}}}{p q} \quad \bullet \quad (M < 1: \checkmark; M > 1: \checkmark)$$

and, because $k > 0$, the identity $\bullet$ gives the formula $\star$ for $\text{cn } t$.

By the given sign identity, the radicals merge on each class of $p q$ as follows:

$$p q \in \mathbb{R}_{>0}: \sqrt{(\text{dn}^2 t - 1) p^2 q^2} = p q \sqrt{\text{dn}^2 t - 1}, \quad \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2} = p q \sqrt{\text{dn}^2 t - (k')^2} \quad \bullet \quad \checkmark$$

$$p q \in i \mathbb{R}_{>0}: \sqrt{(\text{dn}^2 t - 1) p^2 q^2} = -i p q \sqrt{1 - \text{dn}^2 t}, \quad \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2} = -i p q \sqrt{(k')^2 - \text{dn}^2 t} \quad \bullet \quad \checkmark$$

$$p q \in \mathbb{R}_{<0}: \sqrt{(\text{dn}^2 t - 1) p^2 q^2} = -p q \sqrt{\text{dn}^2 t - 1}, \quad \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2} = -p q \sqrt{\text{dn}^2 t - (k')^2} \quad \bullet \quad \checkmark$$

Finally, on each cell below, the range of σ determines $\varepsilon = \text{sgn}(\sin(\sigma))$

, the interval of t is read from Table 1 and the values of $\text{sn } t$ and $\text{cn } t$

from Table 2, and, substituting the identities $\bullet$, the formulas

$\star$ for $\text{sn } t$ and $\text{cn } t$ (for $\text{cn } t$, this is a second, independent

verification of the formula derived above) agree with these values:

$p q \in \mathbb{R}_{>0}, \sigma < \pi \rightarrow \varepsilon = 1, t \in (0, iK')$	$\text{sn } t = \frac{i \varepsilon \sqrt{(\text{dn}^2 t - 1) p^2 q^2}}{k p q} = \frac{i \sqrt{\text{dn}^2 t - 1}}{k}$	$\checkmark$
	$\text{cn } t = \frac{\varepsilon \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2}}{k p q} = \frac{\sqrt{\text{dn}^2 t - 1 + k^2}}{k}$	$\checkmark$
$p q \in \mathbb{R}_{>0}, \sigma > \pi \rightarrow \varepsilon = -1, t \in (2K, 2K + iK')$	$\text{sn } t = \frac{i \varepsilon \sqrt{(\text{dn}^2 t - 1) p^2 q^2}}{k p q} = -\frac{i \sqrt{\text{dn}^2 t - 1}}{k}$	$\checkmark$
	$\text{cn } t = \frac{\varepsilon \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2}}{k p q} = -\frac{\sqrt{\text{dn}^2 t - 1 + k^2}}{k}$	$\checkmark$
$p q \in i \mathbb{R}_{>0}, \sigma < \pi \rightarrow \varepsilon = 1, t \in (K, K + iK')$	$\text{sn } t = \frac{i \varepsilon \sqrt{(\text{dn}^2 t - 1) p^2 q^2}}{k p q} = \frac{\sqrt{1 - \text{dn}^2 t}}{k}$	$\checkmark$
	$\text{cn } t = \frac{\varepsilon \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2}}{k p q} = -\frac{i \sqrt{1 - k^2 - \text{dn}^2 t}}{k}$	$\checkmark$
$p q \in i \mathbb{R}_{>0}, \sigma > \pi \rightarrow \varepsilon = -1, t \in (3K, 3K + iK')$	$\text{sn } t = \frac{i \varepsilon \sqrt{(\text{dn}^2 t - 1) p^2 q^2}}{k p q} = -\frac{\sqrt{1 - \text{dn}^2 t}}{k}$	$\checkmark$
	$\text{cn } t = \frac{\varepsilon \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2}}{k p q} = \frac{i \sqrt{1 - k^2 - \text{dn}^2 t}}{k}$	$\checkmark$
$p q \in \mathbb{R}_{<0}, \sigma < \pi \rightarrow \varepsilon = 1, t \in (2K, 2K + iK')$	$\text{sn } t = \frac{i \varepsilon \sqrt{(\text{dn}^2 t - 1) p^2 q^2}}{k p q} = -\frac{i \sqrt{\text{dn}^2 t - 1}}{k}$	$\checkmark$
	$\text{cn } t = \frac{\varepsilon \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2}}{k p q} = -\frac{\sqrt{\text{dn}^2 t - 1 + k^2}}{k}$	$\checkmark$
$p q \in \mathbb{R}_{<0}, \sigma > \pi \rightarrow \varepsilon = -1, t \in (0, iK')$	$\text{sn } t = \frac{i \varepsilon \sqrt{(\text{dn}^2 t - 1) p^2 q^2}}{k p q} = \frac{i \sqrt{\text{dn}^2 t - 1}}{k}$	$\checkmark$
	$\text{cn } t = \frac{\varepsilon \sqrt{(\text{dn}^2 t - (k')^2) p^2 q^2}}{k p q} = \frac{\sqrt{\text{dn}^2 t - 1 + k^2}}{k}$	$\checkmark$

Together, these give the formulas $\star$.

Section 4

Results of Section 4 are used in the proof of Theorem 1

In[3567]:=

($\star$ ===== $\star$)

```

(*PART 1:SETUP*)
(* =====*)
ClearAll[f, i, j, l, m, c, s, si, sl, fi, fj, fl, fm, M, chi,  $\chi$ , display,
  propEq, standingOf, assumptions, isTrueQ, statusChip, smallChip,
  caseCheck, results, starRef, s4colTest, fj3, fm3, fj4, fm4];
Format[fi, TraditionalForm] = Subscript[f, i];
Format[fl, TraditionalForm] = Subscript[f, l];
Format[chi, TraditionalForm] =  $\chi$ ;
(*Standing hypotheses:M and all four f's are positive,
distinct from 1 and from each other's modulus pairing f#M,
and each pair-product (f-1)(f-M) is positive;
these hold under the given hypotheses.The verification below is branch-wise:
  sign data entering through chi and the regime of M is enumerated explicitly,
so every square root reaching the kernel has a sign-determined
  radicand.Parametrization:the first term of the identity carries zeta1 alone;
in every case the verifying choice is zeta1=chi (the value of chi
  on the branch).All display-only rows are wrapped in HoldForm,
so they cannot evaluate through symbols defined in other sections.*)
standingOf[fjE_, fmE_] := M > 0 && M  $\neq$  1 && And@@ (# > 0 & /@ {fi, fjE, fl, fmE}) &&
  And@@ (#  $\neq$  1 & /@ {fi, fjE, fl, fmE}) && And@@ (#  $\neq$  M & /@ {fi, fjE, fl, fmE}) &&
  And@@ ((# - 1) (# - M) > 0 & /@ {fi, fjE, fl, fmE});
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions  $\rightarrow$  assumptions]];
display[zeta1_, zeta2_, chiv_, fjE_, fmE_] :=
  zeta1 M Sqrt[(fi - 1) (fjE - 1) (fl - 1) (fmE - 1)] + zeta2 (1 - M)
  Sqrt[fi fjE fl fmE] - zeta2 chiv Sqrt[(fi - M) (fjE - M) (fl - M) (fmE - M)];
propEq[zeta1_, zeta2_, chiv_, fjE_, fmE_] := Style[TraditionalForm[
  HoldForm[#] && (display[zeta1, zeta2, chiv, fjE, fmE] == M (1 - M))], 16];
s4colTest = RGBColor[0.10, 0.35, 0.75];
starRef = Style[" $\star$ ", Bold, 14, s4colTest];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background  $\rightarrow$  RGBColor[0.13, 0.55, 0.13], FrameStyle  $\rightarrow$  None,
  RoundingRadius  $\rightarrow$  7, FrameMargins  $\rightarrow$  {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background  $\rightarrow$  RGBColor[0.82, 0.10, 0.10], FrameStyle  $\rightarrow$  None,
  RoundingRadius  $\rightarrow$  7, FrameMargins  $\rightarrow$  {{12, 12}, {4, 4}}];
smallChip[True] := Style[" $\boxtimes$ ", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
(*Two-stage verification per
  case:STAGE 1 (the collapse) identifies the instantiated left-
  hand side with its square-root-free form under the branch assumptions;
STAGE 2 verifies that the collapsed form equals M(1-M).The check
  mark beside the instantiated identity certifies both stages;
the check mark beside the expanding row certifies stage 2;
the chip in the title row is Verified only if both stages hold.**)
caseCheck[title_, col_, condRow_, zeta1_, zeta2_, branchRow_, chiv_,
  fjE_, fmE_, extraAssum_, extraRow_, collapsedHeld_] := Module[

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{okCollapse, okZero, ok}, assumptions = standingOf[fjE, fmE] && extraAssum;
okCollapse =
  isTrueQ[display[zeta1, zeta2, chiv, fjE, fmE], ReleaseHold[collapsedHeld]];
okZero = isTrueQ[ReleaseHold[collapsedHeld], M (1 - M)];
ok = okCollapse && okZero;
Print[
  Panel[Column[{Row[{Style[title, Bold, 15, col], Spacer[16], Style[condRow,
    13, col], Spacer[26], statusChip[ok]}], Row[{Style["testing ", 13],
    starRef, Style[Row[{" with ", Subscript["g", 1], " = ",
      TraditionalForm[zeta1], ", ", Subscript["g", 2], " = ",
      TraditionalForm[zeta2]}], Bold, 13], Spacer[24], branchRow}],
    Row[{propEq[zeta1, zeta2, chiv, fjE, fmE], " ", smallChip[ok]}],
    Row[{Style["expanding the square roots: ", GrayLevel[0.35], 12],
      Style[TraditionalForm[HoldForm[#] && (collapsedHeld == M (1 - M))], 13],
      " ", smallChip[okZero]}],
      Style[Row[{"additionally used: ", extraRow}], GrayLevel[0.35], 12]],
    Spacings → 1], Background → Lighter[col, 0.93], FrameMargins → 12]];
ok];
(* =====*)
(*PART 2:GIVEN PANEL*)
(* =====*)
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[f, i]]], ", ",
    TraditionalForm[HoldForm[M]], " > 0, ", TraditionalForm[HoldForm[
      Subscript[f, i]]], ", ", TraditionalForm[HoldForm[M]], " ≠ 1, ",
      TraditionalForm[HoldForm[Subscript[f, i] ≠ M]], ", ", TraditionalForm[
        HoldForm[(Subscript[f, i] - 1) (Subscript[f, i] - M) > 0]],
        " for each ", TraditionalForm[HoldForm[i]],
        " = 1, ..., 4, ", TraditionalForm[HoldForm[χ ∈ {±1}]]}], 13],
    Style[Row[{"In the cases below, ", TraditionalForm[HoldForm[i]],
      " ", TraditionalForm[HoldForm[j]], " ",
      TraditionalForm[HoldForm[l]], " ", TraditionalForm[HoldForm[m]],
      " ∈ {1, ..., 4} are distinct, that is, ",
      TraditionalForm[HoldForm[{i, j, l, m} = {1, 2, 3, 4}]]}],
      13], Style[Row[{"Each ", smallChip[True],
        " below certifies that the statement displayed beside it has been
        verified by a symbolic computation at that point; a ",
        smallChip[False], " would mean that this verification failed."}],
        12, GrayLevel[0.35]], "", Style[
      Row[{Style["We want to test the following identity ", Bold, 14],
        Style["*", 14, s4colTest],
        Style[Row[{" , for some ", Subscript["g", 1], " ", Subscript["g", 2],
          " ∈ {±1}, in each of the cases below:"}], Bold, 14}]]],
      Style[RawBoxes[FormBox[RowBox[{SubscriptBox["g", "1"], " ", "M", SqrtBox[
        RowBox[{UnderoverscriptBox["", RowBox[{"i", "=", "1"}], "4",
          LimitsPositioning → False], RowBox[{"(", RowBox[{SubscriptBox[
            "f", "i"], "-", "1"}], ")"}]]], "+", SubscriptBox["g", "2"],

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", RowBox[{"(", RowBox[{"1", "-", "M"}], ")"}], SqrtBox[
  RowBox[{"UnderoverscriptBox["\[I]", RowBox[{"i", "=", "1"}], "4",
    LimitsPositioning -> False], SubscriptBox["f", "i"]}], "-",
  SubscriptBox["g", "2"], "", "x", SqrtBox[RowBox[{"UnderoverscriptBox[
    "\[I]", RowBox[{"i", "=", "1"}], "4", LimitsPositioning -> False],
    RowBox[{"(", RowBox[{"SubscriptBox["f", "i"], "-", "M"}], ")"}]]}],
  "=", RowBox[{"M", RowBox[{"(", RowBox[{"1", "-", "M"}], ")"}]]}],
  TraditionalForm]], 16, s4colTest]], Spacings -> 1.1],
Background -> RGBColor[0.99, 0.96, 0.88], FrameMargins ->
14]];
(* =====*)
(*PART 3:CASE-BY-CASE VERIFICATION*)
(* =====*)
results = {};
(*Condition (i):fj=fi, fm=fl;
chi (fi-1) (fl-1)>0;
zeta1=chi=si*sl per branch,
zeta2=1.The four branches enumerate the signs of fi-1 and fl-1;
by the standing pair-products,fi-M and fi-1 have the same sign,
and similarly at l,so chi=si*sl.*)
Do[AppendTo[results, caseCheck["Condition (i):", RGBColor[0.10, 0.30, 0.70],
  Row[{TraditionalForm[HoldForm[Subscript[f, j] == Subscript[f, i]]],
    "", TraditionalForm[HoldForm[Subscript[f, m] == Subscript[f, l]]],
    "", TraditionalForm[
      HoldForm[x (Subscript[f, i] - 1) (Subscript[f, l] - 1) > 0]}], si sl,
1, Style[Row[{"branch: sign(", TraditionalForm[HoldForm[fi - 1]],
  ") = ", si, " sign(", TraditionalForm[HoldForm[fl - 1]], ") = ", sl,
  " -> x = ", si sl, " so \!\(\*SubscriptBox[\(g\), \(\{1\}\)]\) = x"}],
13, GrayLevel[0.3]], si sl, fi, fl, si (fi - 1) > 0 &&
si (fi - M) > 0 && sl (fl - 1) > 0 && sl (fl - M) > 0,
Row[{TraditionalForm[HoldForm[#] &@(si (fi - 1) > 0)], " ",
  TraditionalForm[HoldForm[#] &@(si (fi - M) > 0)],
  " ", TraditionalForm[HoldForm[#] &@(sl (fl - 1) > 0)],
  " ", TraditionalForm[HoldForm[#] &@(sl (fl - M) > 0)],
  " (same-sign facts by the given hypotheses)"}],
  HoldForm[M (fi - 1) (fl - 1) + (1 - M) fi fl - (fi - M) (fl - M)]]],
{si, {1, -1}}, {sl, {1, -1}}];
(*Condition (ii):fi*fj=M, fl*fm=M, that is, fj=M/fi, fm=M/fl;
zeta1=chi(symbolic), zeta2=1;
no sign condition is consumed and the two chi-
dressed terms cancel syntactically.*)
AppendTo[results, caseCheck["Condition (ii):", RGBColor[0.55, 0.25, 0.70],
  Row[{TraditionalForm[HoldForm[Subscript[f, i] Subscript[f, j] == M]],
    "", TraditionalForm[
      HoldForm[Subscript[f, l] Subscript[f, m] == M]}], chi, 1, Style[
  Row[{"single symbolic branch, \!\(\*SubscriptBox[\(g\), \(\{1\}\)]\) = x"}],
13, GrayLevel[0.3]], chi, M / fi, M / fl, True,

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"none ( $\chi$  remains symbolic; no sign condition is consumed)",
HoldForm[chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)] +
  (1 - M) M - chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)]]];
(*Condition (iii): fi+fj-fi*fj=M, fl+fm-fl*fm=M,
that is, fj=(M-fi)/(1-fi), fm=(M-fl)/(1-fl);
chi (1-M)>0;
zeta1=chi=c per branch, zeta2=1. Two branches: chi=+1 with M<1,
and chi=-1 with M>1.*)
fj3 = (M - fi) / (1 - fi); fm3 = (M - fl) / (1 - fl);
Do[AppendTo[results, caseCheck["Condition (iii):",
  RGBColor[0.00, 0.50, 0.50], Row[{TraditionalForm[HoldForm[
    Subscript[f, i] + Subscript[f, j] - Subscript[f, i] Subscript[f, j] == M]],
    ", ", TraditionalForm[HoldForm[
    Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] == M]],
    ", ", TraditionalForm[HoldForm[ $\chi$  (1 - M) > 0]]}], c, 1,
  Style[Row[{"branch:  $\chi$  = ", c, ", ", If[c == 1, "M < 1", "M > 1"],
    ", so  $\backslash!\backslash(*\text{SubscriptBox}[\backslash(\xi\backslash), \backslash(1\backslash)]\backslash) = \chi$ "}],
    13, GrayLevel[0.3]], c, fj3, fm3, If[c == 1, M < 1, M > 1],
  Row[{TraditionalForm[HoldForm[#] &@If[c == 1, M < 1, M > 1]],
    " (this branch of ",
    TraditionalForm[HoldForm[ $\chi$  (1 - M) > 0]], ")"], HoldForm[
    M (1 - M) + (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))] - (1 - M)
    Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))]]], {c, {1, -1}}];
(*Condition (iv): M(fi+fj-1)=fi*fj, M(fl+fm-1)=fl*fm,
that is, fj=M(fi-1)/(fi-M), fm=M(fl-1)/(fl-M);
zeta1=chi=c per branch, and zeta2=-1 if chi(1-M)>0,
zeta2=+1 if chi(1-M)<0, encoded as zeta2=
-c*s with s=sign(1-M). Four branches over chi and the regime of M;
the collapse row uses per-branch literal templates so
the cancelling pair of square-root terms is shown.*)
fj4 = M (fi - 1) / (fi - M); fm4 = M (fl - 1) / (fl - M);
Do[AppendTo[results, caseCheck["Condition (iv):", RGBColor[0.75, 0.35, 0.05],
  Row[{TraditionalForm[HoldForm[M (Subscript[f, i] + Subscript[f, j] - 1) ==
    Subscript[f, i] Subscript[f, j]]], ", ",
    TraditionalForm[HoldForm[M (Subscript[f, l] + Subscript[f, m] - 1) ==
    Subscript[f, l] Subscript[f, m]]}], c, -c s,
  Style[Row[{"branch:  $\chi$  = ", c, ", ", If[s == 1, "M < 1", "M > 1"],
    "  $\rightarrow$  ", If[c s == 1, " $\chi(1-M) > 0$ ", " $\chi(1-M) < 0$ "],
    ",  $\backslash!\backslash(*\text{SubscriptBox}[\backslash(\xi\backslash), \backslash(1\backslash)]\backslash) = \chi$ "}],
    13, GrayLevel[0.3]], c, fj4, fm4, If[s == 1, M < 1, M > 1],
  Row[{TraditionalForm[HoldForm[#] &@If[s == 1, M < 1, M > 1]],
    " (this branch of the rule: ", Subscript[" $\xi$ ", 2], " = -1 if ",
    TraditionalForm[HoldForm[ $\chi$  (1 - M) > 0]], ", ", Subscript[" $\xi$ ", 2],
    " = +1 if ", TraditionalForm[HoldForm[ $\chi$  (1 - M) < 0]], ")"], If[c s == 1,
    HoldForm[(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] -
      (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] + M (1 - M)],
    HoldForm[-(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +

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(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
M (1 - M)]]], {c, {1, -1}}, {s, {1, -1}}];
(* =====*)
(*PART 4:SUMMARY*)
(* =====*)
Print[
  Framed[Style[If[And@@results, "ALL CASES VERIFIED", "SOME CASES FAILED"],
    Bold, White, 16], Background →
    If[And@@results, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
    FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]]];
```

Given hypotheses:

$f_i, M > 0, f_i, M \neq 1, f_i \neq M, (f_i - 1)(f_i - M) > 0$ for each $i = 1, \dots, 4, \chi \in \{\pm 1\}$

In the cases below, $i, j, l, m \in \{1, \dots, 4\}$ are distinct, that is, $\{i, j, l, m\} = \{1, 2, 3, 4\}$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test the following identity ★

, for some $\zeta_1, \zeta_2 \in \{\pm 1\}$, in each of the cases below:

$$\zeta_1 M \sqrt{\prod_{i=1}^4 (f_i - 1)} + \zeta_2 (1 - M) \sqrt{\prod_{i=1}^4 f_i} - \zeta_2 \chi \sqrt{\prod_{i=1}^4 (f_i - M)} = M(1 - M)$$

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

Verified

testing ★ with $\zeta_1 = 1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = 1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M \quad \checkmark$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M \quad \checkmark$

additionally used:

$-1 + f_i > 0, f_i - M > 0, -1 + f_l > 0, f_l - M > 0$ (same-sign facts by the given hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

Verified

testing ★ with $\zeta_1 = -1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = -1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M \quad \checkmark$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M \quad \checkmark$

additionally used:

$-1 + f_i > 0, f_i - M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the given hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

Verified

testing ★ with $\zeta_1 = -1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = -1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M \quad \checkmark$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M \quad \checkmark$

additionally used:

$1 - f_i > 0, -f_i + M > 0, -1 + f_i > 0, f_l - M > 0$ (same-sign facts by the given hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

Verified

testing ★ with $\zeta_1 = 1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = 1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M \quad \checkmark$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M \quad \checkmark$

additionally used:

$1 - f_i > 0, -f_i + M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the given hypotheses)

Condition (ii): $f_i f_j = M, f_l f_m = M$

Verified

testing ★ with $\zeta_1 = \chi, \zeta_2 = 1$ single symbolic branch, $\zeta_1 = \chi$

$$(1 - M) \sqrt{M^2} + \chi M \sqrt{(-1 + f_i)(-1 + f_l) \left(-1 + \frac{M}{f_i}\right) \left(-1 + \frac{M}{f_l}\right)} -$$

$$\chi \sqrt{(f_i - M)(f_l - M) \left(-M + \frac{M}{f_i}\right) \left(-M + \frac{M}{f_l}\right)} = (1 - M) M \quad \checkmark$$

expanding the square roots: $\chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} +$

$$(1 - M) M - \chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} = (1 - M) M \quad \checkmark$$

additionally used: none (χ remains symbolic; no sign condition is consumed)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

Verified

testing ★ with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M < 1$, so $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i + M) (-f_l + M)}{(1-f_i) (1-f_l)}} + M \sqrt{(-1+f_i) (-1+f_l) \left(-1 + \frac{-f_i + M}{1-f_i}\right) \left(-1 + \frac{-f_l + M}{1-f_l}\right)} - \sqrt{(f_i - M) (f_l - M) \left(-M + \frac{-f_i + M}{1-f_i}\right) \left(-M + \frac{-f_l + M}{1-f_l}\right)} = (1-M) M \quad \checkmark$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i (M-f_i) f_l (M-f_l)}{(1-f_i) (1-f_l)}} - (1-M) \sqrt{\frac{f_i (M-f_i) f_l (M-f_l)}{(1-f_i) (1-f_l)}} = (1-M) M \quad \checkmark$$

additionally used: $M < 1$ (this branch of $\chi(1-M) > 0$)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

Verified

testing ★ with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M > 1$, so $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i + M) (-f_l + M)}{(1-f_i) (1-f_l)}} - M \sqrt{(-1+f_i) (-1+f_l) \left(-1 + \frac{-f_i + M}{1-f_i}\right) \left(-1 + \frac{-f_l + M}{1-f_l}\right)} + \sqrt{(f_i - M) (f_l - M) \left(-M + \frac{-f_i + M}{1-f_i}\right) \left(-M + \frac{-f_l + M}{1-f_l}\right)} = (1-M) M \quad \checkmark$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i (M-f_i) f_l (M-f_l)}{(1-f_i) (1-f_l)}} - (1-M) \sqrt{\frac{f_i (M-f_i) f_l (M-f_l)}{(1-f_i) (1-f_l)}} = (1-M) M \quad \checkmark$$

additionally used: $M > 1$ (this branch of $\chi(1-M) > 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

Verified

testing ★ with $\zeta_1 = 1$, $\zeta_2 = -1$ branch: $\chi = 1$, $M < 1 \rightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M) \sqrt{\frac{(-1+f_i) f_j (-1+f_l) f_l M^2}{(f_i-M) (f_l-M)}}\right) + M \sqrt{(-1+f_i) (-1+f_l) \left(-1 + \frac{(-1+f_i) M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l) M}{f_l-M}\right)} + \sqrt{(f_i - M) (f_l - M) \left(-M + \frac{(-1+f_i) M}{f_i-M}\right) \left(-M + \frac{(-1+f_l) M}{f_l-M}\right)} = (1-M) M \quad \checkmark$$

expanding the square roots:

$$(1-M) \sqrt{\frac{f_i (M(f_i-1)) f_l (M(f_l-1))}{(f_i-M) (f_l-M)}} - (1-M) \sqrt{\frac{f_i (M(f_i-1)) f_l (M(f_l-1))}{(f_i-M) (f_l-M)}} + M(1-M) = (1-M) M \quad \checkmark$$

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

Verified

testing ★ with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M > 1 \rightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} + M \sqrt{\left(-1+f_i\right)\left(-1+f_l\right)\left(-1+\frac{(-1+f_i) M}{f_i-M}\right)\left(-1+\frac{(-1+f_l) M}{f_l-M}\right)} - \sqrt{\left(f_i-M\right)\left(f_l-M\right)\left(-M+\frac{(-1+f_i) M}{f_i-M}\right)\left(-M+\frac{(-1+f_l) M}{f_l-M}\right)} = (1-M) M \quad \checkmark$$

expanding the square roots:

$$-(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M) M \quad \checkmark$$

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

Verified

testing ★ with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M < 1 \rightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} - M \sqrt{\left(-1+f_i\right)\left(-1+f_l\right)\left(-1+\frac{(-1+f_i) M}{f_i-M}\right)\left(-1+\frac{(-1+f_l) M}{f_l-M}\right)} + \sqrt{\left(f_i-M\right)\left(f_l-M\right)\left(-M+\frac{(-1+f_i) M}{f_i-M}\right)\left(-M+\frac{(-1+f_l) M}{f_l-M}\right)} = (1-M) M \quad \checkmark$$

expanding the square roots:

$$-(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M) M \quad \checkmark$$

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

Verified

testing ★ with $\zeta_1 = -1$, $\zeta_2 = -1$ branch: $\chi = -1$, $M > 1 \rightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}}\right) - M \sqrt{\left(-1+f_i\right)\left(-1+f_l\right)\left(-1+\frac{(-1+f_i) M}{f_i-M}\right)\left(-1+\frac{(-1+f_l) M}{f_l-M}\right)} - \sqrt{\left(f_i-M\right)\left(f_l-M\right)\left(-M+\frac{(-1+f_i) M}{f_i-M}\right)\left(-M+\frac{(-1+f_l) M}{f_l-M}\right)} = (1-M) M \quad \checkmark$$

expanding the square roots:

$$(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M) M \quad \checkmark$$

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

ALL CASES VERIFIED

Section 5

Results of Section 5 are used in the proof of Lemma 10

In[3270]:=

```
(* =====*)
(*PART 1:SETUP*)
(* =====*)
ClearAll[gi, gj, kk, kq, kk2, kp2, SS, DD, sig, em, snD, cnD, dnD, chkR, statusChip,
  smallChip, s9colTest, s9colHdr, dupOK, relOK, finOK1, finOK2, finOK3, okAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s9colTest = RGBColor[0.10, 0.35, 0.75];
s9colHdr = RGBColor[0.15, 0.35, 0.45];
(*Every check below is a rational identity in a fixed set of atoms,
decided by Together;
no simplification search is involved.Atoms:gi and gj are free;
sig stands for sn t_i and appears only through even powers;
SS is the solved value of sn^2 t_i;
kk2 and kp2 are the resulting values of k^2 and k'^2;
kq is a generic symbol for k^2 in the duplication checks;
em stands for (-1)^m and enters every identity linearly,
so a real symbol suffices;
kk stands for k itself and appears only to the first power in denominators,
where it cancels.Divisions occurring in the displayed identities are by k,
g_i,g_j,sn(2t_i),and 1-k^2 sn^4 t_i,all nonzero by the given
hypotheses.All display-only rows are wrapped in HoldForm.*)
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
dupOK = Module[{dd = 1 - kq sig^4},
  chkR[2 sig (-I gj sig) gi / dd, -2 I gi gj sig^2 / dd] &&
  chkR[((-I gj sig)^2 - sig^2 gi^2) / dd, -(gi^2 + gj^2) sig^2 / dd] &&
  chkR[(gi^2 - kq sig^2 (-I gj sig)^2) / dd, (gi^2 + kq gj^2 sig^4) / dd]];
SS = 1 / (1 - gj^2);
kk2 = (1 - gi^2) (1 - gj^2);
kp2 = 1 - kk2;
relOK = chkR[SS + (-I gj)^2 SS, 1] && chkR[1 - kk2 SS, gi^2];
DD = 1 - kk2 SS^2;
snD = -2 I gi gj SS / DD;
cnD = -(gi^2 + gj^2) SS / DD;
```

```

dnD = (gi^2 + kk2 gj^2 SS^2) / DD;
finOK1 = chkR[em / (kk snD), I em (gi^2 - gj^2) / (2 kk gi gj)];
finOK2 = chkR[em I dnD / (kk snD), em (gi^2 gj^2 - kp2) / (2 kk gi gj)];
finOK3 = chkR[I cnD / snD, (gi^2 + gj^2) / (2 gi gj)];
okAll = dupOK && relOK && finOK1 && finOK2 && finOK3;
(* =====*)
(*PART 2:GIVEN PANEL*)
(* =====*)
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
Style[Row[{"k ∈ (0, 1), ", TraditionalForm[HoldForm[k'^2 = 1 - k^2]]}],
14], Style[Row[{"For ", TraditionalForm[HoldForm[i]], ", ",
TraditionalForm[HoldForm[j]], " ∈ {1, ..., 4} :"}], Bold, 14], Style[
Row[{TraditionalForm[HoldForm[m ∈ ℤ]], ", ", TraditionalForm[HoldForm[
Subscript[t, i] + Subscript[t, j] = 2 m K + I K']}], 14], Style[
Row[{TraditionalForm[HoldForm[Subscript[g, i] = dn[Subscript[t, i]]],
", ", TraditionalForm[HoldForm[Subscript[g, j] = dn[Subscript[t, j]]],
", ", TraditionalForm[HoldForm[Subscript[g, i]]],
", ", TraditionalForm[HoldForm[Subscript[g, j]]],
" ≠ 0, ", TraditionalForm[HoldForm[
cn[Subscript[t, i] = -I Subscript[g, j] sn[Subscript[t, i]]]], 14],
Style[Row[{TraditionalForm[HoldForm[sn[2 Subscript[t, i] ≠ 0]], ", ",
TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, i]]^4 ≠ 0]]}], 14], ""],
Style["Given formulas:", Bold, 14], Style[Row[{TraditionalForm[HoldForm[
{sn, cn, dn}[v + 2 m K] = {(-1)^m sn[v], (-1)^m cn[v], dn[v]}]],
", ", TraditionalForm[HoldForm[{sn, cn, dn}[v - I K'] =
{1 / (k sn[v]), I dn[v] / (k sn[v]), I cn[v] / sn[v]}]], " ",
Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]]}],
14], Style[Row[{TraditionalForm[HoldForm[sn[2 v] =
2 sn[v] cn[v] dn[v] / (1 - k^2 sn[v]^4)]], ", ", TraditionalForm[
HoldForm[cn[2 v] = (cn[v]^2 - sn[v]^2 dn[v]^2) / (1 - k^2 sn[v]^4)]],
", ", TraditionalForm[HoldForm[
dn[2 v] = (dn[v]^2 - k^2 sn[v]^2 cn[v]^2) / (1 - k^2 sn[v]^4)]],
" ", Style["the duplication formulas", Italic, 13,
RGBColor[0.10, 0.30, 0.70]]}], 14],
Style[Row[{TraditionalForm[HoldForm[sn[v]^2 + cn[v]^2 = 1]],
", ", TraditionalForm[HoldForm[dn[v]^2 = 1 - k^2 sn[v]^2]],
" ", Style["the Pythagorean relations",
Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
14], Style[Row[{"Each ", smallChip[True],
" below certifies that the statement displayed beside it has been
verified by a symbolic computation at that point; a ",
smallChip[False], " would mean that this verification failed."}],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test the following formula ", Bold, 14],
Style["*", 14, s9colTest], Style[" :", Bold, 14]]}], Style[
TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[t, i] - Subscript[t, j]] =
{I (-1)^m (Subsuperscript[g, i, 2] - Subsuperscript[g, j, 2]) /

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(2 k Subscript[g, i] Subscript[g, j]),
(-1)^m (Subsuperscript[g, i, 2] Subsuperscript[g, j, 2] - k'^2) /
(2 k Subscript[g, i] Subscript[g, j]),
(Subsuperscript[g, i, 2] + Subsuperscript[g, j, 2]) /
(2 Subscript[g, i] Subscript[g, j])]]], 15, s9colTest]],
Spacings → 1.1], Background → RGBColor[0.99, 0.96, 0.88],
FrameMargins →
14]]];
(* =====*)
(*PART 3:VERIFICATION*)
(* =====*)
Print[Panel[Column[
  Row[{Style["The derivation: the formula for ", Bold, 15, s9colHdr],
    Style[TraditionalForm[HoldForm[{sn, cn, dn}[
      Subscript[t, i] - Subscript[t, j]]], Bold, 15, s9colHdr], Spacer[20],
    statusChip[okAll]]], Style["Verification:", Bold, 13], Style[
Row[{"By ", TraditionalForm[HoldForm[Subscript[t, i] - Subscript[t, j] ==
  2 Subscript[t, i] - (Subscript[t, i] + Subscript[t, j]) ==
  -2 m K + (2 Subscript[t, i] - I K')]], " and ",
  Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]],
  ", at ", TraditionalForm[HoldForm[v == 2 Subscript[t, i]]], " :"}],
13], Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[t, i] -
  Subscript[t, j]] == {(-1)^m / (k sn[2 Subscript[t, i]]),
  (-1)^m I dn[2 Subscript[t, i]] / (k sn[2 Subscript[t, i]]),
  I cn[2 Subscript[t, i]] / sn[2 Subscript[t, i]]}], " "},
  Style["*", 15, RGBColor[0.10, 0.30, 0.70]]], 14], Style[
Row[{"Substituting ", TraditionalForm[HoldForm[cn[Subscript[t, i]] ==
  -I Subscript[g, j] sn[Subscript[t, i]]], " and ",
  TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
  " into ", Style["the duplication formulas",
  Italic, 13, RGBColor[0.10, 0.30, 0.70]], " at ",
  TraditionalForm[HoldForm[v == Subscript[t, i]]], " we get"}], 13],
Style[Row[{TraditionalForm[HoldForm[sn[2 Subscript[t, i]] ==
  -2 I Subscript[g, i] Subscript[g, j]
  sn[Subscript[t, i]]^2 / (1 - k^2 sn[Subscript[t, i]]^4)],
  ", ", TraditionalForm[HoldForm[cn[2 Subscript[t, i]] ==
  -(Subsuperscript[g, i, 2] + Subsuperscript[g, j, 2])
  sn[Subscript[t, i]]^2 / (1 - k^2 sn[Subscript[t, i]]^4)],
  ", ", TraditionalForm[HoldForm[dn[2 Subscript[t, i]] ==
  (Subsuperscript[g, i, 2] + k^2 Subsuperscript[g, j, 2]
  sn[Subscript[t, i]]^4) / (1 - k^2 sn[Subscript[t, i]]^4)],
  " "}, Style["●", 14, RGBColor[0.55, 0.33, 0.10]],
  " ", smallChip[dupOK]]], 13],
Style[Row[{"By ", TraditionalForm[HoldForm[cn[Subscript[t, i]] ==
  -I Subscript[g, j] sn[Subscript[t, i]]], " ",
  TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
  ", and ", Style["the Pythagorean relations",

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      Italic, 13, RGBColor[0.75, 0.35, 0.05]], " :")], 13],
Style[Row[{TraditionalForm[HoldForm[sn[Subscript[t, i]]^2 ==
      1 / (1 - Subsuperscript[g, j, 2])]], " ", TraditionalForm[HoldForm[
      k^2 == (1 - Subsuperscript[g, i, 2]) (1 - Subsuperscript[g, j, 2])]],
"      ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]],
"      ", smallChip[relOK]}], 13],
Style[Row[{"Substituting the identities ",
      Style["●", 13, RGBColor[0.55, 0.33, 0.10]], " and ",
      Style["●", 13, RGBColor[0.00, 0.45, 0.25]], " and the equality ",
      TraditionalForm[HoldForm[k'^2 == 1 - k^2]], " into ",
      Style["★", 13, RGBColor[0.10, 0.30, 0.70]], " , we get"}], 13],
Style[Row[{TraditionalForm[HoldForm[(-1)^m / (k sn[2 Subscript[t, i]]) ==
      I (-1)^m (Subsuperscript[g, i, 2] - Subsuperscript[g, j, 2]) /
      (2 k Subscript[g, i] Subscript[g, j])]],
"      ", smallChip[finOK1]}], 14], Style[Row[{TraditionalForm[
      HoldForm[(-1)^m I dn[2 Subscript[t, i]] / (k sn[2 Subscript[t, i]]) ==
      (-1)^m (Subsuperscript[g, i, 2] Subsuperscript[g, j, 2] - k'^2) /
      (2 k Subscript[g, i] Subscript[g, j])]],
"      ", smallChip[finOK2]}], 14], Style[Row[{TraditionalForm[
      HoldForm[I cn[2 Subscript[t, i]] / sn[2 Subscript[t, i]] ==
      (Subsuperscript[g, i, 2] + Subsuperscript[g, j, 2]) /
      (2 Subscript[g, i] Subscript[g, j])]], "      ", smallChip[finOK3]}],
14], Style[Row[{"Together, these give the formula ",
      Style["★", 13, s9colTest], " ."}], 13]],
Spacings → 1], Background → Lighter[s9colHdr, 0.93],
FrameMargins →
12]];

```

Given hypotheses:

$$k \in (0, 1), \quad (k')^2 = 1 - k^2$$

For $i, j \in \{1, \dots, 4\}$:

$$m \in \mathbb{Z}, \quad t_i + t_j = 2mK + iK'$$

$$g_i = \operatorname{dn}(t_i), \quad g_j = \operatorname{dn}(t_j), \quad g_i, g_j \neq 0, \quad \operatorname{cn}(t_i) = -i g_j \operatorname{sn}(t_i)$$

$$\operatorname{sn}(2t_i) \neq 0, \quad 1 - k^2 \operatorname{sn}(t_i)^4 \neq 0$$

Given formulas:

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(v + 2mK) = \{(-1)^m \operatorname{sn}(v), (-1)^m \operatorname{cn}(v), \operatorname{dn}(v)\}$$

$$, \quad \{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(v - iK') = \left\{ \frac{1}{k \operatorname{sn}(v)}, \frac{i \operatorname{dn}(v)}{k \operatorname{sn}(v)}, \frac{i \operatorname{cn}(v)}{\operatorname{sn}(v)} \right\} \quad \text{the shift formulas}$$

$$\operatorname{sn}(2v) = \frac{2 \operatorname{sn}(v) \operatorname{cn}(v) \operatorname{dn}(v)}{1 - k^2 \operatorname{sn}(v)^4}, \quad \operatorname{cn}(2v) = \frac{\operatorname{cn}(v)^2 - \operatorname{sn}(v)^2 \operatorname{dn}(v)^2}{1 - k^2 \operatorname{sn}(v)^4}$$

$$, \quad \operatorname{dn}(2v) = \frac{\operatorname{dn}(v)^2 - k^2 \operatorname{sn}(v)^2 \operatorname{cn}(v)^2}{1 - k^2 \operatorname{sn}(v)^4} \quad \text{the duplication formulas}$$

$$\operatorname{sn}(v)^2 + \operatorname{cn}(v)^2 = 1, \quad \operatorname{dn}(v)^2 = 1 - k^2 \operatorname{sn}(v)^2 \quad \text{the Pythagorean relations}$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test the following formula  :

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(t_i - t_j) = \left\{ \frac{i (-1)^m (g_i^2 - g_j^2)}{2k g_i g_j}, \frac{(-1)^m (g_i^2 g_j^2 - (k')^2)}{2k g_i g_j}, \frac{g_i^2 + g_j^2}{2g_i g_j} \right\}$$

The derivation: the formula for $\{\text{sn}, \text{cn}, \text{dn}\}(t_i - t_j)$

Verified

Verification:

By $t_i - t_j = 2t_i - (t_i + t_j) = -2mK + (2t_i - iK')$ and *the shift formulas*, at $v = 2t_i$:

$$\{\text{sn}, \text{cn}, \text{dn}\}(t_i - t_j) = \left\{ \frac{(-1)^m}{k \text{sn}(2t_i)}, \frac{(-1)^m i \text{dn}(2t_i)}{k \text{sn}(2t_i)}, \frac{i \text{cn}(2t_i)}{\text{sn}(2t_i)} \right\} \quad \star$$

Substituting $\text{cn}(t_i) = -i g_j \text{sn}(t_i)$ and

$\text{dn}(t_i) = g_i$ into *the duplication formulas*, at $v = t_i$, we get

$$\text{sn}(2t_i) = -\frac{2i g_i g_j \text{sn}(t_i)^2}{1 - k^2 \text{sn}(t_i)^4}, \quad \text{cn}(2t_i) = -\frac{(g_i^2 + g_j^2) \text{sn}(t_i)^2}{1 - k^2 \text{sn}(t_i)^4}, \quad \text{dn}(2t_i) = \frac{g_i^2 + k^2 g_j^2 \text{sn}(t_i)^4}{1 - k^2 \text{sn}(t_i)^4} \quad \bullet \quad \checkmark$$

By $\text{cn}(t_i) = -i g_j \text{sn}(t_i)$, $\text{dn}(t_i) = g_i$, and *the Pythagorean relations*:

$$\text{sn}(t_i)^2 = \frac{1}{1 - g_j^2}, \quad k^2 = (1 - g_i^2)(1 - g_j^2) \quad \bullet \quad \checkmark$$

Substituting the identities $\bullet$ and $\bullet$ and the equality $(k')^2 = 1 - k^2$ into $\star$, we get

$$\frac{(-1)^m}{k \text{sn}(2t_i)} = \frac{i (-1)^m (g_i^2 - g_j^2)}{2 k g_i g_j} \quad \checkmark$$

$$\frac{(-1)^m i \text{dn}(2t_i)}{k \text{sn}(2t_i)} = \frac{(-1)^m (g_i^2 g_j^2 - (k')^2)}{2 k g_i g_j} \quad \checkmark$$

$$\frac{i \text{cn}(2t_i)}{\text{sn}(2t_i)} = \frac{g_i^2 + g_j^2}{2 g_i g_j} \quad \checkmark$$

Together, these give the formula $\star$.

Section 6

Results of Section 6 are used in the proof of Lemma 11

In[3231]:=

```
( * ===== * )
(*PART 1:SETUP*)
( * ===== * )
ClearAll[gi, gj, kk, kp2, mi, mj, eI, eJ, chkR, statusChip, smallChip,
s10colTest, clSym, mkL, braceGrp, pairs3, pairsLoc, resOf, prodOf,
cmpOf, locOf, lblPair, resList, prodList, cmpGrid, snList, cnList,
dnList, locList, PiT, SPt, r1t, r1pt, r2t, r3t, r4t, xiT, epsT, kkT,
DG, Dden, asmSn, asmCn, asmDn, asmLoc, ok3, okLoc, resultsAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s10colTest = RGBColor[0.10, 0.35, 0.75];
```

```

(*Every check below is a rational identity in a fixed set of radical atoms,
decided by Together;
no simplification search is involved.Class
regions:each of p_i q_i and p_j q_j lies in one of the three classes R_{>0},
i R_{>0},R_{<0},written p q=m,i m,-m with an atom m>0;
the sign s of p^2 q^2 is then+1,-1,+1,and,by the given sign identities,
s is also the sign of g^2-1 and of g^2-k'^2. On each class the atoms A=
Sqrt[s(g^2-1)] and B=Sqrt[s(g^2-k'^2)] are real principal
roots of positive radicands,and every radical occurring in the
given and tested formulas resolves into a product of these atoms;
each resolution is certified by squaring,which is a rational identity,
and both sides are nonnegative by construction.The verification is
layered:the product identities are verified on each class region;
the assembly of the product identities into the tested
formulas is verified once,with the radicals as free symbols;
and the tested formulas are also verified end-to-
end on each class region.k'^2 enters only through the atoms,so it stays
a free symbol kp2.e_i and e_j enter every identity in degree at most one,
so they stay symbolic reals,covering both sign choices.eps_i and eps_j enter
only through the product eps_i eps_j and its square,so evaluating at eps_i=
eps with eps in {1,-1} and eps_j=1 covers all four sign choices.The first
equality of the second tested chain is analytic and enters as a given;
its remaining equalities are verified here.Divisions occurring in the
identities are by k,p_l q_l,and k^2-(g_i^2-1)(g_j^2-1),all nonzero by
the given hypotheses.All display-only rows are wrapped in HoldForm.*)
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
clSym[1] = Subscript["R", "> 0"];
clSym[2] = Row[{"i ", Subscript["R", "> 0"]];
clSym[3] = Subscript["R", "< 0"];
lblPair[{ci_, cj_}] :=
  Row[{TraditionalForm[HoldForm[Subscript[p, i] Subscript[q, i]]],
    " ∈ ", clSym[ci], ", ", TraditionalForm[
      HoldForm[Subscript[p, j] Subscript[q, j]]], " ∈ ", clSym[cj]}}];
braceGrp[items_List, sz_ : 13, col_ : GrayLevel[0]] := DisplayForm[
  RowBox[{GridBox[({ToBoxes[Style[#, sz, col]]} & /@ items), GridBoxAlignment →
    {"Columns" → {{Left}}}, GridBoxSpacings → {"Rows" → {{1.8}}},
    StyleBox[""]", FontColor → GrayLevel[0.55], SpanMaxSize → Infinity]]]];
mkL[c_, g_, m_] := <|"pq" → Switch[c, 1, m, 2, I m, 3, -m],
  "s" → If[c == 2, -1, 1], "A" → If[c == 2, Sqrt[1 - g^2], Sqrt[g^2 - 1]],
  "B" → If[c == 2, Sqrt[kp2 - g^2], Sqrt[g^2 - kp2]]|>;
pairs3 = Tuples[{1, 2, 3}, 2];
pairsLoc = Select[pairs3, mkL[#[[1]], gi, mi] ["s"] mkL[#[[2]], gj, mj] ["s"] == 1 &];
resOf[{ci_, cj_}] := Module[
  {Li = mkL[ci, gi, mi], Lj = mkL[cj, gj, mj], xi, snI, snJ}, xi = Li ["s"] Lj ["s"];
  snI = I mi Li ["A"] / (kk Li ["pq"]);
  snJ = I mj Lj ["A"] / (kk Lj ["pq"]);
  chkR[(mi Li ["A"])^2, (gi^2 - 1) Li ["pq"]^2] &&

```

```

chkR[(mj Lj["A"])^2, (gj^2 - 1) Lj["pq"]^2] &&
chkR[(mi Li["B"])^2, (gi^2 - kp2) Li["pq"]^2] && chkR[(mj Lj["B"])^2,
  (gj^2 - kp2) Lj["pq"]^2] && chkR[(mi mj)^2, Li["pq"]^2 Lj["pq"]^2 xi] &&
chkR[(Li["A"] Lj["B"])^2, (gi^2 - 1) (gj^2 - kp2) xi] &&
chkR[(Lj["A"] Li["B"])^2, (gj^2 - 1) (gi^2 - kp2) xi] &&
chkR[(Li["B"] Lj["B"])^2, (gi^2 - kp2) (gj^2 - kp2) xi] &&
chkR[(Li["A"] Lj["A"])^2, (gi^2 - 1) (gj^2 - 1) xi] &&
chkR[(Li["A"] Li["B"] Lj["A"] Lj["B"])^2,
  (gi^2 - 1) (gi^2 - kp2) (gj^2 - 1) (gj^2 - kp2)] &&
chkR[1 - kk^2 snI^2 snJ^2, (kk^2 - (gi^2 - 1) (gj^2 - 1)) / kk^2]]];
prodOf[{ci_, cj_}, ep_] := Module[{Li = mkL[ci, gi, mi], Lj = mkL[cj, gj, mj],
  xi, sni, cni, snj, cnj, SPv, PPv}, xi = Li["s"] Lj["s"];
sni = I ep mi Li["A"] / (kk Li["pq"]);
cni = ep mi Li["B"] / (kk Li["pq"]);
snj = I mj Lj["A"] / (kk Lj["pq"]);
cnj = mj Lj["B"] / (kk Lj["pq"]);
SPv = mi mj; PPv = Li["pq"] Lj["pq"];
chkR[sni cnj, I ep SPv Li["A"] Lj["B"] / (kk^2 PPv)] &&
chkR[snj cni, I ep SPv Lj["A"] Li["B"] / (kk^2 PPv)] &&
chkR[cni cnj, ep SPv Li["B"] Lj["B"] / (kk^2 PPv)] &&
chkR[sni snj, -ep SPv Li["A"] Lj["A"] / (kk^2 PPv)] &&
chkR[sni snj cni cnj, -xi Li["A"] Li["B"] Lj["A"] Lj["B"] / kk^4]];
cmpOf[{ci_, cj_}, ep_] := Module[{Li = mkL[ci, gi, mi], Lj = mkL[cj, gj, mj], xi,
  sni, cni, snj, cnj, den2, SV, dTar, h1V, h2V, h3V}, xi = Li["s"] Lj["s"];
sni = I ep mi Li["A"] / (kk Li["pq"]);
cni = ep mi Li["B"] / (kk Li["pq"]);
snj = I mj Lj["A"] / (kk Lj["pq"]);
cnj = mj Lj["B"] / (kk Lj["pq"]);
den2 = 1 - kk^2 sni^2 snj^2;
SV = ep mi mj / (Li["pq"] Lj["pq"]);
dTar = kk^2 - (gi^2 - 1) (gj^2 - 1);
h1V = (eI gj Li["A"] Lj["B"] + eJ gi Lj["A"] Li["B"]) / dTar;
h2V = (Li["B"] Lj["B"] + eI eJ gi gj Li["A"] Lj["A"]) / dTar;
h3V = (kk^2 gi gj + eI eJ xi Li["A"] Li["B"] Lj["A"] Lj["B"]) / dTar;
{chkR[(eI sni cnj gj + eJ snj cni gi) / den2, I SV h1V],
  chkR[(cni cnj - eI eJ sni snj gi gj) / den2, SV h2V],
  chkR[(gi gj - kk^2 eI eJ sni snj cni cnj) / den2, h3V]}}];
locOf[{ci_, cj_}, ep_] := Module[{Li = mkL[ci, gi, mi],
  Lj = mkL[cj, gj, mj], kkL, sni, snj}, kkL = Li["A"] Lj["A"];
sni = I ep mi Li["A"] / (kkL Li["pq"]);
snj = I mj Lj["A"] / (kkL Lj["pq"]);
chkR[kkL^2, (gi^2 - 1) (gj^2 - 1)] &&
chkR[(mi mj)^2, Li["pq"]^2 Lj["pq"]^2] &&
chkR[-kkL sni snj, ep mi mj / (Li["pq"] Lj["pq"])]];
resList = resOf /@ pairs3;
prodList = Table[And@@Table[prodOf[pr, ep], {ep, {1, -1}}], {pr, pairs3}];
cmpGrid = Table[cmpOf[pr, ep], {pr, pairs3}, {ep, {1, -1}}];

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snList = Table[And@@cmpGrid[ip, All, 1], {ip, Length[pairs3]};
cnList = Table[And@@cmpGrid[ip, All, 2], {ip, Length[pairs3]};
dnList = Table[And@@cmpGrid[ip, All, 3], {ip, Length[pairs3]};
locList = Table[And@@Table[locOf[pr, ep], {ep, {1, -1}}, {pr, pairsLoc}];
(*Assembly checks:
the product identities are substituted into the addition-formula expansions,
with the radicals SPt (for Sqrt[p^2q^2p^2q^2 xi]),r1t,r1pt,
r2t,r3t,r4t (for the composite radicals),
the product PiT=p_i q_i p_j q_j,the sign xiT,
and the product epsT=eps_i eps_j all free symbols;
each check is a single rational identity,independent of the class region.*)
DG = (gi^2 - 1) (gj^2 - 1);
Dden = (kk^2 - DG) / kk^2;
asmSn = chKR[
(eI (I epsT SPt r1t / (kk^2 PiT)) gj + eJ (I epsT SPt r1pt / (kk^2 PiT)) gi) /
Dden, I (epsT SPt / PiT) (eI gj r1t + eJ gi r1pt) / (kk^2 - DG)];
asmCn =
chKR[(epsT SPt r2t / (kk^2 PiT)) - eI eJ (- (epsT SPt r3t / (kk^2 PiT))) gi gj) /
Dden, (epsT SPt / PiT) (r2t + eI eJ gi gj r3t) / (kk^2 - DG)];
asmDn = chKR[(gi gj - kk^2 eI eJ (- (xiT r4t / kk^4))) / Dden,
(kk^2 gi gj + eI eJ xiT r4t) / (kk^2 - DG)];
asmLoc = chKR[-kkT (- (epsT SPt kkT / (kkT^2 PiT))), epsT SPt / PiT];
ok3 = (And@@resList) && (And@@prodList) && asmSn && asmCn &&
asmDn && (And@@snList) && (And@@cnList) && (And@@dnList);
okLoc = asmLoc && (And@@locList);
resultsAll = {ok3, okLoc};
(* =====*)
(*PART 2:GIVEN PANEL*)
(* =====*)
Print[Panel[
Column[{Style["Given hypotheses:", Bold, 14], Style["k, k' ∈ (0, 1)", 14],
Style[Row[{"For ", TraditionalForm[HoldForm[i]], ", ",
TraditionalForm[HoldForm[j]], " ∈ {1, ..., 4}, ", TraditionalForm[
HoldForm[i ≠ j]], " :"}], Bold, 14], Style[Row[{TraditionalForm[
HoldForm[Subscript[g, i] = dn[Subscript[t, i]]], ", ",
TraditionalForm[HoldForm[Subscript[g, j] = dn[Subscript[t, j]]],
" ∈ (0, k') ∪ (1, ∞), ", TraditionalForm[HoldForm[Subscript[p, i]]],
", ", TraditionalForm[HoldForm[Subscript[q, i]]],
", ", TraditionalForm[HoldForm[Subscript[p, j]]], ", ",
TraditionalForm[HoldForm[Subscript[q, j]]], " ∈ ",
Subscript["ℝ", "> 0"], " ∪ ℤ ", Subscript["ℝ", "> 0"]}], 14],
Style[Row[{TraditionalForm[HoldForm[Sign[Subsuperscript[g, i, 2] - 1] ==
Sign[Subsuperscript[g, i, 2] - Superscript[k', 2]] ==
Sign[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]]], " ≠ 0, ",
TraditionalForm[HoldForm[Sign[Subsuperscript[g, j, 2] - 1] ==
Sign[Subsuperscript[g, j, 2] - Superscript[k', 2]] ==
Sign[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]],

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" ≠ 0", " ", Style["the sign identities", Italic,
13, RGBColor[0.00, 0.50, 0.50]]], 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, ij] =
Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]],
" , " , TraditionalForm[HoldForm[Subscript[e, i]]], " , ",
TraditionalForm[HoldForm[Subscript[e, j]]], " , ",
TraditionalForm[HoldForm[Subscript[ε, i]]], " , ",
TraditionalForm[HoldForm[Subscript[ε, j]]], " , ",
TraditionalForm[HoldForm[Subscript[ξ, ij]]], " ∈ {±1}"]], 14],
"", Style["Given formulas and expressions:", Bold, 14],
Grid[{{braceGrp[{TraditionalForm[HoldForm[sn[Subscript[t, l]] =
I Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - 1)
Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] / (k
Subscript[p, l] Subscript[q, l])]],
TraditionalForm[HoldForm[cn[Subscript[t, l]] = Subscript[ε, l]
Sqrt[(Subsuperscript[g, l, 2] - Superscript[k', 2])
Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] /
(k Subscript[p, l] Subscript[q, l])]]}], 14],
Style[Row[{TraditionalForm[HoldForm[l]], " ∈ {" , TraditionalForm[
HoldForm[i]], " , " , TraditionalForm[HoldForm[j]], "}" }], 13]]},
Alignment → {Left, Center}, Spacings → {1.5, 0}],
Style[TraditionalForm[HoldForm[Subscript[S, ij] = Subscript[ε, i]
Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
Subsuperscript[p, j, 2] Subsuperscript[q, j, 2] Subscript[ξ, ij]] /
(Subscript[p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]],
14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] =
(Subscript[e, i] Subscript[g, j] Sqrt[(Subsuperscript[g, i, 2] - 1)
(Subsuperscript[g, j, 2] - Superscript[k', 2])
Subscript[ξ, ij]] + Subscript[e, j] Subscript[g, i]
Sqrt[(Subsuperscript[g, j, 2] - 1) (Subsuperscript[g, i, 2] -
Superscript[k', 2]) Subscript[ξ, ij]]) /
(k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1))]],
14], Style[TraditionalForm[HoldForm[Subscript[h, ij2] =
(Sqrt[(Subsuperscript[g, i, 2] - Superscript[k', 2])
(Subsuperscript[g, j, 2] - Superscript[k', 2]) Subscript[ξ, ij]] +
Subscript[e, i] Subscript[e, j] Subscript[g, i]
Subscript[g, j] Sqrt[(Subsuperscript[g, i, 2] - 1)
(Subsuperscript[g, j, 2] - 1) Subscript[ξ, ij]]) /
(k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1))]],
14], Style[TraditionalForm[HoldForm[Subscript[h, ij3] =
(k^2 Subscript[g, i] Subscript[g, j] + Subscript[e, i] Subscript[e, j]
Subscript[ξ, ij] Sqrt[(Subsuperscript[g, i, 2] - 1)
(Subsuperscript[g, i, 2] - Superscript[k', 2]) (Subsuperscript[
g, j, 2] - 1) (Subsuperscript[g, j, 2] - Superscript[k', 2]))] /
(k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1))]],
14], Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn][-v] =

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      {-sn[v], cn[v], dn[v]]}], "    ", Style["the parity formulas",
      Italic, 13, RGBColor[0.75, 0.35, 0.05]]], 14],
Style[Row[{TraditionalForm[HoldForm[sn[v + w] == (sn[v] cn[w] dn[w] +
      sn[w] cn[v] dn[v]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
      "    ", Style["the first addition formula", Italic, 13,
      RGBColor[0.10, 0.30, 0.70]]], 14],
Style[Row[{TraditionalForm[HoldForm[cn[v + w] == (cn[v] cn[w] -
      sn[v] sn[w] dn[v] dn[w]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
      "    ", Style["the second addition formula", Italic, 13,
      RGBColor[0.10, 0.30, 0.70]]], 14],
Style[Row[{TraditionalForm[HoldForm[dn[v + w] == (dn[v] dn[w] -
      k^2 sn[v] sn[w] cn[v] cn[w]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
      "    ", Style["the third addition formula", Italic, 13,
      RGBColor[0.10, 0.30, 0.70]]],
14], Style[Row[{"Each    ", smallChip[True],
      " below certifies that the statement displayed beside it has been
      verified by a symbolic computation at that point; a    ",
      smallChip[False], " would mean that this verification failed."}],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test these formulas    ", Bold, 14],
      Style["*"], 14, s10colTest], Style[" :"], Bold, 14}}],
Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[
      Subscript[e, i] Subscript[t, i] + Subscript[e, j] Subscript[t, j]] ==
      {I Subscript[S, ij] Subscript[h, ij1], Subscript[S, ij]
      Subscript[h, ij2], Subscript[h, ij3}}]], "    ",
      Style[Row[{"in the case    ", TraditionalForm[HoldForm[k^2 ≠
      (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]}],
      Italic, 13, GrayLevel[0.35]]], 15, s10colTest],
Style[Row[{TraditionalForm[HoldForm[(-1)^m == -k sn[Subscript[t, i]]
      sn[Subscript[t, j]] == Subscript[ε, i] Subscript[ε, j] Sqrt[
      Subsuperscript[p, i, 2] Subsuperscript[q, i, 2] Subsuperscript[p,
      j, 2] Subsuperscript[q, j, 2]] / (Subscript[p, i] Subscript[q,
      i] Subscript[p, j] Subscript[q, j]) == Subscript[S, ij]]],
      "    ", Style[Row[{"in the case    ", TraditionalForm[HoldForm[k^2 ==
      (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]}],
      Italic, 13, GrayLevel[0.35]]], 15, s10colTest]],
Spacings → 1.1], Background → RGBColor[
0.99,
0.96,
0.88], FrameMargins → 14]];
(* =====*)
(*PART 3:CASE k^2≠(gi^2-1)(gj^2-1)*)
(* =====*)
Print[Panel[
Column[{Row[{Style["The derivation: the formula for    ", Bold, 15, RGBColor[
0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[{sn, cn, dn}[
      Subscript[e, i] Subscript[t, i] + Subscript[e, j] Subscript[t, j]]],

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Bold, 15, RGBColor[0.10, 0.30, 0.70]], Style[" in the case ",
Bold, 15, RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[
  k^2 ≠ (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)],
Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok3]]],
Style["Given hypotheses, in addition to those above:", Bold, 13], Style[
Row[{TraditionalForm[HoldForm[±Subscript[t, i]]], ", ", TraditionalForm[
  HoldForm[±Subscript[t, j]]], " are not poles, ", TraditionalForm[
  HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2 =
    (k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)) /
    k^2 ≠ 0]]], 13], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[±Subscript[t, i]]],
  " and ", TraditionalForm[HoldForm[±Subscript[t, j]]],
  " are not poles and ", TraditionalForm[
    HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2 ≠ 0]],
  ", ", Style["the addition formulas", Italic, 13,
    RGBColor[0.10, 0.30, 0.70]], ", at ",
  TraditionalForm[HoldForm[v = Subscript[e, i] Subscript[t, i]]],
  ", ", TraditionalForm[HoldForm[w = Subscript[e, j] Subscript[t, j]]],
  ", hold pointwise and, together with ", Style["the parity formulas",
    Italic, 13, RGBColor[0.75, 0.35, 0.05]], " and ", TraditionalForm[
    HoldForm[dn[Subscript[t, l]] = Subscript[g, l]]], " , give"}], 13],
Grid[{{braceGrp[{TraditionalForm[HoldForm[sn[Subscript[e, i]
  Subscript[t, i] + Subscript[e, j] Subscript[t, j]] =
    (Subscript[e, i] sn[Subscript[t, i]] cn[Subscript[t, j]]
    Subscript[g, j] + Subscript[e, j] sn[Subscript[t, j]]
    cn[Subscript[t, i]] Subscript[g, i]) / (1 - k^2 sn[Subscript[
      t, i]]^2 sn[Subscript[t, j]]^2)], TraditionalForm[
    HoldForm[cn[Subscript[e, i] Subscript[t, i] + Subscript[e, j]
    Subscript[t, j]] = (cn[Subscript[t, i]] cn[Subscript[t, j]] -
    Subscript[e, i] Subscript[e, j] sn[Subscript[t, i]]
    sn[Subscript[t, j]] Subscript[g, i] Subscript[g, j]) /
    (1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2)],
    TraditionalForm[HoldForm[dn[Subscript[e, i] Subscript[t, i] +
    Subscript[e, j] Subscript[t, j]] = (Subscript[g, i] Subscript[g,
    j] - k^2 Subscript[e, i] Subscript[e, j] sn[Subscript[t, i]]
    sn[Subscript[t, j]] cn[Subscript[t, i]] cn[Subscript[t, j]]) /
    (1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2)]}],
    14], Style["*"], 15, RGBColor[0.10, 0.30, 0.70]]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}],
Style[Row[{"By ", Style["the given sign identities",
  Italic, 13, RGBColor[0.00, 0.50, 0.50]],
  " , every radicand below is positive and the radicals
  resolve as follows:"}], 13], Grid[
{{braceGrp[{Row[{TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, l, 2] -
  1) Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] = Sqrt[
    Abs[Subsuperscript[g, l, 2] - 1] Sqrt[Abs[Subsuperscript[p,
    l, 2] Subsuperscript[q, l, 2]]]]], ", ", TraditionalForm[

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HoldForm[Sqrt[(Subsuperscript[g, l, 2] - Superscript[k', 2])
  Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] ==
  Sqrt[Abs[Subsuperscript[g, l, 2] - Superscript[k', 2]]]
  Sqrt[Abs[Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]]]],
",  ", TraditionalForm[HoldForm[l]], " ∈ {" , TraditionalForm[
  HoldForm[i]], " , TraditionalForm[HoldForm[j]], "}" ]],
TraditionalForm[HoldForm[Sqrt[Subsuperscript[p, i, 2]
  Subsuperscript[q, i, 2] Subsuperscript[p, j,
  2] Subsuperscript[q, j, 2] Subscript[ξ, ij]] ==
  Sqrt[Abs[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]]]
  Sqrt[Abs[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]]]],
Row[{TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1)
  (Subsuperscript[g, j, 2] - Superscript[k', 2])
  Subscript[ξ, ij]] == Sqrt[Abs[Subsuperscript[g, i, 2] - 1]]
  Sqrt[Abs[Subsuperscript[g, j, 2] - Superscript[k', 2]]]]],
",  ", TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, j, 2] - 1)
  (Subsuperscript[g, i, 2] - Superscript[k', 2])
  Subscript[ξ, ij]] == Sqrt[Abs[Subsuperscript[g, j, 2] - 1]]
  Sqrt[Abs[Subsuperscript[g, i, 2] - Superscript[k', 2]]]]]]],
Row[{TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] -
  Superscript[k', 2]) (Subsuperscript[g, j, 2] -
  Superscript[k', 2]) Subscript[ξ, ij]] ==
  Sqrt[Abs[Subsuperscript[g, i, 2] - Superscript[k', 2]]]
  Sqrt[Abs[Subsuperscript[g, j, 2] - Superscript[k', 2]]]]],
",  ", TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1)
  (Subsuperscript[g, j, 2] - 1) Subscript[ξ, ij]] ==
  Sqrt[Abs[Subsuperscript[g, i, 2] - 1]] Sqrt[
  Abs[Subsuperscript[g, j, 2] - 1]]]]], TraditionalForm[
  HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[
    g, i, 2] - Superscript[k', 2]) (Subsuperscript[g, j, 2] - 1)
    (Subsuperscript[g, j, 2] - Superscript[k', 2))] == Sqrt[Abs[
      Subsuperscript[g, i, 2] - 1]] Sqrt[Abs[Subsuperscript[g, i, 2] -
      Superscript[k', 2]]] Sqrt[Abs[Subsuperscript[g, j, 2] - 1]]
      Sqrt[Abs[Subsuperscript[g, j, 2] - Superscript[k', 2]]]]]]],
13], Style["●", 14, RGBColor[0.00, 0.45, 0.25]]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}],
Style[Row[{"Multiplying the given formulas for ",
  TraditionalForm[HoldForm[sn[Subscript[t, l]]]],
  " and ", TraditionalForm[HoldForm[cn[Subscript[t, l]]]],
  " , and using the identities " ,
  Style["●", 13, RGBColor[0.00, 0.45, 0.25]],
  " , we get the product identities:"}], 13],
Grid[{{braceGrp[{TraditionalForm[HoldForm[sn[Subscript[t, i]]
  cn[Subscript[t, j]] == I Subscript[ε, i] Subscript[ε, j] Sqrt[
  Subsuperscript[p, i, 2] Subsuperscript[q, i, 2] Subsuperscript[
  p, j, 2] Subsuperscript[q, j, 2] Subscript[ξ, ij]]
  Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] -

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Superscript[k', 2]) Subscript[ξ, ij]] / (k^2 Subscript[p,
i] Subscript[q, i] Subscript[p, j] Subscript[q, j]]],
TraditionalForm[HoldForm[sn[Subscript[t, j]] cn[Subscript[t, i]] =
I Subscript[ε, i] Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2]
Subsuperscript[q, i, 2] Subsuperscript[p, j, 2]
Subsuperscript[q, j, 2] Subscript[ξ, ij]]
Sqrt[(Subsuperscript[g, j, 2] - 1) (Subsuperscript[g, i, 2] -
Superscript[k', 2]) Subscript[ξ, ij]] / (k^2 Subscript[p,
i] Subscript[q, i] Subscript[p, j] Subscript[q, j]]],
TraditionalForm[HoldForm[cn[Subscript[t, i]] cn[Subscript[t, j]] =
Subscript[ε, i] Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2]
Subsuperscript[q, i, 2] Subsuperscript[p, j, 2] Subsuperscript[
q, j, 2] Subscript[ξ, ij]] Sqrt[(Subsuperscript[g, i, 2] -
Superscript[k', 2]) (Subsuperscript[g, j, 2] - Superscript[
k', 2]) Subscript[ξ, ij]] / (k^2 Subscript[p, i]
Subscript[q, i] Subscript[p, j] Subscript[q, j]]],
TraditionalForm[HoldForm[sn[Subscript[t, i]] sn[Subscript[t, j]] =
-Subscript[ε, i] Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2]
Subsuperscript[q, i, 2] Subsuperscript[p, j, 2]
Subsuperscript[q, j, 2] Subscript[ξ, ij]]
Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)
Subscript[ξ, ij]] / (k^2 Subscript[p, i]
Subscript[q, i] Subscript[p, j] Subscript[q, j]]],
TraditionalForm[HoldForm[sn[Subscript[t, i]] sn[Subscript[t, j]]
cn[Subscript[t, i]] cn[Subscript[t, j]] = -Subscript[ξ, ij]
Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, i, 2] -
Superscript[k', 2]) (Subsuperscript[g, j, 2] - 1)
(Subsuperscript[g, j, 2] - Superscript[k', 2])] / k^4]], 13],
Style["●", 14, RGBColor[0.55, 0.33, 0.10]]}], Alignment →
{Left, Center}, Spacings →
{1.5, 0}],
Style[Row[{"which are verified separately on each class region of ",
"( ", TraditionalForm[
HoldForm[Subscript[p, i] Subscript[q, i]]], ", ",
TraditionalForm[HoldForm[Subscript[p, j] Subscript[q, j]]],
" )", " , for both values of ",
TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]],
" :"}], 13],
Grid[Join[{{Style["", 11], Style[Row[{"the identities ", Style["●",
12, RGBColor[0.00, 0.45, 0.25]]}], 11, Bold, GrayLevel[0.3]],
Style[Row[{"the product identities ", Style["●", 12,
RGBColor[0.55, 0.33, 0.10]]}], 11, Bold, GrayLevel[0.3]]}],
Table[{Style[lblPair[pairs3[[ip]]], 12, GrayLevel[0.3]],
smallChip[resList[ip]],
smallChip[prodList[ip]]}, {ip, Length[pairs3]}]],
Alignment → Left, Spacings → {2, 0.7}], "", Style[
Row[

```

```

{"Thus, substituting the product identities ",
 Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
 " and the given equality for ",
 TraditionalForm[
 HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2]],
 " into the expansions ", Style["*", 13,
 RGBColor[0.10, 0.30, 0.70]], " , with ",
 TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]],
 " , ", TraditionalForm[HoldForm[Subscript[e, i]]],
 " , ", TraditionalForm[HoldForm[Subscript[e, j]]],
 " , and the radicals of the identities ",
 Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
 " symbolic, that is, for all their real values
 at once, we get the first formula of ",
 Style["*", 13, s10colTest], " : sn ", smallChip[asmSn],
 " cn ", smallChip[asmCn],
 " dn ", smallChip[asmDn]]], 13], "",
 Style[Row[{"The first formula of ", Style["*", 13, s10colTest],
 " is also verified end-to-end separately
 on each class region, with ",
 TraditionalForm[HoldForm[Subscript[e, i]]], " , ",
 TraditionalForm[HoldForm[Subscript[e, j]]],
 " symbolic and for both values of ",
 TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]],
 " :"}], 13],
 Grid[Join[{{Style["", 11], Style["sn", 11, Bold, GrayLevel[0.3]],
 Style["cn", 11, Bold, GrayLevel[0.3]],
 Style["dn", 11, Bold, GrayLevel[0.3]]}},
 Table[{Style[lblPair[pairs3[[ip]], 12, GrayLevel[0.3]],
 smallChip[snList[[ip]], smallChip[cnList[[ip]],
 smallChip[dnList[[ip]]]}, {ip, Length[pairs3]}]],
 Alignment → Left, Spacings → {2, 0.7}}],
 Spacings → 1], Background →
 Lighter[
 RGBColor[
 0.10,
 0.30,
 0.70],
 0.93], FrameMargins → 12]]];

(* =====*)
(*PART 4:CASE k^2==(gi^2-1)(gj^2-1)*)
(* =====*)
Print[Panel[Column[
 {Row[{Style["The derivation: the sign identity in the case ", Bold, 15,
 RGBColor[0.45, 0.15, 0.60]], Style[TraditionalForm[HoldForm[k^2 ==
 (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]], Bold,
 15, RGBColor[0.45, 0.15, 0.60]], Spacer[20], statusChip[okLoc]}],

```

```

Style["Given hypotheses, in addition to those above:", Bold, 13],
Style[Row[{TraditionalForm[
  HoldForm[Subscript[t, i] + Subscript[t, j] == 2 m K + I K']]], " ",
  TraditionalForm[HoldForm[m ∈ ℤ]], " ", TraditionalForm[
  HoldForm[(-1)^m == -k sn[Subscript[t, i]] sn[Subscript[t, j]]]],
13], Style[Row[{TraditionalForm[HoldForm[
  k^2 == (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)],
" , hence ", TraditionalForm[HoldForm[
  k == Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]],
" and ", TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]]}], 13],
Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]],
" , because ", TraditionalForm[HoldForm[
  k^2 == (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1) > 0]],
" and, by ", Style["the given sign identities", Italic, 12,
  RGBColor[0.00, 0.50, 0.50]], " , the factors have the signs of ",
TraditionalForm[HoldForm[Subsuperscript[p, i, 2]
  Subsuperscript[q, i, 2]]], " and ", TraditionalForm[
  HoldForm[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]],
" ; hence these signs coincide, and exactly the
  five class regions below occur."}],
12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
Style[Row[{"The first equality of the second formula of ",
  Style["*", 13, s10colTest], " is given above."}], 13],
Style[Row[{"By ", Style["the given sign identities", Italic,
  13, RGBColor[0.00, 0.50, 0.50]], " , every radicand below is
  positive and the radicals resolve as follows:"}], 13],
Grid[{{braceGrp[{TraditionalForm[HoldForm[Sqrt[(Subsuperscript[
  g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)] ==
  Sqrt[Abs[Subsuperscript[g, i, 2] - 1]] Sqrt[Abs[
  Subsuperscript[g, j, 2] - 1]]], TraditionalForm[
  HoldForm[Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i,
  2] Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]] ==
  Sqrt[Abs[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]]]
  Sqrt[Abs[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]]]}],
13], Style["●", 14, RGBColor[0.45, 0.15, 0.60]]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}],
Style[Row[{"Substituting ", TraditionalForm[HoldForm[
  Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)] == k]],
" and the identities ", Style["●", 13, RGBColor[0.45, 0.15, 0.60]],
" into the fourth product identity ",
Style["●", 13, RGBColor[0.55, 0.33, 0.10]], " , at ",
TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]],
" , and using the given expression for ",
TraditionalForm[HoldForm[Subscript[S, ij]]], " at ",
TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]],
" , we get the second and third equalities of the second formula
  of ", Style["*", 13, s10colTest], " , with ",

```

```

TraditionalForm[HoldForm[Subscript[ $\epsilon$ , i] Subscript[ $\epsilon$ , j]]],
" and the radicals of the identities ",
Style["●", 13, RGBColor[0.45, 0.15, 0.60]],
" symbolic, that is, for all their real values at once ",
smallChip[asmLoc]], 13], "",
Style[Row[{"The second and third equalities of the second formula of ",
Style["*", 13, s10colTest],
" are also verified end-to-end separately on each class region:"}],
13], Grid[Table[{Style[lblPair[pairsLoc[[ip]]], 12, GrayLevel[0.3]],
Style[Row[{"for both values of ", TraditionalForm[
HoldForm[Subscript[ $\epsilon$ , i] Subscript[ $\epsilon$ , j]]}], 12, GrayLevel[0.3]],
smallChip[locList[[ip]]], {ip, Length[pairsLoc]}],
Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
Background → Lighter[RGBColor[0.45, 0.15, 0.60], 0.93],
FrameMargins →
12]];
(* =====*)
(*PART 5:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And@@resultsAll, "ALL CLAIMS VERIFIED",
"SOME CLAIMS FAILED"], Bold, White, 16], Background →
If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];

```

Given hypotheses:

$$k, k' \in (0, 1)$$

For $i, j \in \{1, \dots, 4\}$, $i \neq j$:

$$g_i = \text{dn}(t_i), \quad g_j = \text{dn}(t_j) \in (0, k') \cup (1, \infty), \quad p_i, q_i, p_j, q_j \in \mathbb{R}_{>0} \cup i \mathbb{R}_{>0}$$

$$\text{sgn}(g_i^2 - 1) = \text{sgn}(g_i^2 - k'^2) = \text{sgn}(p_i^2 q_i^2) \neq 0,$$

$$\text{sgn}(g_j^2 - 1) = \text{sgn}(g_j^2 - k'^2) = \text{sgn}(p_j^2 q_j^2) \neq 0 \quad \text{the sign identities}$$

$$\xi_{ij} = \text{sgn}(p_i^2 q_i^2 p_j^2 q_j^2), \quad e_i, e_j, \varepsilon_i, \varepsilon_j, \xi_{ij} \in \{\pm 1\}$$

Given formulas and expressions:

$$\left. \begin{aligned} \text{sn}(t_l) &= \frac{i \varepsilon_l \sqrt{(g_l^2 - 1) p_l^2 q_l^2}}{k p_l q_l} \\ \text{cn}(t_l) &= \frac{\varepsilon_l \sqrt{(g_l^2 - k'^2) p_l^2 q_l^2}}{k p_l q_l} \end{aligned} \right\} \quad l \in \{i, j\}$$

$$S_{ij} = \frac{\varepsilon_i \varepsilon_j \sqrt{p_i^2 q_i^2 p_j^2 q_j^2} \xi_{ij}}{p_i q_i p_j q_j}$$

$$h_{ij1} = \frac{e_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - k'^2)} \xi_{ij} + e_j g_i \sqrt{(g_j^2 - 1)(g_i^2 - k'^2)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij2} = \frac{\sqrt{(g_i^2 - k'^2)(g_j^2 - k'^2)} \xi_{ij} + e_i e_j g_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - 1)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij3} = \frac{k^2 g_i g_j + e_i e_j \xi_{ij} \sqrt{(g_i^2 - 1)(g_i^2 - k'^2)(g_j^2 - 1)(g_j^2 - k'^2)}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$\{\text{sn}, \text{cn}, \text{dn}\}(-v) = \{-\text{sn}(v), \text{cn}(v), \text{dn}(v)\} \quad \text{the parity formulas}$$

$$\text{sn}(v + w) = \frac{\text{sn}(v) \text{cn}(w) \text{dn}(w) + \text{sn}(w) \text{cn}(v) \text{dn}(v)}{1 - k^2 \text{sn}(v)^2 \text{sn}(w)^2} \quad \text{the first addition formula}$$

$$\text{cn}(v + w) = \frac{\text{cn}(v) \text{cn}(w) - \text{sn}(v) \text{sn}(w) \text{dn}(v) \text{dn}(w)}{1 - k^2 \text{sn}(v)^2 \text{sn}(w)^2} \quad \text{the second addition formula}$$

$$\text{dn}(v + w) = \frac{\text{dn}(v) \text{dn}(w) - k^2 \text{sn}(v) \text{sn}(w) \text{cn}(v) \text{cn}(w)}{1 - k^2 \text{sn}(v)^2 \text{sn}(w)^2} \quad \text{the third addition formula}$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test these formulas ★ :

$$\{\text{sn}, \text{cn}, \text{dn}\}(e_i t_i + e_j t_j) = \{i S_{ij} h_{ij1}, S_{ij} h_{ij2}, h_{ij3}\} \quad \text{in the case } k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$$

$$(-1)^m = -k \text{sn}(t_i) \text{sn}(t_j) = \frac{\varepsilon_i \varepsilon_j \sqrt{p_i^2 q_i^2 p_j^2 q_j^2}}{p_i q_i p_j q_j} = S_{ij} \quad \text{in the case } k^2 = (g_i^2 - 1)(g_j^2 - 1)$$

The derivation: the formula for $\{\text{sn}, \text{cn}, \text{dn}\}(e_i t_i + e_j t_j)$

in the case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$

Verified

Given hypotheses, in addition to those above:

$$\pm t_i, \pm t_j \text{ are not poles, } 1 - k^2 \text{sn}(t_i)^2 \text{sn}(t_j)^2 = \frac{k^2 - (g_i^2 - 1)(g_j^2 - 1)}{k^2} \neq 0$$

Verification:

Since $\pm t_i$ and $\pm t_j$ are not poles and $1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2 \neq 0$, *the addition formulas*, at $v = e_i t_i$, $w = e_j t_j$, hold pointwise and, together with *the parity formulas* and $\operatorname{dn}(t_i) = g_i$, give

$$\left. \begin{aligned} \operatorname{sn}(e_i t_i + e_j t_j) &= \frac{e_i \operatorname{sn}(t_i) \operatorname{cn}(t_j) g_i + e_j \operatorname{sn}(t_j) \operatorname{cn}(t_i) g_j}{1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2} \\ \operatorname{cn}(e_i t_i + e_j t_j) &= \frac{\operatorname{cn}(t_i) \operatorname{cn}(t_j) - e_i e_j \operatorname{sn}(t_i) \operatorname{sn}(t_j) g_i g_j}{1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2} \\ \operatorname{dn}(e_i t_i + e_j t_j) &= \frac{g_i g_j - k^2 e_i e_j \operatorname{sn}(t_i) \operatorname{sn}(t_j) \operatorname{cn}(t_i) \operatorname{cn}(t_j)}{1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2} \end{aligned} \right\} \star$$

By *the given sign identities*, every radicand below is positive and the radicals resolve as follows:

$$\begin{aligned} \sqrt{(g_i^2 - 1) p_i^2 q_i^2} &= \sqrt{|g_i^2 - 1|} \sqrt{|p_i^2 q_i^2|}, \quad \sqrt{(g_i^2 - k'^2) p_i^2 q_i^2} = \sqrt{|g_i^2 - k'^2|} \sqrt{|p_i^2 q_i^2|}, \quad i \in \{i, j\} \\ \sqrt{p_i^2 q_i^2 p_j^2 q_j^2 \xi_{ij}} &= \sqrt{|p_i^2 q_i^2|} \sqrt{|p_j^2 q_j^2|} \\ \sqrt{(g_i^2 - 1) (g_j^2 - k'^2) \xi_{ij}} &= \sqrt{|g_i^2 - 1|} \sqrt{|g_j^2 - k'^2|}, \quad \sqrt{(g_j^2 - 1) (g_i^2 - k'^2) \xi_{ij}} = \sqrt{|g_j^2 - 1|} \sqrt{|g_i^2 - k'^2|} \\ \sqrt{(g_i^2 - k'^2) (g_j^2 - k'^2) \xi_{ij}} &= \sqrt{|g_i^2 - k'^2|} \sqrt{|g_j^2 - k'^2|}, \quad \sqrt{(g_i^2 - 1) (g_j^2 - 1) \xi_{ij}} = \sqrt{|g_i^2 - 1|} \sqrt{|g_j^2 - 1|} \\ \sqrt{(g_i^2 - 1) (g_i^2 - k'^2) (g_j^2 - 1) (g_j^2 - k'^2)} &= \sqrt{|g_i^2 - 1|} \sqrt{|g_i^2 - k'^2|} \sqrt{|g_j^2 - 1|} \sqrt{|g_j^2 - k'^2|} \end{aligned}$$

Multiplying the given formulas for $\operatorname{sn}(t_i)$ and $\operatorname{cn}(t_i)$

, and using the identities $\bullet$, we get the product identities:

$$\left. \begin{aligned} \operatorname{sn}(t_i) \operatorname{cn}(t_j) &= \frac{i \varepsilon_i \varepsilon_j \sqrt{p_i^2 q_i^2 p_j^2 q_j^2 \xi_{ij}} \sqrt{(g_i^2 - 1) (g_j^2 - k'^2) \xi_{ij}}}{k^2 p_i q_i p_j q_j} \\ \operatorname{sn}(t_j) \operatorname{cn}(t_i) &= \frac{i \varepsilon_i \varepsilon_j \sqrt{p_i^2 q_i^2 p_j^2 q_j^2 \xi_{ij}} \sqrt{(g_j^2 - 1) (g_i^2 - k'^2) \xi_{ij}}}{k^2 p_i q_i p_j q_j} \\ \operatorname{cn}(t_i) \operatorname{cn}(t_j) &= \frac{\varepsilon_i \varepsilon_j \sqrt{p_i^2 q_i^2 p_j^2 q_j^2 \xi_{ij}} \sqrt{(g_i^2 - k'^2) (g_j^2 - k'^2) \xi_{ij}}}{k^2 p_i q_i p_j q_j} \\ \operatorname{sn}(t_i) \operatorname{sn}(t_j) &= -\frac{\varepsilon_i \varepsilon_j \sqrt{p_i^2 q_i^2 p_j^2 q_j^2 \xi_{ij}} \sqrt{(g_i^2 - 1) (g_j^2 - 1) \xi_{ij}}}{k^2 p_i q_i p_j q_j} \\ \operatorname{sn}(t_i) \operatorname{sn}(t_j) \operatorname{cn}(t_i) \operatorname{cn}(t_j) &= -\frac{\xi_{ij} \sqrt{(g_i^2 - 1) (g_i^2 - k'^2) (g_j^2 - 1) (g_j^2 - k'^2)}}{k^4} \end{aligned} \right\} \bullet$$

which are verified separately on each class region of $(p_i q_i, p_j q_j)$, for both values of $\varepsilon_i \varepsilon_j$:

	the identities $\bullet$	the product identities $\bullet$
$p_i q_i \in \mathbb{R}_{>0}, \quad p_j q_j \in \mathbb{R}_{>0}$	☒	☒
$p_i q_i \in \mathbb{R}_{>0}, \quad p_j q_j \in i \mathbb{R}_{>0}$	☒	☒
$p_i q_i \in \mathbb{R}_{>0}, \quad p_j q_j \in \mathbb{R}_{<0}$	☒	☒
$p_i q_i \in i \mathbb{R}_{>0}, \quad p_j q_j \in \mathbb{R}_{>0}$	☒	☒
$p_i q_i \in i \mathbb{R}_{>0}, \quad p_j q_j \in i \mathbb{R}_{>0}$	☒	☒
$p_i q_i \in i \mathbb{R}_{>0}, \quad p_j q_j \in \mathbb{R}_{<0}$	☒	☒
$p_i q_i \in \mathbb{R}_{<0}, \quad p_j q_j \in \mathbb{R}_{>0}$	☒	☒
$p_i q_i \in \mathbb{R}_{<0}, \quad p_j q_j \in i \mathbb{R}_{>0}$	☒	☒
$p_i q_i \in \mathbb{R}_{<0}, \quad p_j q_j \in \mathbb{R}_{<0}$	☒	☒

Thus, substituting the product identities $\bullet$ and the given equality for $1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2$ into the expansions $\star$, with $\varepsilon_i \varepsilon_j$, e_i , e_j , and the radicals of the identities $\bullet$ symbolic, that is, for all their real values at once, we get the first formula of $\star$:

$$\star: \operatorname{sn} \checkmark \operatorname{cn} \checkmark \operatorname{dn} \checkmark$$

The first formula of $\star$ is also verified end-to-end separately on each class region, with e_i , e_j symbolic and for both values of $\varepsilon_i \varepsilon_j$:

	sn	cn	dn
$p_i q_i \in \mathbb{R}_{>0}, p_j q_j \in \mathbb{R}_{>0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in \mathbb{R}_{>0}, p_j q_j \in i \mathbb{R}_{>0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in \mathbb{R}_{>0}, p_j q_j \in \mathbb{R}_{<0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in i \mathbb{R}_{>0}, p_j q_j \in \mathbb{R}_{>0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in i \mathbb{R}_{>0}, p_j q_j \in i \mathbb{R}_{>0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in i \mathbb{R}_{>0}, p_j q_j \in \mathbb{R}_{<0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in \mathbb{R}_{<0}, p_j q_j \in \mathbb{R}_{>0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in \mathbb{R}_{<0}, p_j q_j \in i \mathbb{R}_{>0}$	$\checkmark$	$\checkmark$	$\checkmark$
$p_i q_i \in \mathbb{R}_{<0}, p_j q_j \in \mathbb{R}_{<0}$	$\checkmark$	$\checkmark$	$\checkmark$

The derivation: the sign identity in the case

$$k^2 = (g_i^2 - 1)(g_j^2 - 1)$$

Verified

Given hypotheses, in addition to those above:

$$t_i + t_j = 2mK + iK', \quad m \in \mathbb{Z}, \quad (-1)^m = -k \operatorname{sn}(t_i) \operatorname{sn}(t_j)$$

$$k^2 = (g_i^2 - 1)(g_j^2 - 1), \quad \text{hence} \quad k = \sqrt{(g_i^2 - 1)(g_j^2 - 1)} \quad \text{and} \quad \xi_{ij} = 1$$

Here $\xi_{ij} = 1$, because $k^2 = (g_i^2 - 1)(g_j^2 - 1) > 0$ and, by

the given sign identities, the factors have the signs of $p_i^2 q_i^2$ and $p_j^2 q_j^2$; hence these signs coincide, and exactly the five class regions below occur.

Verification:

The first equality of the second formula of $\star$ is given above.

By the given sign identities

, every radicand below is positive and the radicals resolve as follows:

$$\left. \begin{aligned} \sqrt{(g_i^2 - 1)(g_j^2 - 1)} &= \sqrt{|g_i^2 - 1|} \sqrt{|g_j^2 - 1|} \\ \sqrt{p_i^2 q_i^2 p_j^2 q_j^2} &= \sqrt{|p_i^2 q_i^2|} \sqrt{|p_j^2 q_j^2|} \end{aligned} \right\} \bullet$$

Substituting $\sqrt{(g_i^2 - 1)(g_j^2 - 1)} = k$ and the identities $\bullet$ into the fourth product identity

$\bullet$, at $\xi_{ij} = 1$, and using the given expression for S_{ij} at $\xi_{ij} = 1$

, we get the second and third equalities of the second formula of $\star$, with $\varepsilon_i \varepsilon_j$

and the radicals of the identities $\bullet$ symbolic, that is, for all their real values at once $\checkmark$

The second and third equalities of the second formula of

$\star$ are also verified end-to-end separately on each class region:

$p_i q_i \in \mathbb{R}_{>0}, \quad p_j q_j \in \mathbb{R}_{>0}$	for both values of $\varepsilon_i \varepsilon_j$	$\checkmark$
$p_i q_i \in \mathbb{R}_{>0}, \quad p_j q_j \in \mathbb{R}_{<0}$	for both values of $\varepsilon_i \varepsilon_j$	$\checkmark$
$p_i q_i \in i\mathbb{R}_{>0}, \quad p_j q_j \in i\mathbb{R}_{>0}$	for both values of $\varepsilon_i \varepsilon_j$	$\checkmark$
$p_i q_i \in \mathbb{R}_{<0}, \quad p_j q_j \in \mathbb{R}_{>0}$	for both values of $\varepsilon_i \varepsilon_j$	$\checkmark$
$p_i q_i \in \mathbb{R}_{<0}, \quad p_j q_j \in \mathbb{R}_{<0}$	for both values of $\varepsilon_i \varepsilon_j$	$\checkmark$

ALL CLAIMS VERIFIED

Section 7

Results of Section 7 are used in the proof of Remark 5

In[3591]:=

```
( * =====*)
(*PART 1:SETUP*)
( * =====*)
```

```

ClearAll[gi, gj, k, kp, ei, ej, xi, h1of, h2of, h3of, baseAssum,
  wConfigs, ePairs, dirForward, dirBackward, branchOK, smallChip,
  statusChip, panelH, resultsAll, condOf, vars, hsym, stmtOf, s5colTest];
Format[gi, TraditionalForm] = Subscript[g, i];
Format[gj, TraditionalForm] = Subscript[g, j];
Format[kp, TraditionalForm] = k';
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☐", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s5colTest = RGBColor[0.10, 0.35, 0.75];
vars = {gi, gj, k, kp};
(*The three h's in the case  $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$ ,
with xi a numeric sign per branch.*)
h1of[e1_, e2_, x_] := (e1 gj Sqrt[(gi^2 - 1) (gj^2 - kp^2) x] +
  e2 gi Sqrt[(gj^2 - 1) (gi^2 - kp^2) x]) / (k^2 - (gi^2 - 1) (gj^2 - 1));
h2of[e1_, e2_, x_] := (Sqrt[(gi^2 - kp^2) (gj^2 - kp^2) x] +
  e1 e2 gi gj Sqrt[(gi^2 - 1) (gj^2 - 1) x]) / (k^2 - (gi^2 - 1) (gj^2 - 1));
h3of[e1_, e2_, x_] :=
  (k^2 gi gj + e1 e2 x Sqrt[(gi^2 - 1) (gi^2 - kp^2) (gj^2 - 1) (gj^2 - kp^2)]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
(*Standing assumptions:k,k' with  $k'^2=1-k^2$ ;
the case guard;
window assumptions added per branch.*)
baseAssum = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 ≠ (gi^2 - 1) (gj^2 - 1);
wConfigs =
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\), \!\(\(*SubscriptBox[\(g\), \(\j\)]\)\)
    ∈ (1, ∞)", gi > 1 && gj > 1, 1},
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\) ∈ (1, ∞), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) ∈ (0, k')", gi > 1 && 0 < gj < kp, -1},
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\) ∈ (0, k'), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) ∈ (1, ∞)", 0 < gi < kp && gj > 1, -1},
  {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) ∈ (0, k')", 0 < gi < kp && 0 < gj < kp, 1}};
ePairs = {{1, 1}, {1, -1}, {-1, 1}, {-1, -1}};
(*Two-directional branch check:no zero off the condition locus,
and no nonzero value on it.A condition False
means the claim on this branch is pure nonvanishing.*)
dirForward[h_, cond_, assum_] :=
  Reduce[assum && h == 0 && ! cond, vars, Reals] === False;
dirBackward[h_, cond_, assum_] :=
  If[cond === False, True, Reduce[assum && cond && h ≠ 0, vars, Reals] === False];
branchOK[h_, cond_, assum_] :=

```

```

dirForward[h, cond, assum] && dirBackward[h, cond, assum];
(*Per-branch condition of each claim;
False encodes "never vanishes".*)
condOf[1, e1_, e2_] := If[e1 == -e2, gi == gj, False];
condOf[2, e1_, e2_] := If[e1 == -e2, gi gj == kp, False];
condOf[3, e1_, e2_] :=
  If[e1 == e2, k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1), False];
hsym[n_] := Subscript["h", "ij" <> ToString[n]];
stmtOf[n_, e1_, e2_] :=
  Module[{c = condOf[n, e1, e2]}, If[c === False, Row[{hsym[n], " ≠ 0"}],
    Row[{hsym[n], " = 0  if and only if  ", TraditionalForm[c]}]]];
(* =====*)
(*PART 2:GIVEN PANEL*)
(* =====*)
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
  Style[Row[{"k ∈ (0, 1),  ", TraditionalForm[HoldForm[k'^2 == 1 - k^2]}]],
  13], Style[Row[{"For  ", TraditionalForm[HoldForm[i]],
    ", ", TraditionalForm[HoldForm[j]], " ∈ {1, ..., 4},  ",
    TraditionalForm[HoldForm[i ≠ j]], " :"}], Bold, 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[g, i]]], ", ",
    TraditionalForm[HoldForm[Subscript[g, j]]],
    " ∈ (0, k') ∪ (1, ∞),  ", TraditionalForm[HoldForm[Subscript[ξ, ij] ==
      Sign[(Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1)]],
    " ,  ", TraditionalForm[HoldForm[Subscript[e, i]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, j]]], ", ",
    TraditionalForm[HoldForm[Subscript[ξ, ij]]], " ∈ {±1}"}], 13], ""],
  Style["Given formulas, in the case \!\(\(*SuperscriptBox[\(k\),
    \((2\)]\)\) ≠ \!\(\(*SubsuperscriptBox[\(g\), \((i\), \((2\)]\)\)
    - 1)\!\(\(*SubsuperscriptBox[\(g\), \((j\), \((2\)]\)\) - 1):",
    Bold, 14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] ==
    (Subscript[e, i] Subscript[g, j] Sqrt[(Subscript[g, i]^2 - 1)
      (Subscript[g, j]^2 - k'^2) Subscript[ξ, ij]] +
      Subscript[e, j] Subscript[g, i] Sqrt[(Subscript[g, j]^2 - 1)
      (Subscript[g, i]^2 - k'^2) Subscript[ξ, ij]]) /
      (k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))], 14],
  Style[TraditionalForm[HoldForm[Subscript[h, ij2] ==
    (Sqrt[(Subscript[g, i]^2 - k'^2) (Subscript[g, j]^2 - k'^2)
      Subscript[ξ, ij]] + Subscript[e, i] Subscript[e, j]
      Subscript[g, i] Subscript[g, j] Sqrt[(Subscript[g, i]^2 - 1)
      (Subscript[g, j]^2 - 1) Subscript[ξ, ij]]) /
      (k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))], 14],
  Style[TraditionalForm[HoldForm[Subscript[h, ij3] ==
    (k^2 Subscript[g, i] Subscript[g, j] + Subscript[e, i] Subscript[e, j]
      Subscript[ξ, ij] Sqrt[(Subscript[g, i]^2 - 1) (Subscript[g, i]^2 -
      k'^2) (Subscript[g, j]^2 - 1) (Subscript[g, j]^2 - k'^2))] /
      (k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))], 14],
  Style["and, in the case \!\(\(*SuperscriptBox[\(k\), \((2\)]\)\)

```

```

= (\!\(\*SubsuperscriptBox[\(g\), \(\i\), \((2\)]\))
- 1) (\!\(\*SubsuperscriptBox[\(g\), \(\j\), \((2\)]\))
- 1) with \!\(\*SubscriptBox[\(e\), \(\i\)]\)) =
-\!\(\*SubscriptBox[\(e\), \(\j\)]\)): ", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] =
Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
(2 k Subscript[g, i] Subscript[g, j])]],
" ", TraditionalForm[HoldForm[Subscript[h, ij2] =
(Subscript[g, i]^2 Subscript[g, j]^2 - k'^2) /
(2 k Subscript[g, i] Subscript[g, j])]], " ", TraditionalForm[
HoldForm[Subscript[h, ij3] = (Subscript[g, i]^2 + Subscript[g, j]^2) /
(2 Subscript[g, i] Subscript[g, j])]]}],
14], Style[Row[{"Each ", smallChip[True],
" below certifies that the statement displayed beside it has been
verified by a symbolic computation at that point; a ",
smallChip[False], " would mean that this verification failed."}],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test the following claims ", Bold, 14],
Style["*", 14, s5colTest], Style[
" , over all window configurations of \!\(\*SubscriptBox[\(g\),
\(\i\)]\), \!\(\*SubscriptBox[\(g\),
\(\j\)]\)) and all \!\(\*SubscriptBox[\(e\),
\(\i\)]\), \!\(\*SubscriptBox[\(e\), \(\j\)]\))
, in both directions:", Bold, 14]}]],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] = 0]],
" if and only if ",
TraditionalForm[HoldForm[Subscript[g, i] = Subscript[g, j]]],
" and ", TraditionalForm[
HoldForm[Subscript[e, i] = -Subscript[e, j]]}], Bold, 14, s5colTest],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij2] = 0]],
" if and only if ",
TraditionalForm[HoldForm[Subscript[g, i] Subscript[g, j] = k']],
" and ", TraditionalForm[
HoldForm[Subscript[e, i] = -Subscript[e, j]]}], Bold, 14, s5colTest],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij3] = 0]],
" if and only if ", TraditionalForm[
HoldForm[k^2 (Subscript[g, i]^2 + Subscript[g, j]^2) =
k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1)], " and ",
TraditionalForm[HoldForm[Subscript[e, i] = Subscript[e, j]]}],
Bold, 14, s5colTest]], Spacings -> 1.1],
Background -> RGBColor[0.99, 0.96, 0.88], FrameMargins ->
14]]];
(* =====*)
(*PART 3:CASE k^2≠(gi^2-1)(gj^2-1)*)
(* =====*)
resultsAll = {};
panelH[n_, hfun_, col_, title_] := Module[{grid, ok, headRow, stmtRow},

```

```

grid = Table[branchOK[hfun[ep[[1]], ep[[2]], w[[3]], condOf[n, ep[[1]], ep[[2]],
  baseAssum&&w[[2]], {w, wConfigs}, {ep, ePairs}];
ok = And@@Flatten[grid];
AppendTo[resultsAll, ok];
headRow = Prepend[
  Table[Style[Row[{"(", ep[[1]], " ", ep[[2]], ")"}], 12, Bold], {ep, ePairs}],
  Style["(!\\(\\*SubscriptBox[(e\\), \\(i\\)]\\), \\(j\\)]\\) →", 12, GrayLevel[0.3]]];
stmtRow = Prepend[Table[Style[stmtOf[n, ep[[1]], ep[[2]], 11, GrayLevel[0.25]],
  {ep, ePairs}], Style["tested →", 11, GrayLevel[0.3]]];
Print[Panel[
  Column[{Row[{Style[title, Bold, 15, col], Spacer[20], statusChip[ok]}],
    Grid[Join[{headRow}, {stmtRow}, Table[Prepend[smallChip/@grid[[iw]],
      Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]], {iw, 4}]],
      Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
    Background → Lighter[col, 0.93], FrameMargins → 12]]];];
panelH[1, h1of, RGBColor[0.10, 0.30, 0.70],
  "First claim (case \\(\\*SuperscriptBox[(k\\),
    \\(2\\)]\\) ≠ \\(\\(\\*SubsuperscriptBox[(g\\), \\(i\\),
    \\(2\\)]\\)-1) \\(\\(\\*SubsuperscriptBox[(g\\), \\(j\\), \\(2\\)]\\)-1)):"];
panelH[2, h2of, RGBColor[0.55, 0.25, 0.70],
  "Second claim (case \\(\\(\\*SuperscriptBox[(k\\),
    \\(2\\)]\\) ≠ \\(\\(\\*SubsuperscriptBox[(g\\), \\(i\\),
    \\(2\\)]\\)-1) \\(\\(\\*SubsuperscriptBox[(g\\), \\(j\\), \\(2\\)]\\)-1)):"];
panelH[3, h3of, RGBColor[0.00, 0.50, 0.50],
  "Third claim (case \\(\\(\\*SuperscriptBox[(k\\),
    \\(2\\)]\\) ≠ \\(\\(\\*SubsuperscriptBox[(g\\), \\(i\\),
    \\(2\\)]\\)-1) \\(\\(\\*SubsuperscriptBox[(g\\), \\(j\\), \\(2\\)]\\)-1)):"];
(* =====*)
(*PART 4:CASE k^2==(gi^2-1)(gj^2-1) WITH ei=-ej*)
(* =====*)
Module[{assum7, ok1, ok2, ok3, ok4, ok},
  assum7 = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 == (gi^2 - 1) (gj^2 - 1) &&
    ((gi > 1 && gj > 1) || (0 < gi < kp && 0 < gj < kp));
  ok1 = And@@Table[
    branchOK[e1 (gi^2 - gj^2) / (2 k gi gj), gi == gj, assum7], {e1, {1, -1}}];
  ok2 = branchOK[(gi^2 gj^2 - kp^2) / (2 k gi gj), gi gj == kp, assum7];
  ok3 = Reduce[assum7 && (gi^2 + gj^2) / (2 gi gj) == 0, vars, Reals] == False;
  ok4 = Reduce[assum7 && k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1),
    vars, Reals] == False;
  ok = ok1 && ok2 && ok3 && ok4;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["Case \\(\\(\\*SuperscriptBox[(k\\), \\(2\\)]\\) =
      \\(\\(\\*SubsuperscriptBox[(g\\), \\(i\\),
      \\(2\\)]\\)-1) \\(\\(\\*SubsuperscriptBox[(g\\), \\(j\\),
      \\(2\\)]\\)-1), with \\(\\(\\*SubscriptBox[(e\\), \\(i\\)]\\)

```

```

= -\!\(\(*SubscriptBox[\(e\), \(\j\)]\)":", Bold, 15,
RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]]],
Column[{Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
(2 k Subscript[g, i] Subscript[g, j])]], ": ",
hsym[1], " = 0 if and only if ", TraditionalForm[HoldForm[
Subscript[g, i] == Subscript[g, j]]], " ", smallChip[ok1]]], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij2] ==
(Subscript[g, i]^2 Subscript[g, j]^2 - k'^2) /
(2 k Subscript[g, i] Subscript[g, j])]], ": ",
hsym[2], " = 0 if and only if ", TraditionalForm[
HoldForm[Subscript[g, i] Subscript[g, j] == k']],
" ", smallChip[ok2]]], 13], Style[
Row[{TraditionalForm[HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 +
Subscript[g, j]^2) / (2 Subscript[g, i] Subscript[g, j])]],
" ≥ 1, so ", hsym[3], " ≠ 0 ", smallChip[ok3]]], 13],
Style[Row[{"condition of the third claim unsatisfiable here ",
smallChip[ok4]]], 13]], Spacings → 0.8]],
Spacings → 1], Background → Lighter[RGBColor[0.75, 0.35,
0.05],
0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 5:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And@@resultsAll, "ALL CLAIMS VERIFIED",
"SOME CLAIMS FAILED"], Bold, White, 16], Background →
If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];

```

Given hypotheses:

$$k \in (0, 1), \quad (k')^2 = 1 - k^2$$

For $i, j \in \{1, \dots, 4\}$, $i \neq j$:

$$g_i, g_j \in (0, k') \cup (1, \infty), \quad \xi_{ij} = \text{sgn}((g_i^2 - 1)(g_j^2 - 1)), \quad e_i, e_j, \xi_{ij} \in \{\pm 1\}$$

Given formulas, in the case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$:

$$h_{ij1} = \frac{e_i g_i \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)} \xi_{ij} + e_j g_j \sqrt{(g_j^2 - 1)(g_i^2 - (k')^2)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij2} = \frac{\sqrt{(g_i^2 - (k')^2)(g_j^2 - (k')^2)} \xi_{ij} + e_i e_j g_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - 1)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij3} = \frac{k^2 g_i g_j + e_i e_j \xi_{ij} \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)(g_i^2 - 1)(g_j^2 - (k')^2)}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

and, in the case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$ with $e_i = -e_j$:

$$h_{ij1} = \frac{e_i (g_i^2 - g_j^2)}{2 k g_i g_j} \quad h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2 k g_i g_j} \quad h_{ij3} = \frac{g_i^2 + g_j^2}{2 g_i g_j}$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test the following claims 

, over all window configurations of g_i, g_j and all e_i, e_j , in both directions:

$h_{ij1} = 0$ if and only if $g_i = g_j$ and $e_i = -e_j$

$h_{ij2} = 0$ if and only if $g_i g_j = k'$ and $e_i = -e_j$

$h_{ij3} = 0$ if and only if $k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1)$ and $e_i = e_j$

First claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij1} \neq 0$	$h_{ij1} = 0$ if and only if $g_i = g_j$	$h_{ij1} = 0$ if and only if $g_i = g_j$	$h_{ij1} \neq 0$
$g_i, g_j \in (1, \infty)$				
$g_i \in (1, \infty), g_j \in (0, k')$				
$g_i \in (0, k'), g_j \in (1, \infty)$				
$g_i, g_j \in (0, k')$				

Second claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):**Verified**

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij2} \neq 0$	$h_{ij2} = 0$ if and only if $g_i g_j = k'$	$h_{ij2} = 0$ if and only if $g_i g_j = k'$	$h_{ij2} \neq 0$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Third claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):**Verified**

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij3} = 0$ if and only if $k^2 (g_i^2 + g_j^2) =$ $k^2 - (g_i^2 - 1)(g_j^2 - 1)$	$h_{ij3} \neq 0$	$h_{ij3} \neq 0$	$h_{ij3} = 0$ if and only if $k^2 (g_i^2 + g_j^2) =$ $k^2 - (g_i^2 - 1)(g_j^2 - 1)$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$, with $e_i = -e_j$:**Verified**

$$h_{ij1} = \frac{e_i (g_i^2 - g_j^2)}{2 k g_i g_j}: h_{ij1} = 0 \text{ if and only if } g_i = g_j \quad \checkmark$$

$$h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2 k g_i g_j}: h_{ij2} = 0 \text{ if and only if } g_i g_j = k' \quad \checkmark$$

$$h_{ij3} = \frac{g_i^2 + g_j^2}{2 g_i g_j} \geq 1, \text{ so } h_{ij3} \neq 0 \quad \checkmark$$

condition of the third claim unsatisfiable here $\checkmark$

ALL CLAIMS VERIFIED

Section 8

Results of Section 8 are used in the proof of Lemma 12:
Case 1

In[2438]:=

```
( * ===== *)
(*PART 1:INITIALIZATION*)
( * ===== *)
```

```

ClearAll[UU, XX, YY, ZZ, tt, wConfigs, sTriples, dataOf, lhsOf, derOf, assumOf,
  derCheck, resultsAll, statusChip, smallChip, lblZ, lblY, lblX, s6colTest];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s6colTest = RGBColor[0.10, 0.35, 0.75];
(*Sign regions:
  two z_l-windows (from g_l in (0,k') or (1,infinity)) and the sign of y_l;
  the sign of x_l is then forced by u x_l y_l z_l>0. The window z_l<
  -1 carries 1+u z_l>0. On each region,p_l and q_l take their
  explicit radical values and the nested square roots collapse;
  the identity is checked there for all 8 sign combinations of e',e_0,eps_l.*)
lblZ[1] = TraditionalForm[HoldForm[Subscript[z, l] > 0]];
lblZ[2] = TraditionalForm[HoldForm[Subscript[z, l] < -1]];
lblY[1] = TraditionalForm[HoldForm[Subscript[y, l] > 0]];
lblY[-1] = TraditionalForm[HoldForm[Subscript[y, l] < 0]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, l] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, l] < 0]];
wConfigs = {{Row[{lblZ[1], " ", " ", lblY[1], " ", hence " ", lblX[1]}], ZZ, 1, 1},
  {Row[{lblZ[1], " ", " ", lblY[-1], " ", hence " ", lblX[-1]}], ZZ, -1, -1},
  {Row[{lblZ[2], " ", " ", lblY[1], " ", hence " ", lblX[-1]}], -1 - ZZ, 1, -1},
  {Row[{lblZ[2], " ", " ", lblY[-1], " ", hence " ", lblX[1]}], -1 - ZZ, -1, 1}};
sTriples = Tuples[{1, -1}, 3];
dataOf[{lab_, zl_, sy_, sx_}] := Module[{yl = sy YY, x1 = sx XX,
  kk = Sqrt[UU], kp = Sqrt[1 - UU], gl, pl, ql, Dd}, gl = Sqrt[1 + 1 / zl];
  pl = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  ql = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  Dd = -(1 - x1 tt^2) (1 - UU x1 tt^2);
  {zl, yl, x1, kk, kp, gl, pl, ql, Dd}];
lhsOf[{zl_, yl_, x1_, kk_, kp_, gl_, pl_, ql_, Dd_}, ep_, e0_, eps_] :=
  Module[{snl, cnl, RR = Sqrt[x1 Dd]},
    snl = I eps Sqrt[(gl^2 - 1) p1^2 ql^2] / (kk p1 ql);
    cnl = eps Sqrt[(gl^2 - kp^2) p1^2 ql^2] / (kk p1 ql);
    (zl ql cnl gl (tt / p1) + ep zl ql snl (e0 I RR / (XX p1))) /
    (zl + x1 tt^2)];
derOf[{zl_, yl_, x1_, kk_, kp_, gl_, pl_, ql_, Dd_}, ep_, e0_, eps_] :=
  eps (tt Sqrt[UU x1 yl (1 + zl) (1 + UU zl)] - ep e0 Sqrt[UU yl zl Dd]) /
  (yl UU (zl + x1 tt^2));
assumOf[{lab_, zl_, sy_, sx_}] := 0 < UU < 1 && XX > 0 && YY > 0 && ZZ > 0 &&
  Element[tt, Reals] && sx XX (- (1 - sx XX tt^2) (1 - UU sx XX tt^2)) ≥ 0 &&
  1 + UU zl > 0 && zl + sx XX tt^2 ≠ 0;
derCheck[w_, {ep_, e0_, eps_}] := Module[{d = dataOf[w]},

```

```

TimeConstrained[FullSimplify[lhsOf[d, ep, e0, eps] - derOf[d, ep, e0, eps] == 0,
  assumOf[w]], 180, False] === True];
(* =====*)
(*PART 2:GIVEN PANEL*)
(* =====*)
Print[Panel[Column[{Style[Row[{"Given hypotheses, Case 1  (" ,
  TraditionalForm[HoldForm[u > 0]], " ) :"}], Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[0 < u < 1]], " ,  ",
    TraditionalForm[HoldForm[k = Sqrt[u]]], " ,  ", TraditionalForm[
      HoldForm[k' = Sqrt[1 - u]]], " ,  ", TraditionalForm[HoldForm[k]],
      " ,  ", TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 14],
  Style[Row[{"For  ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
    Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
      " ,  ", TraditionalForm[HoldForm[Subscript[y, i]]], " ,  ",
      TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
      " ,  ", TraditionalForm[HoldForm[Subscript[x, i]]],
      " ,  ", TraditionalForm[HoldForm[Subscript[y, i]]],
      " ,  ", TraditionalForm[HoldForm[Subscript[z, i]]],
      " ≠ -1, 0,  ", TraditionalForm[HoldForm[Subscript[g, i] =
        dn[Subscript[t, i] = Sqrt[1 + 1 / Subscript[z, i]]]], 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[p, i]]], " ,  ",
    TraditionalForm[HoldForm[Subscript[q, i]]], " ∈ ",
    Subscript["ℝ", "> 0"], " ∪ i ", Subscript["ℝ", "> 0"],
    " ,  ", TraditionalForm[HoldForm[Subscript[e, 0]]],
    " ,  ", TraditionalForm[HoldForm[Derivative[1][e]]],
    " ,  ", TraditionalForm[HoldForm[Subscript[e, i]]],
    " ,  ", TraditionalForm[HoldForm[Subscript[ε, i]]], " ,  ",
    TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"}], 14],
  Style[Row[{TraditionalForm[HoldForm[u Subscript[x, i]
    Subscript[y, i] Subscript[z, i] > 0]], " ,  ",
    TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
    " ,  ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
  Style[Row[{TraditionalForm[HoldForm[∂[Subscript[x, 1], t] =
    - (1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
    " ,  ", TraditionalForm[HoldForm[
      R = Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]]], " ,  ",
      TraditionalForm[HoldForm[Subscript[x, 1] ∂[Subscript[x, 1], t] ≥ 0]]}],
    14], "", Style["Given formulas:", Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[sn[v + w] == (sn[v] cn[w] dn[w] +
    sn[w] cn[v] dn[v]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
    " ,  ", Style["the first addition formula", Italic, 13,
      RGBColor[0.10, 0.30, 0.70]]}], 14], Style[
    Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] == {-sn[v], cn[v], dn[v]}]],
      " ,  ", Style["the parity formulas",
        Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
    14], Style[Row[{"Each  ", smallChip[True],
      " below certifies that the statement displayed beside it has been

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        verified by a symbolic computation at that point; a ",
        smallChip[False], " would mean that this verification failed."]],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test these formulas ", Bold, 14],
Style["*", 14, s6colTest], Style[" :", Bold, 14]}]],
Style[TraditionalForm[HoldForm[Subscript[W, 2] == Subscript[ε, 2]
(t Sqrt[u Subscript[x, 1] Subscript[y, 2] (1 + Subscript[z, 2])
(1 + u Subscript[z, 2])] + Subscript[e, 0] Subscript[e, 2]
Sqrt[u Subscript[y, 2] Subscript[z, 2] D[Subscript[x, 1], t]] /
(Subscript[y, 2] u (Subscript[z, 2] + Subscript[x, 1] t^2))]], 15,
s6colTest], Style[TraditionalForm[HoldForm[Subscript[W, 1] == Subscript[ε,
1] (t Sqrt[u Subscript[x, 1] Subscript[y, 1] (1 + Subscript[z, 1])
(1 + u Subscript[z, 1])] - Subscript[e, 0] Subscript[e, 1]
Sqrt[u Subscript[y, 1] Subscript[z, 1] D[Subscript[x, 1], t]] /
(Subscript[y, 1] u (Subscript[z, 1] + Subscript[x, 1] t^2))]],
15, s6colTest], Style[Row[{TraditionalForm[HoldForm[
U == Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
(Abs[Subscript[x, 1]] Subscript[h, 232] Subscript[h, 233]
t + Subscript[e, 0] Subscript[h, 231]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] /
(Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]] (1 + Subscript[ε, 23]
k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))]],
" ", Style["in cases (a) and (b)", Italic, 13, GrayLevel[0.35]]}],
15, s6colTest], Style[Row[{TraditionalForm[HoldForm[U == Subscript[ε, 2]
Subscript[ε, 3] Sign[u Subscript[z, 2]] Sign[Subscript[x, 1]] /
(k Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]] t)]},
" ", Style["in case (c)", Italic, 13, GrayLevel[0.35]]}],
15, s6colTest]], Spacings → 1.1],
Background → RGBColor[0.99, 0.96, 0.88], FrameMargins →
14]]];
(* =====*)
(*PART 3:THE W-FORMULAS*)
(* =====*)
resultsAll = {};
Module[{grid, regionOK, ok},
grid = Table[derCheck[w, s], {w, wConfigs}, {s, sTriples}];
regionOK = And@@@grid;
ok = And@@regionOK;
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formulas for ", Bold, 15,
RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[
HoldForm[Subscript[W, 2]]], Bold, 15, RGBColor[0.10, 0.30, 0.70]],
Style[" and ", Bold, 15, RGBColor[0.10, 0.30, 0.70]],
Style[TraditionalForm[HoldForm[Subscript[W, 1]]], Bold,
15, RGBColor[0.10, 0.30, 0.70]], Style[" for ", Bold, 15,
RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u > 0]],

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    Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]]],
Style["Given hypotheses and formulas, in addition to those above:",
    Bold, 13], Style[
    Row[{"For ", TraditionalForm[HoldForm[l]], " = 1, 2 :"}], Bold, 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1]
    Subsuperscript[p, 1, 2] == 1]], ", ", TraditionalForm[
    HoldForm[Subscript[y, l] Subsuperscript[q, l, 2] == 1]], ", ",
    TraditionalForm[HoldForm[Subscript[p, l] == Subscript[p, 1]]]}], 13],
Style[TraditionalForm[HoldForm[sn[Subscript[t, l]] == I Subscript[ε, l]
    Sqrt[(Subsuperscript[g, l, 2] - 1) Subsuperscript[p, l, 2]
    Subsuperscript[q, l, 2]] / (k Subscript[p, l] Subscript[q, l])]],
14], Style[TraditionalForm[HoldForm[cn[Subscript[t, l]] ==
    Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - k'^2)
    Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] /
    (k Subscript[p, l] Subscript[q, l])]], 14],
Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
    ", ", TraditionalForm[HoldForm[Subscript[p, 1] cn[ς] dn[ς] ==
    Subscript[e, 0] I R / Abs[Subscript[x, 1]]]}], 14],
Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Derivative[1][e]
    Subscript[t, l]] == {Derivative[1][e] sn[Subscript[t, l]],
    cn[Subscript[t, l]], dn[Subscript[t, l]]}],
    " ", Style["by the parity formulas", Italic, 13,
    RGBColor[0.75, 0.35, 0.05]]}], 14], Style[Row[{TraditionalForm[
    HoldForm[sn[Subscript[t, l]]^2 == (1 - Subsuperscript[g, l, 2]) / k^2 ==
    - (1 / (u Subscript[z, l]))}], ", ",
    TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, l]]^2 sn[ς]^2 ==
    (Subscript[z, l] + Subscript[x, 1] t^2) / Subscript[z, l] ≠ 0]],
    ", by the choice of ", TraditionalForm[HoldForm[t]]}],
14], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[±Subscript[t, l]]],
    " and ", TraditionalForm[HoldForm[ς]], " are not poles and ",
    TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, l]]^2 sn[ς]^2 ≠ 0]],
    ", ", Style["the first addition formula", Italic, 13, RGBColor[0.10,
    0.30, 0.70]]], ", at ", TraditionalForm[HoldForm[v = ς]], ", ",
    TraditionalForm[HoldForm[w = Derivative[1][e] Subscript[t, l]]],
    ", holds pointwise and, together with ",
    Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
    ", gives"}], 13], Style[TraditionalForm[
    HoldForm[Subscript[q, l] sn[ς + Derivative[1][e] Subscript[t, l]] ==
    (Subscript[z, l] Subscript[q, l] cn[Subscript[t, l]]
    dn[Subscript[t, l]] sn[ς] + Derivative[1][e] Subscript[z, l]
    Subscript[q, l] sn[Subscript[t, l]] cn[ς] dn[ς]) /
    (Subscript[z, l] + Subscript[x, 1] t^2)], 14],
Style["Using the given formulas, we get", 13],
Style[Row[{TraditionalForm[
    HoldForm[Subscript[q, l] sn[ς + Derivative[1][e] Subscript[t, l]] ==
    Subscript[ε, l] (t Sqrt[u Subscript[x, 1] Subscript[y, l]

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(1 + Subscript[z, l]) (1 + u Subscript[z, l]) -
Derivative[1][e] Subscript[e, 0] Sqrt[u Subscript[
y, l] Subscript[z, l] D[Subscript[x, 1], t]] /
(Subscript[y, l] u (Subscript[z, l] + Subscript[x, 1] t^2))]],
"      ", Style["★", 15, RGBColor[0.10, 0.30, 0.70]]], 14],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 2]], ", ",
TraditionalForm[HoldForm[Derivative[1][e] = -Subscript[e, 2]]],
" in ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]]],
" gives the first formula of ",
Style["★", 13, s6colTest], " ."}], 13],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 1]], ", ",
TraditionalForm[HoldForm[Derivative[1][e] = Subscript[e, 1]]],
" in ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]]],
" gives the second formula of ",
Style["★", 13, s6colTest], " ."}], 13], "",
Style[Row[{"The identity ", Style["★", 13, RGBColor[0.10, 0.30, 0.70]]],
" is verified separately on each sign region:"}], 13],
Grid[Table[{Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for all 8 sign combinations of ",
TraditionalForm[HoldForm[Derivative[1][e]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 0]]], ", ",
TraditionalForm[HoldForm[Subscript[ε, l]]], " ∈ {±1}"}],
12, GrayLevel[0.3]], smallChip[regionOK[[iw]]], {iw, 4}],
Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
FrameMargins →
12]]];];

(* =====*)
(*PART 4:U-FORMULA IN CASES (a) AND (b)*)
(* =====*)
ClearAll[X3, RR, h1, h2, h3, ee0, ep2, ep3, kk, w3Configs,
thAssum, thLhs, thRhs, thCheck, lblX3, lblY2, lblXi, lblUZ];
lblX3[1] = TraditionalForm[HoldForm[Subscript[x, 3] > 0]];
lblX3[-1] = TraditionalForm[HoldForm[Subscript[x, 3] < 0]];
lblY2[1] = TraditionalForm[HoldForm[Subscript[y, 2] > 0]];
lblY2[-1] = TraditionalForm[HoldForm[Subscript[y, 2] < 0]];
lblXi[1] = TraditionalForm[HoldForm[Subscript[ξ, 23] = 1]];
lblXi[-1] = TraditionalForm[HoldForm[Subscript[ξ, 23] = -1]];
lblUZ[1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] = 1]];
lblUZ[-1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] = -1]];
w3Configs =
Table[{Row[{lblX[sx], ", ", lblX3[s3], ", ", lblY2[sy], ", hence ",
lblXi[sx s3], ", ", lblUZ[sx sy]]], sx, s3, sy},
{sx, {1, -1}}, {s3, {1, -1}}, {sy, {1, -1}}];
w3Configs = Flatten[w3Configs, 2];
thCheck[{lab_, sx_, s3_, sy_}] :=
Module[{x1 = sx XX, x3 = s3 X3, xi = sx s3, suz = sx sy, p1, p3, q2, S23,

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denom, lhs, rhs, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
denom = 1 + xi kk^2 x1 h1^2 tt^2;
lhs = p3 (S23 h2 h3 (tt / p1) - I S23 h1 (ee0 I RR / (XX p1))) / denom;
rhs = ep2 ep3 suz (XX h2 h3 tt + ee0 h1 RR) / (Sqrt[XX X3] denom);
assum = XX > 0 && X3 > 0 && YY > 0 && 0 < kk < 1 && RR ≥ 0 &&
Element[tt | h1 | h2 | h3 | ee0 | ep2 | ep3, Reals] && denom ≠ 0;
TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] === True];
Module[{grid, ok}, grid = thCheck /@ w3Configs;
ok = And @@ grid;
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
" in cases (a) and (b) for ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
Style[TraditionalForm[HoldForm[u > 0]], Bold, 15,
RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]}],
Style["Given hypotheses and formulas, in addition to those above:",
Bold, 13], Style[Row[{TraditionalForm[HoldForm[Subscript[x, 3]
Subsuperscript[p, 3, 2] == 1]], ", ", TraditionalForm[
HoldForm[Subscript[y, 2] Subsuperscript[q, 2, 2] == 1]], ", ",
TraditionalForm[HoldForm[Subscript[p, 2] == Subscript[p, 1]]],
", ", TraditionalForm[HoldForm[Subscript[q, 3] == Subscript[q, 2]]],
", ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]], " ∈ ℝ"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] ==
Sign[Subscript[x, 1] Subscript[x, 3]]],
", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
Sign[Subscript[x, 1] Subscript[y, 2]]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
", ", TraditionalForm[HoldForm[Subscript[p, 1] cn[ς] dn[ς] ==
Subscript[e, 0] I R / Abs[Subscript[x, 1]]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2]
Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] Subsuperscript[
p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /
(Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2])]],
", ", TraditionalForm[HoldForm[Subsuperscript[S, 23, 2] ==
Subscript[ξ, 23]]]]}], 13],
Style[TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[e, 2]
Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
{I Subscript[S, 23] Subscript[h, 231], Subscript[S, 23]
Subscript[h, 232], Subscript[h, 233]}]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[Subscript[e, 2]

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Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
I Subscript[S, 23] Subscript[h, 231]]], ", ",
TraditionalForm[HoldForm[sn[Subscript[e, 2] Subscript[t, 2] +
Subscript[e, 3] Subscript[t, 3]]^2 ==
-Subscript[ξ, 23] Subsuperscript[h, 231, 2]]], ", ",
TraditionalForm[HoldForm[cn[Subscript[e, 2] Subscript[t, 2] +
Subscript[e, 3] Subscript[t, 3]] dn[Subscript[e, 2]
Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
Subscript[S, 23] Subscript[h, 232] Subscript[h, 233]]]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς]^2 == Subscript[x, 1] t^2]],
", ", TraditionalForm[
HoldForm[1 - k^2 sn[ς]^2 sn[Subscript[e, 2] Subscript[t, 2] +
Subscript[e, 3] Subscript[t, 3]]^2 == 1 + Subscript[ξ, 23]
k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2 ≠ 0]],
", by the choice of ", TraditionalForm[HoldForm[t]]}],
13], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[± (Subscript[e, 2]
Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3])]], " and ",
TraditionalForm[HoldForm[ς]], " are not poles and ",
TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 sn[Subscript[e, 2]
Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 ≠ 0]],
", ", Style["the first addition formula", Italic, 13,
RGBColor[0.10, 0.30, 0.70]],
", at ", TraditionalForm[HoldForm[v = ς]], ", ",
TraditionalForm[HoldForm[w = -Subscript[e, 2] Subscript[t, 2] -
Subscript[e, 3] Subscript[t, 3]]],
", holds pointwise and, together with ", Style[
"the parity formulas", Italic, 13,
RGBColor[0.75, 0.35, 0.05]], ", gives"}], 13],
Style[TraditionalForm[HoldForm[Subscript[p, 3] sn[ς - Subscript[e, 2]
Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] ==
(Subscript[p, 3] (cn[Subscript[e, 2] Subscript[t, 2] +
Subscript[e, 3] Subscript[t, 3]] dn[Subscript[e, 2]
Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] sn[ς] -
sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3]
Subscript[t, 3]] cn[ς] dn[ς])) / (1 + Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[
Row[{TraditionalForm[HoldForm[Subscript[p, 3] sn[ς - Subscript[e, 2]
Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] ==
Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
(Abs[Subscript[x, 1] Subscript[h, 232] Subscript[h, 233] t +
Subscript[e, 0] Subscript[h, 231]
Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]]) / (Sqrt[
Abs[Subscript[x, 1] Subscript[x, 3]] (1 + Subscript[ξ, 23]
k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))}],

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"      ", Style["♦", 14, RGBColor[0.55, 0.25, 0.70]]}], 14],
Style[Row[{"which is the third formula of ",
  Style["*", 13, s6colTest], " ."}], 13], "",
Style[Row[{"The identity ", Style["♦", 13, RGBColor[0.55, 0.25, 0.70]],
  " is verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[h, 231]]],
  ", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
  ", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
  ", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
  ", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
  ", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
  " symbolic, that is, for all their real values at once,
  which covers all their sign choices:"}], 13],
Style[Row[{"In case (a), both values of ", TraditionalForm[
  HoldForm[Subscript[ξ, 23]]], " occur, because the sign of ",
  TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3]]],
  " is not constrained; in case (b), ",
  TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]], ", because ",
  TraditionalForm[HoldForm[k^2 == (Subsuperscript[g, 2, 2] - 1)
    (Subsuperscript[g, 3, 2] - 1) > 0]], " and ", TraditionalForm[
  HoldForm[Subsuperscript[g, i, 2] - 1 == 1 / Subscript[z, i]]], ", so ",
  TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
  " . The eight sign regions cover both cases."}],
12, GrayLevel[0.35]], Grid[Table[{Style[w3Configs[[iw, 1]],
  12, GrayLevel[0.3]], smallChip[grid[[iw]]]}, {iw, 8}],
Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.55, 0.25, 0.70],
0.93], FrameMargins → 12]]];];

(* =====*)
(*PART 5:U-FORMULA IN CASE (c) FOR u>0*)
(* =====*)
ClearAll[cCheck];
cCheck[{lab_, sx_, s3_, sy_}] :=
Module[{p1, p3, q2, xi = sx s3, suz = sx sy, S23, lhs, rhs, assum},
  p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
  q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
  lhs = S23 p1 p3 / (kk tt);
  rhs = suz ep2 ep3 sx / (kk Sqrt[XX X3] tt);
  assum =
    XX > 0 && X3 > 0 && YY > 0 && 0 < kk < 1 && Element[tt | ep2 | ep3, Reals] && tt ≠ 0;
  TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] == True];
Module[{cConfigs, grid, ok}, cConfigs = Select[w3Configs, #[[2]] #[[3]] == 1 &];
grid = cCheck /@ cConfigs;
ok = And@@grid;
AppendTo[resultsAll, ok];

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Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
    15, RGBColor[0.00, 0.50, 0.50]], Style[" in case (c) for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u > 0]],
    Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]]},
  Style["Given hypotheses and formulas, in addition to those above:",
  Bold, 13], Style[Row[{TraditionalForm[
    HoldForm[Subscript[ξ, 23] == Sign[Subscript[x, 1] Subscript[x, 3]]],
    ", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
    Sign[Subscript[x, 1] Subscript[y, 2]]]]}], 13],
  Style[TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2]
    Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] Subsuperscript[
    p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /
    (Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2])]], 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[t, 2] + Subscript[t, 3] ==
    2 m K + I K']], ", ", TraditionalForm[HoldForm[m ∈ ℤ]], ", ",
    TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]]], 13],
  Style[Row[{TraditionalForm[HoldForm[sn[v + 2 m K] == (-1)^m sn[v]]],
    ", ", TraditionalForm[HoldForm[sn[v ± I K'] == 1 / (k sn[v])]],
    " ", Style["the shift formulas", Italic,
    13, RGBColor[0.00, 0.50, 0.50]]}], 13],
  Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
    ", ", TraditionalForm[HoldForm[t ≠ 0]],
    ", by the choice of ", TraditionalForm[HoldForm[t]]}], 13], "",
  Style["Verification:", Bold, 13], Style[Row[{"Since ", TraditionalForm[
    HoldForm[Subscript[e, 2] == Subscript[e, 3]]], " and ",
    TraditionalForm[HoldForm[(-1)^(m Subscript[e, 2]) == (-1)^m]],
    ", ", Style["the shift formulas", Italic, 13,
    RGBColor[0.00, 0.50, 0.50]], " give"}], 13],
  Style[TraditionalForm[HoldForm[Subscript[p, 3]
    sn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
    (-1)^m Subscript[p, 3] / (k sn[ς]) ==
    Subscript[S, 23] Subscript[p, 1] Subscript[p, 3] / (k t)]], 14],
  Style["Using the given hypotheses and formulas, we get", 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] sn[ς - Subscript[e,
    2] (Subscript[t, 2] + Subscript[t, 3])] == Subscript[ε, 2]
    Subscript[ε, 3] Sign[u Subscript[z, 2]] Sign[Subscript[x, 1]] /
    (k Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]] t)]],
    " ", Style["▲", 14, RGBColor[0.00, 0.50, 0.50]]}], 14],
  Style[Row[{"which is the fourth formula of ",
    Style["*"], 13, s6colTest], " ."}], 13], "",
  Style[Row[{"The identity ", Style["▲", 13, RGBColor[0.00, 0.50, 0.50]],
    " is verified separately on each of the four sign
    regions realizable in case (c), with ",
    TraditionalForm[HoldForm[Subscript[ε, 2]]], ", ",

```

```

TraditionalForm[HoldForm[Subscript[ $\varepsilon$ , 3]]],
" symbolic, that is, for all their real values at once,
  which covers all their sign choices:"]], 13],
Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ $\xi$ , 23] == 1]],
", because ", TraditionalForm[HoldForm[k^2 ==
  (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
" and ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
  1 / Subscript[z, i]]], ", so ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
" ."}], 12, GrayLevel[0.35]],
Grid[Table[{Style[cConfigs[[iw], 1]], 12, GrayLevel[0.3]],
  smallChip[grid[[iw]]], {iw, 4}],
  Alignment -> Left, Spacings -> {2, 0.7}], Spacings -> 1],
Background -> Lighter[RGBColor[0.00, 0.50, 0.50], 0.93],
FrameMargins ->
  12]]];];

(* =====*)
(*PART 6:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And@@resultsAll, "CASE 1: ALL CLAIMS VERIFIED",
  "SOME CLAIMS FAILED"], Bold, White, 16], Background ->
  If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle -> None, RoundingRadius -> 10, FrameMargins -> {{18, 18}, {8, 8}}]];

```

Given hypotheses, Case 1 ($u > 0$) :

$$0 < u < 1, \quad k = \sqrt{u}, \quad k' = \sqrt{1-u}, \quad k, k' \in (0, 1)$$

For $i = 1, 2, 3, 4$:

$$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0, \quad g_i = \operatorname{dn}(t_i) = \sqrt{1 + \frac{1}{z_i}}$$

$$p_i, q_i \in \mathbb{R}_{>0} \cup i \mathbb{R}_{>0}, \quad e_0, e', e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$$

$$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$$

$$\mathcal{D}(x_1, t) = -(1 - x_1 t^2) (1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$$

Given formulas:

$$\operatorname{sn}(v + w) = \frac{\operatorname{sn}(v) \operatorname{cn}(w) \operatorname{dn}(w) + \operatorname{sn}(w) \operatorname{cn}(v) \operatorname{dn}(v)}{1 - k^2 \operatorname{sn}(v)^2 \operatorname{sn}(w)^2} \quad \text{the first addition formula}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(-v) = \{-\operatorname{sn}(v), \operatorname{cn}(v), \operatorname{dn}(v)\} \quad \text{the parity formulas}$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test these formulas ★ :

$$W_2 = \frac{\varepsilon_2 (t \sqrt{u x_1 y_2 (1+z_2) (1+u z_2)} + e_0 e_2 \sqrt{u y_2 z_2 \mathcal{D}(x_1, t)})}{y_2 u (z_2 + x_1 t^2)}$$

$$W_1 = \frac{\varepsilon_1 (t \sqrt{u x_1 y_1 (1+z_1) (1+u z_1)} - e_0 e_1 \sqrt{u y_1 z_1 \mathcal{D}(x_1, t)})}{y_1 u (z_1 + x_1 t^2)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (1 + \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \text{in cases (a) and (b)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) \operatorname{sgn}(x_1)}{k \sqrt{|x_1 x_3|} t} \quad \text{in case (c)}$$

The derivation: the formulas for W_2 and W_1 for $u > 0$

Verified

Given hypotheses and formulas, in addition to those above:

For $l = 1, 2$:

$$x_l p_l^2 = 1, \quad y_l q_l^2 = 1, \quad p_l = p_1$$

$$\operatorname{sn}(t_l) = \frac{i \varepsilon_l \sqrt{(g_l^2 - 1) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{cn}(t_l) = \frac{\varepsilon_l \sqrt{(g_l^2 - (k')^2) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{sn}(\zeta) = \frac{t}{p_1}, \quad p_1 \operatorname{cn}(\zeta) \operatorname{dn}(\zeta) = \frac{e_0 i R}{|x_1|}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(e' t_l) = \{e' \operatorname{sn}(t_l), \operatorname{cn}(t_l), \operatorname{dn}(t_l)\} \quad \text{by the parity formulas}$$

$$\operatorname{sn}(t_l)^2 = \frac{1 - g_l^2}{k^2} = -\frac{1}{u z_l}, \quad 1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 = \frac{z_l + x_1 t^2}{z_l} \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm t_l$ and ζ are not poles and $1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 \neq 0$, the first addition formula, at $v = \zeta$, $w = e' t_l$, holds pointwise and, together with the parity formulas, gives

$$q_l \operatorname{sn}(\zeta + e' t_l) = \frac{z_l q_l \operatorname{cn}(t_l) \operatorname{dn}(t_l) \operatorname{sn}(\zeta) + e' z_l q_l \operatorname{sn}(t_l) \operatorname{cn}(\zeta) \operatorname{dn}(\zeta)}{z_l + x_1 t^2}$$

Using the given formulas, we get

$$q_l \operatorname{sn}(\zeta + e' t_l) = \frac{\varepsilon_l (t \sqrt{u x_1 y_l (1 + z_l) (1 + u z_l)} - e' e_0 \sqrt{u y_l z_l \mathcal{D}(x_1, t)})}{y_l u (z_l + x_1 t^2)} \quad \star$$

Setting $l = 2$, $e' = -e_2$ in $\star$ gives the first formula of $\star$.

Setting $l = 1$, $e' = e_1$ in $\star$ gives the second formula of $\star$.

The identity $\star$ is verified separately on each sign region:

$z_l > 0, y_l > 0$, hence $x_1 > 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓
$z_l > 0, y_l < 0$, hence $x_1 < 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓
$z_l < -1, y_l > 0$, hence $x_1 < 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓
$z_l < -1, y_l < 0$, hence $x_1 > 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓

The derivation: the formula for U in cases (a) and (b) for $u > 0$

Verified

Given hypotheses and formulas, in addition to those above:

$$x_3 p_3^2 = 1, \quad y_2 q_2^2 = 1, \quad p_2 = p_1, \quad q_3 = q_2, \quad h_{231}, h_{232}, h_{233} \in \mathbb{R}$$

$$\xi_{23} = \text{sgn}(x_1 x_3), \quad \text{sgn}(u z_2) = \text{sgn}(x_1 y_2)$$

$$\text{sn}(\varsigma) = \frac{t}{p_1}, \quad p_1 \text{cn}(\varsigma) \text{dn}(\varsigma) = \frac{e_0 i R}{|x_1|}$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}, \quad S_{23}^2 = \xi_{23}$$

$$\{\text{sn}, \text{cn}, \text{dn}\}(e_2 t_2 + e_3 t_3) = \{i S_{23} h_{231}, S_{23} h_{232}, h_{233}\}$$

$$\text{sn}(e_2 t_2 + e_3 t_3) = i S_{23} h_{231}, \quad \text{sn}(e_2 t_2 + e_3 t_3)^2 = -\xi_{23} h_{231}^2$$

$$, \quad \text{cn}(e_2 t_2 + e_3 t_3) \text{dn}(e_2 t_2 + e_3 t_3) = S_{23} h_{232} h_{233}$$

$$\text{sn}(\varsigma)^2 = x_1 t^2, \quad 1 - k^2 \text{sn}(\varsigma)^2 \text{sn}(e_2 t_2 + e_3 t_3)^2 = 1 + \xi_{23} k^2 x_1 h_{231}^2 t^2 \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm(e_2 t_2 + e_3 t_3)$ and ς are not poles and

$$1 - k^2 \text{sn}(\varsigma)^2 \text{sn}(e_2 t_2 + e_3 t_3)^2 \neq 0, \quad \text{the first addition formula, at } v = \varsigma,$$

$w = -e_2 t_2 - e_3 t_3$, holds pointwise and, together with the parity formulas, gives

$$p_3 \text{sn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{p_3 (\text{cn}(e_2 t_2 + e_3 t_3) \text{dn}(e_2 t_2 + e_3 t_3) \text{sn}(\varsigma) - \text{sn}(e_2 t_2 + e_3 t_3) \text{cn}(\varsigma) \text{dn}(\varsigma))}{1 + \xi_{23} k^2 x_1 h_{231}^2 t^2}$$

Using the given hypotheses and formulas, we get

$$p_3 \text{sn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (1 + \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \blacklozenge$$

which is the third formula of $\star$.

The identity $\blacklozenge$ is verified separately on each sign region, with $h_{231}, h_{232}, h_{233}, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

In case (a), both values of ξ_{23} occur, because the sign of $z_2 z_3$

is not constrained; in case (b), $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$

and $g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$. The eight sign regions cover both cases.

$$x_1 > 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = -1, \quad \text{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = -1, \quad \text{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = -1, \quad \text{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = -1, \quad \text{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \text{sgn}(u z_2) = 1 \quad \checkmark$$

The derivation: the formula for U in case (c) for $u > 0$ **Verified****Given hypotheses and formulas, in addition to those above:**

$$\xi_{23} = \operatorname{sgn}(x_1 x_3), \quad \operatorname{sgn}(u z_2) = \operatorname{sgn}(x_1 y_2)$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{\rho_1^2 \rho_3^2 q_2^4 \xi_{23}}}{\rho_1 \rho_3 q_2^2}$$

$$t_2 + t_3 = 2mK + iK', \quad m \in \mathbb{Z}, \quad (-1)^m = S_{23}, \quad e_2 = e_3$$

$$\operatorname{sn}(v + 2mK) = (-1)^m \operatorname{sn}(v), \quad \operatorname{sn}(v \pm iK') = \frac{1}{k \operatorname{sn}(v)} \quad \text{the shift formulas}$$

$$\operatorname{sn}(\varsigma) = \frac{t}{\rho_1}, \quad t \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $e_2 = e_3$ and $(-1)^{m e_2} = (-1)^m$, the shift formulas give

$$\rho_3 \operatorname{sn}(\varsigma - e_2(t_2 + t_3)) = \frac{(-1)^m \rho_3}{k \operatorname{sn}(\varsigma)} = \frac{S_{23} \rho_1 \rho_3}{k t}$$

Using the given hypotheses and formulas, we get

$$\rho_3 \operatorname{sn}(\varsigma - e_2(t_2 + t_3)) = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) \operatorname{sgn}(x_1)}{k \sqrt{|x_1 x_3|} t} \quad \blacktriangle$$

which is the fourth formula of $\star$.

The identity $\blacktriangle$

is verified separately on each of the four sign regions realizable in case (c), with $\varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$.

$$x_1 > 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = 1 \quad \checkmark$$

CASE 1: ALL CLAIMS VERIFIED

Section 9

Results of Section 9 are used in the proof of Lemma 12: Case 2

In[2477]:=

```
(* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[UU, XX, YY, ZZ, tt, wConfigs, sTriples, dataOf, lhsOf, derOf, assumOf,
  derCheck, resultsAll, statusChip, smallChip, lblZ, lblY, lblX, s7colTest];
```

```

statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s7colTest = RGBColor[0.10, 0.35, 0.75];
(*Case 2:  $u < 0$ . We write  $u = -UU$  with  $UU > 0$ ,
so  $M = 1 + UU$ ,  $k' = 1/\text{Sqrt}[1 + UU]$ ,  $k = k' \text{Sqrt}[UU]$ ,
 $g_l = 1/\text{Sqrt}[1 + 1/z_l]$ . Sign regions: two  $z_l$ -windows and the sign of  $y_l$ ;
the sign of  $x_l$  is forced by  $u x_l y_l z_l > 0$ ,
which for  $u < 0$  means  $x_l y_l z_l < 0$ . The window  $z_l > 0$  carries  $1 + u z_l > 0$ , that is,
 $1 - UU z_l > 0$ . On each region,  $p_l$  and  $q_l$  take their explicit radical values;
the identity is checked there for all 8 sign combinations of  $e', e_0, \text{eps}_l$ .)
lblZ[1] = TraditionalForm[HoldForm[Subscript[z, l] > 0]];
lblZ[2] = TraditionalForm[HoldForm[Subscript[z, l] < -1]];
lblY[1] = TraditionalForm[HoldForm[Subscript[y, l] > 0]];
lblY[-1] = TraditionalForm[HoldForm[Subscript[y, l] < 0]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, l] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, l] < 0]];
wConfigs = {{Row[{lblZ[1], " ", lblY[1], " hence ", lblX[-1]}], ZZ, 1, -1},
  {Row[{lblZ[1], " ", lblY[-1], " hence ", lblX[1]}], ZZ, -1, 1},
  {Row[{lblZ[2], " ", lblY[1], " hence ", lblX[1]}], -1 - ZZ, 1, 1},
  {Row[{lblZ[2], " ", lblY[-1], " hence ", lblX[-1]}], -1 - ZZ, -1, -1}};
sTriples = Tuples[{1, -1}, 3];
dataOf[{lab_, zl_, sy_, sx_}] :=
Module[{yl = sy YY, x1 = sx XX, kk = Sqrt[UU] / Sqrt[1 + UU],
  kp = 1 / Sqrt[1 + UU], gl, pl, ql, Dd}, gl = 1 / Sqrt[1 + 1 / zl];
pl = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
ql = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
Dd = -(1 - x1 tt^2) (1 + UU x1 tt^2);
{zl, yl, x1, kk, kp, gl, pl, ql, Dd}];
lhsOf[{zl_, yl_, x1_, kk_, kp_, gl_, pl_, ql_, Dd_}, ep_, e0_, eps_] :=
Module[{snl, cnl, RR = Sqrt[x1 Dd]},
  snl = I eps Sqrt[(gl^2 - 1) p1^2 ql^2] / (kk p1 ql);
  cnl = eps Sqrt[(gl^2 - kp^2) p1^2 ql^2] / (kk p1 ql);
  ((1 + zl) ql cnl (tt / p1) +
    ep (1 + zl) ql snl gl (e0 I kp RR / (XX p1))) / (zl + x1 tt^2)];
derOf[{zl_, yl_, x1_, kk_, kp_, gl_, pl_, ql_, Dd_}, ep_, e0_, eps_] :=
Module[{uu = -UU}, eps (tt Sqrt[uu x1 yl (1 + zl) (1 + uu zl)] -
  ep e0 Sqrt[uu yl zl Dd]) / (yl uu (zl + x1 tt^2))];
assumOf[{lab_, zl_, sy_, sx_}] := UU > 0 && XX > 0 && YY > 0 && ZZ > 0 &&
  Element[tt, Reals] && sx XX (- (1 - sx XX tt^2) (1 + UU sx XX tt^2)) ≥ 0 &&
  1 - UU zl > 0 && zl + sx XX tt^2 ≠ 0;
derCheck[w_, {ep_, e0_, eps_}] := Module[{d = dataOf[w]},

```

```

TimeConstrained[FullSimplify[lhsOf[d, ep, e0, eps] - derOf[d, ep, e0, eps] == 0,
  assumOf[w]], 180, False] === True];
(* =====*)
(*PART 2:GIVEN PANEL*)
(* =====*)
Print[Panel[Column[{Style[Row[{"Given hypotheses, Case 2  ("},
  TraditionalForm[HoldForm[u < 0]], ") :"}], Bold, 14],
Style[Row[{TraditionalForm[HoldForm[u < 0]], " ",
  TraditionalForm[HoldForm[k' == 1 / Sqrt[1 - u]]], " ", TraditionalForm[
  HoldForm[k == k' Sqrt[-u]]], " ", TraditionalForm[HoldForm[k]],
  " ", TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 14],
Style[Row[{"For ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
  " ", TraditionalForm[HoldForm[Subscript[y, i]]], " ",
  TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
  " ", TraditionalForm[HoldForm[Subscript[x, i]]],
  " ", TraditionalForm[HoldForm[Subscript[y, i]]],
  " ", TraditionalForm[HoldForm[Subscript[z, i]]],
  " ≠ -1, 0, ", TraditionalForm[HoldForm[Subscript[g, i] ==
  dn[Subscript[t, i] == 1 / Sqrt[1 + 1 / Subscript[z, i]]]]], 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, i]]], " ",
  TraditionalForm[HoldForm[Subscript[q, i]]], " ∈ ",
  Subscript["ℝ", "> 0"], " ∪ i ", Subscript["ℝ", "> 0"],
  " ", TraditionalForm[HoldForm[Subscript[e, 0]]],
  " ", TraditionalForm[HoldForm[Derivative[1][e]]],
  " ", TraditionalForm[HoldForm[Subscript[e, i]]],
  " ", TraditionalForm[HoldForm[Subscript[ε, i]]], " ",
  TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"}], 14],
Style[Row[{TraditionalForm[HoldForm[u Subscript[x, i]
  Subscript[y, i] Subscript[z, i] > 0]], " ",
  TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
  " ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
Style[Row[{TraditionalForm[HoldForm[∂[Subscript[x, 1], t] ==
  -(1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
  " ", TraditionalForm[HoldForm[
  R == Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]]], " ",
  TraditionalForm[HoldForm[Subscript[x, 1] ∂[Subscript[x, 1], t] ≥ 0]]}],
14], "", Style["Given formulas:", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[cn[v + w] == (cn[v] cn[w] -
  sn[v] sn[w] dn[v] dn[w]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
  " ", Style["the second addition formula", Italic, 13,
  RGBColor[0.10, 0.30, 0.70]]], 14], Style[
Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] == {-sn[v], cn[v], dn[v]}]],
  " ", Style["the parity formulas",
  Italic, 13, RGBColor[0.75, 0.35, 0.05]]],
14], Style[Row[{"Each ", smallChip[True],
  " below certifies that the statement displayed beside it has been

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        verified by a symbolic computation at that point; a ",
        smallChip[False], " would mean that this verification failed."}],
12, GrayLevel[0.35]], "", Style[
Row[{Style["We want to test these formulas ", Bold, 14],
Style["*", 14, s7colTest], Style[" :", Bold, 14]}]],
Style[TraditionalForm[HoldForm[Subscript[W, 2] ==
Subscript[ε, 2] (t Sqrt[u Subscript[x, 1] Subscript[y, 2]
(1+Subscript[z, 2]) (1+u Subscript[z, 2])) +
Subscript[e, 0] Subscript[e, 2] Sqrt[u Subscript[y, 2]
Subscript[z, 2] D[Subscript[x, 1], t]] /
(Subscript[y, 2] u (Subscript[z, 2] + Subscript[x, 1] t^2))]],
15, s7colTest], Style[TraditionalForm[HoldForm[Subscript[W, 1] ==
Subscript[ε, 1] (t Sqrt[u Subscript[x, 1] Subscript[y, 1]
(1+Subscript[z, 1]) (1+u Subscript[z, 1])) -
Subscript[e, 0] Subscript[e, 1] Sqrt[u Subscript[y, 1]
Subscript[z, 1] D[Subscript[x, 1], t]] / (Subscript[y, 1] u
(Subscript[z, 1] + Subscript[x, 1] t^2))]], 15, s7colTest],
Style[Row[{TraditionalForm[HoldForm[U = Subscript[ε, 2] Subscript[ε, 3]
Sign[u Subscript[z, 2]] (Abs[Subscript[x, 1]] Subscript[h,
232] t + Subscript[e, 0] k' Subscript[h, 231] Subscript[
h, 233] Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]]) /
(Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]])
(Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))]], " ",
Style["in cases (a) and (b)", Italic, 13, GrayLevel[0.35]]}],
15, s7colTest], Style[Row[{TraditionalForm[HoldForm[
U = Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
Subscript[e, 0] Subscript[e, 2] k' Sqrt[Subscript[x, 1] D[
Subscript[x, 1], t]] / (k Sqrt[Abs[Subscript[x, 1]
Subscript[x, 3]]) (1 - Subscript[x, 1] t^2))]],
" ", Style["in case (c)", Italic, 13, GrayLevel[0.35]]}],
15, s7colTest]], Spacings → 1.1],
Background → RGBColor[0.99, 0.96, 0.88],
FrameMargins →
14]];
(* =====*)
(*PART 3:THE PARAMETER NORMALIZATION*)
(* =====*)
resultsAll = {};
ClearAll[nCheck];
nCheck[sx_] :=
Module[{x1 = sx XX, p1, Dd, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
Dd = -(1 - x1 tt^2) (1 + UU x1 tt^2);
assum = UU > 0 && XX > 0 && Element[tt, Reals];
TimeConstrained[
FullSimplify[p1^2 (1 - x1 tt^2) (1 / (1 + UU) + UU / (1 + UU) x1 tt^2) -
(- (1 / (1 + UU)) x1 Dd / x1^2) == 0, assum], 180, False] == True];

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```

Module[{grid, ok}, grid = nCheck /@ {1, -1};
ok = And@@grid;
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The parameter normalization for ", Bold, 15,
    RGBColor[0.75, 0.35, 0.05]], Style[TraditionalForm[HoldForm[u < 0]],
    Bold, 15, RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]]},
  Style["Given hypotheses and formulas, in addition to those above:",
  Bold, 13], Style[Row[{TraditionalForm[HoldForm[sn[v]^2 + cn[v]^2 == 1]],
  ", ", TraditionalForm[HoldForm[dn[v]^2 == k'^2 + k^2 cn[v]^2]],
  " ", Style["the Pythagorean relations", Italic, 13,
  RGBColor[0.75, 0.35, 0.05]]}], 13], Style[Row[{TraditionalForm[
  HoldForm[Subscript[x, 1] Subsuperscript[p, 1, 2] == 1]], ", ",
  TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]], ", ",
  TraditionalForm[HoldForm[cn[ς]^2 == Subscript[x, 1] t^2]]}],
  13], "", Style["Verification:", Bold, 13],
  Style[Row[{"By ", Style["the Pythagorean relations",
    Italic, 13, RGBColor[0.75, 0.35, 0.05]], " and ",
    TraditionalForm[HoldForm[k^2 == -u k'^2]], ", we have"}], 13],
  Style[TraditionalForm[HoldForm[(Subscript[p, 1] sn[ς] dn[ς])^2 ==
    Subsuperscript[p, 1, 2] (1 - cn[ς]^2) (k'^2 + k^2 cn[ς]^2) ==
    - (k'^2 Subscript[x, 1] ∂[Subscript[x, 1], t] / Subsuperscript[
    x, 1, 2]) == (I k' R / Abs[Subscript[x, 1]])^2]], 14],
  Style[Row[{"By ", Style["the parity formulas", Italic, 13, RGBColor[
    0.75, 0.35, 0.05]], ", replacing ", TraditionalForm[HoldForm[ς]],
    " by ", TraditionalForm[HoldForm[-ς]], " preserves ",
    TraditionalForm[HoldForm[cn[ς]]], " and reverses the sign of ",
    TraditionalForm[HoldForm[sn[ς] dn[ς]]], " ."}], 13],
  Style[Row[{"Thus, we may fix ", Style[TraditionalForm[
    HoldForm[Subscript[p, 1] sn[ς] dn[ς] ==
    -Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]],
    RGBColor[0.75, 0.35, 0.05]], ", so that the two values of ",
    TraditionalForm[HoldForm[Subscript[e, 0]]],
    " correspond to the two admissible choices of ",
    TraditionalForm[HoldForm[ς]], " ."}], 13],
  Style[Row[{"The middle equality of the displayed chain, that is, ",
    TraditionalForm[HoldForm[Subsuperscript[p, 1, 2]
    (1 - cn[ς]^2) (k'^2 + k^2 cn[ς]^2) == - (k'^2 Subscript[x,
    1] ∂[Subscript[x, 1], t] / Subsuperscript[x, 1, 2])]],
    " , is verified on each sign region:"}], 13],
  Grid[{{Style[lblX[1], 12, GrayLevel[0.3]], smallChip[grid[[1]]]},
    {Style[lblX[-1], 12, GrayLevel[0.3]], smallChip[grid[[2]]]}},
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
  Background → Lighter[RGBColor[0.75, 0.35, 0.05], 0.93],
  FrameMargins → 12]]];];
(* =====*)
(*PART 4:THE W-FORMULAS*)

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```
(* =====*)
Module[{grid, regionOK, ok},
  grid = Table[derCheck[w, s], {w, wConfigs}, {s, sTriples}];
  regionOK = And@@@grid;
  ok = And@@regionOK;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["The derivation: the formulas for ", Bold, 15,
      RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[
        HoldForm[Subscript[W, 2]]], Bold, 15, RGBColor[0.10, 0.30, 0.70]],
      Style[" and ", Bold, 15, RGBColor[0.10, 0.30, 0.70]],
      Style[TraditionalForm[HoldForm[Subscript[W, 1]]], Bold,
        15, RGBColor[0.10, 0.30, 0.70]], Style[" for ", Bold, 15,
          RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u < 0]],
            Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]]],
      Style["Given hypotheses and formulas, in addition to those above:",
        Bold, 13], Style[
        Row[{"For ", TraditionalForm[HoldForm[l]], " = 1, 2 :"}], Bold, 13],
      Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1]
        Subsuperscript[p, 1, 2] == 1]], ", ", TraditionalForm[
        HoldForm[Subscript[y, l] Subsuperscript[q, l, 2] == 1]], ", ",
        TraditionalForm[HoldForm[Subscript[p, l] == Subscript[p, 1]]]], 13],
      Style[TraditionalForm[HoldForm[sn[Subscript[t, l]] ==
        I Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - 1)
          Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] /
          (k Subscript[p, l] Subscript[q, l])]], 14],
      Style[TraditionalForm[HoldForm[cn[Subscript[t, l]] ==
        Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - k'^2)
          Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] /
          (k Subscript[p, l] Subscript[q, l])]], 14],
      Style[Row[{TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]],
        ", ", TraditionalForm[HoldForm[Subscript[p, 1] sn[ς] dn[ς] ==
          -Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]], 14],
      Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Derivative[1][e]
        Subscript[t, l]] == {Derivative[1][e] sn[Subscript[t, l]],
          cn[Subscript[t, l]], dn[Subscript[t, l]]}],
        " ", Style["by the parity formulas", Italic, 13,
          RGBColor[0.75, 0.35, 0.05]]], 14], Style[Row[{TraditionalForm[
        HoldForm[sn[Subscript[t, l]]^2 == (1 - Subsuperscript[g, l, 2]) / k^2 ==
          -((1 - u) / (u (1 + Subscript[z, l])))], ", ",
        TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, l]]^2 sn[ς]^2 ==
          (Subscript[z, l] + Subscript[x, 1] t^2) / (1 + Subscript[z, l]) ≠ 0]],
        ", by the choice of ", TraditionalForm[HoldForm[t]]],
      14], "", Style["Verification:", Bold, 13],
      Style[Row[{"Since ", TraditionalForm[HoldForm[±Subscript[t, l]]],
        " and ", TraditionalForm[HoldForm[ς]],
        " are not poles and ", TraditionalForm[
```

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HoldForm[1 - k^2 sn[Subscript[t, l]]^2 sn[c]^2 ≠ 0]], " ",
Style["the second addition formula", Italic, 13, RGBColor[0.10,
0.30, 0.70]], " at ", TraditionalForm[HoldForm[v = c]], " ",
TraditionalForm[HoldForm[w = Derivative[1][e] Subscript[t, l]]],
", holds pointwise and, together with ",
Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
", gives"}], 13], Style[TraditionalForm[
HoldForm[Subscript[q, l] cn[c + Derivative[1][e] Subscript[t, l]] =
((1 + Subscript[z, l]) Subscript[q, l] cn[Subscript[t, l]] cn[c] -
Derivative[1][e] (1 + Subscript[z, l]) Subscript[q, l]
sn[Subscript[t, l]] dn[Subscript[t, l]] sn[c] dn[c]) /
(Subscript[z, l] + Subscript[x, 1] t^2)], 14],
Style["Using the given formulas, we get", 13],
Style[Row[{TraditionalForm[
HoldForm[Subscript[q, l] cn[c + Derivative[1][e] Subscript[t, l]] =
Subscript[ε, l] (t Sqrt[u Subscript[x, 1] Subscript[y, l]
(1 + Subscript[z, l]) (1 + u Subscript[z, l])) -
Derivative[1][e] Subscript[e, 0] Sqrt[u Subscript[y, l]
Subscript[z, l] ∂[Subscript[x, 1], t]] / (Subscript[y, l]
u (Subscript[z, l] + Subscript[x, 1] t^2))}],
", Style["*", 15, RGBColor[0.10, 0.30, 0.70]]}], 14],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 2]], " ",
TraditionalForm[HoldForm[Derivative[1][e] = -Subscript[e, 2]]],
" in ", Style["*", 13, RGBColor[0.10, 0.30, 0.70]],
" gives the first formula of ",
Style["*", 13, s7colTest], " ."}], 13],
Style[Row[{"Setting ", TraditionalForm[HoldForm[l = 1]], " ",
TraditionalForm[HoldForm[Derivative[1][e] = Subscript[e, 1]]],
" in ", Style["*", 13, RGBColor[0.10, 0.30, 0.70]],
" gives the second formula of ",
Style["*", 13, s7colTest], " ."}], 13], "",
Style[Row[{"The identity ", Style["*", 13, RGBColor[0.10, 0.30, 0.70]],
" is verified separately on each sign region:"}], 13],
Grid[Table[{Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for all 8 sign combinations of ",
TraditionalForm[HoldForm[Derivative[1][e]]], " ",
TraditionalForm[HoldForm[Subscript[e, 0]]], " ",
TraditionalForm[HoldForm[Subscript[ε, l]]], " ∈ {±1}"}],
12, GrayLevel[0.3]], smallChip[regionOK[[iw]]], {iw, 4}],
Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
FrameMargins →
12]]];];
(* =====*)
(*PART 5:U-FORMULA IN CASES (a) AND (b)*)
(* =====*)
ClearAll[X3, RR, h1, h2, h3, ee0, ep2, ep3,

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kk, kp, w3Configs, thCheck, lblX3, lblY2, lblXi, lblUZ];
kk = Sqrt[UU] / Sqrt[1 + UU]; kp = 1 / Sqrt[1 + UU];
lblX3[1] = TraditionalForm[HoldForm[Subscript[x, 3] > 0]];
lblX3[-1] = TraditionalForm[HoldForm[Subscript[x, 3] < 0]];
lblY2[1] = TraditionalForm[HoldForm[Subscript[y, 2] > 0]];
lblY2[-1] = TraditionalForm[HoldForm[Subscript[y, 2] < 0]];
lblXi[1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]];
lblXi[-1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == -1]];
lblUZ[1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == 1]];
lblUZ[-1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == -1]];
w3Configs =
  Table[{Row[{lblX[sx], " ", " ", lblX3[s3], " ", " ", lblY2[sy], " ", "hence ",
    lblXi[sx s3], " ", " ", lblUZ[sx sy]}], sx, s3, sy},
    {sx, {1, -1}}, {s3, {1, -1}}, {sy, {1, -1}}];
w3Configs = Flatten[w3Configs, 2];
thCheck[{lab_, sx_, s3_, sy_}] :=
  Module[{x1 = sx XX, xi = sx s3, suz = sx sy, p1, p3, q2, S23, denom,
    lhs, rhs, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
  q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
  denom = h3^2 - xi kk^2 x1 h1^2 tt^2;
  lhs =
    p3 (S23 h2 (tt / p1) + I S23 h1 h3 (-ee0 I kp RR / (XX p1))) / denom;
  rhs = ep2 ep3 suz (XX h2 tt + ee0 kp h1 h3 RR) / (Sqrt[XX X3] denom);
  assum = XX > 0 && X3 > 0 && YY > 0 && UU > 0 && RR ≥ 0 &&
    Element[tt | h1 | h2 | h3 | ee0 | ep2 | ep3, Reals] && denom ≠ 0;
  TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] === True];
Module[{grid, ok}, grid = thCheck /@ w3Configs;
ok = And @@ grid;
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
    Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
    " in cases (a) and (b) for ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
    Style[TraditionalForm[HoldForm[u < 0]], Bold, 15,
    RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]}],
  Style["Given hypotheses and formulas, in addition to those above:",
  Bold, 13], Style[Row[{TraditionalForm[HoldForm[Subscript[x, 3]
    Subsuperscript[p, 3, 2] == 1]], ", ", TraditionalForm[
    HoldForm[Subscript[y, 2] Subsuperscript[q, 2, 2] == 1]], ", ",
    TraditionalForm[HoldForm[Subscript[p, 2] == Subscript[p, 1]]],
    ", ", TraditionalForm[HoldForm[Subscript[q, 3] == Subscript[q, 2]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
    TraditionalForm[HoldForm[Subscript[h, 233]]], " ∈ ℝ"}], 13],

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Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] ==
    Sign[Subscript[x, 1] Subscript[x, 3]]],
    ", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
    Sign[Subscript[x, 1] Subscript[y, 2]]]]], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2]
    Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] Subsuperscript[
    p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /
    (Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2])]],
    ", ", TraditionalForm[HoldForm[Subsuperscript[S, 23, 2] ==
    Subscript[ξ, 23]]]], 13],
Style[TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
    {I Subscript[S, 23] Subscript[h, 231], Subscript[S, 23]
    Subscript[h, 232], Subscript[h, 233]}]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
    I Subscript[S, 23] Subscript[h, 231]]], ", ",
    TraditionalForm[HoldForm[sn[Subscript[e, 2] Subscript[t, 2] +
    Subscript[e, 3] Subscript[t, 3]]^2 ==
    -Subscript[ξ, 23] Subsuperscript[h, 231, 2]]], ", ",
    TraditionalForm[HoldForm[sn[Subscript[e, 2] Subscript[t, 2] +
    Subscript[e, 3] Subscript[t, 3]] dn[Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
    I Subscript[S, 23] Subscript[h, 231] Subscript[h, 233]]],
    ", ", TraditionalForm[HoldForm[cn[Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
    Subscript[S, 23] Subscript[h, 232]]]], 13],
Style[Row[{TraditionalForm[HoldForm[Subsuperscript[h, 233, 2] -
    Subscript[ξ, 23] k^2 Subsuperscript[h, 231, 2] == 1]],
    " (by ", TraditionalForm[HoldForm[dn[v]^2 + k^2 sn[v]^2 == 1]],
    " at ", TraditionalForm[HoldForm[v == Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]], " )"]], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1]
    Subsuperscript[p, 1, 2] == 1]], ", ",
    TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]], ", ",
    TraditionalForm[HoldForm[Subscript[p, 1] sn[ς] dn[ς] ==
    -Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς]^2 == 1 - cn[ς]^2 ==
    1 - Subscript[x, 1] t^2]], ", ",
    TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 sn[Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 ==
    Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
    Subscript[x, 1] Subsuperscript[h, 231, 2] t^2 ≠ 0]],
    ", by the choice of ", TraditionalForm[HoldForm[t]]],
    13], "", Style["Verification:", Bold, 13],
Style[Row[{"Since ", TraditionalForm[HoldForm[± (Subscript[e, 2]
    Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3])]], " and ",

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TraditionalForm[HoldForm[ϕ]], " are not poles and ",
TraditionalForm[HoldForm[1 - k^2 sn[ϕ]^2 sn[Subscript[e, 2]
Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 ≠ 0]],
", ", Style["the second addition formula", Italic, 13,
RGBColor[0.10, 0.30, 0.70]],
", at ", TraditionalForm[HoldForm[v = ϕ]], ", ",
TraditionalForm[HoldForm[w = -Subscript[e, 2] Subscript[t, 2] -
Subscript[e, 3] Subscript[t, 3]]],
", holds pointwise and, together with ", Style[
"the parity formulas", Italic, 13,
RGBColor[0.75, 0.35, 0.05]], ", gives"}], 13],
Style[TraditionalForm[HoldForm[Subscript[p, 3] cn[ϕ - Subscript[e, 2]
Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] =
(Subscript[p, 3] (cn[Subscript[e, 2] Subscript[t, 2] +
Subscript[e, 3] Subscript[t, 3]] cn[ϕ] +
sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3]
Subscript[t, 3]] dn[Subscript[e, 2] Subscript[t, 2] +
Subscript[e, 3] Subscript[t, 3]] sn[ϕ] dn[ϕ])) /
(Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[
Row[{TraditionalForm[HoldForm[Subscript[p, 3] cn[ϕ - Subscript[e, 2]
Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] =
Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
(Abs[Subscript[x, 1]] Subscript[h, 232] t +
Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]]) /
(Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]])
(Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)]}],
", Style["♦", 14, RGBColor[0.55, 0.25, 0.70]]}], 14],
Style[Row[{"which is the third formula of ",
Style["*", 13, s7colTest], " ."}], 13], "",
Style[Row[{"The identity ", Style["♦", 13, RGBColor[0.55, 0.25, 0.70]],
" is verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],
Style[Row[{"In case (a), both values of ", TraditionalForm[
HoldForm[Subscript[ξ, 23]]], " occur, because the sign of ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3]]],

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" is not constrained; in case (b), ",
TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
", because ", TraditionalForm[HoldForm[k^2 ==
  (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
", ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
  - (1 / (1 + Subscript[z, i]))]], ", and ", TraditionalForm[
  HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]], ", so ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
" . The eight sign regions cover both cases."}],
12, GrayLevel[0.35]], Grid[Table[{Style[w3Configs[[iw, 1]],
  12, GrayLevel[0.3]], smallChip[grid[[iw]]]}, {iw, 8}],
Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.55, 0.25, 0.70],
  0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 6:U-FORMULA IN CASE (c) FOR u<0*)
(* =====*)
ClearAll[cCheck, e0s, e2s];
cCheck[{lab_, sx_, s3_, sy_}] :=
Module[{x1 = sx XX, xi = sx s3, suz = sx sy, p1, p3, q2, S23, lhs, rhs, assum},
  p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
  p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
  q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
  S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
  lhs = S23 e2s I p3 (-e0s I kp RR / (XX p1)) / (kk (1 - x1 tt^2));
  rhs = suz ep2 ep3 e0s e2s kp RR / (kk Sqrt[XX X3] (1 - x1 tt^2));
  assum = XX > 0 && X3 > 0 && YY > 0 && UU > 0 &&
    Element[tt | RR | e0s | e2s | ep2 | ep3, Reals] && 1 - x1 tt^2 ≠ 0;
  TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] == True];
Module[{cConfigs, grid, ok}, cConfigs = Select[w3Configs, #[[2]] #[[3]] == 1 &];
grid = cCheck /@ cConfigs;
ok = And@@grid;
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
    15, RGBColor[0.00, 0.50, 0.50]], Style[" in case (c) for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u < 0]],
    Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
    Style["Given hypotheses and formulas, in addition to those above:",
    Bold, 13], Style[Row[{TraditionalForm[
      HoldForm[Subscript[ξ, 23] == Sign[Subscript[x, 1] Subscript[x, 3]]],
      ", ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
        Sign[Subscript[x, 1] Subscript[y, 2]]]]], 13],
    Style[TraditionalForm[HoldForm[Subscript[S, 23] == Subscript[ε, 2]
      Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2] Subsuperscript[
        p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /

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(Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2]]], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[t, 2] + Subscript[t, 3] ==
2 K m + I K']], ", ", TraditionalForm[HoldForm[m ∈ ℤ]], ", ",
TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]]], 13],
Style[Row[{TraditionalForm[HoldForm[cn[v + 2 m K] == (-1)^m cn[v]]],
", ", TraditionalForm[
HoldForm[cn[v ∈ I K'] == ±I dn[v] / (k sn[v])]], " ",
Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]]}],
13], Style[Row[{TraditionalForm[HoldForm[
Subscript[x, 1] Subsuperscript[p, 1, 2] == 1]], ", ",
TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[p, 1] sn[ς] dn[ς] ==
-Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]], 13],
Style[Row[{TraditionalForm[HoldForm[sn[ς]^2 == 1 - cn[ς]^2 ==
1 - Subscript[x, 1] t^2]], ", ",
TraditionalForm[HoldForm[1 - Subscript[x, 1] t^2 ≠ 0]],
", by the choice of ", TraditionalForm[HoldForm[t]]], 13], ""],
Style["Verification:", Bold, 13], Style[Row[{"Since ", TraditionalForm[
HoldForm[Subscript[e, 2] == Subscript[e, 3]]], " and ",
TraditionalForm[HoldForm[(-1)^(m Subscript[e, 2]) == (-1)^m]],
", ", Style["the shift formulas", Italic, 13,
RGBColor[0.00, 0.50, 0.50]], " give"}], 13],
Style[TraditionalForm[HoldForm[Subscript[p, 3]
cn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
(-1)^m Subscript[e, 2] I Subscript[p, 3] dn[ς] / (k sn[ς]) ==
Subscript[S, 23] Subscript[e, 2] I
Subscript[p, 3] sn[ς] dn[ς] / (k sn[ς]^2)]], 14],
Style["Using the given hypotheses and formulas, we get", 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3]
cn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
Subscript[e, 0] Subscript[e, 2] k' Sqrt[Subscript[x, 1]
D[Subscript[x, 1], t]] / (k Sqrt[Abs[Subscript[x, 1]
Subscript[x, 3]]] (1 - Subscript[x, 1] t^2))]],
", ", Style["▲", 14, RGBColor[0.00, 0.50, 0.50]]}], 14],
Style[Row[{"which is the fourth formula of ",
Style["★", 13, s7colTest], " ."}], 13], ""],
Style[Row[{"The identity ", Style["▲", 13, RGBColor[0.00, 0.50, 0.50]],
" is verified separately on each of the four sign
regions realizable in case (c), with ",
TraditionalForm[HoldForm[Subscript[e, 0]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[ε, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[ε, 3]]], ", ", TraditionalForm[HoldForm[R]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"}], 13],

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Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
  ", because ", TraditionalForm[HoldForm[k^2 ==
    (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
  ", ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
    - (1 / (1 + Subscript[z, i]))]], ", and ", TraditionalForm[
    HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]], ", so ",
    TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
    " ."}], 12, GrayLevel[0.35]],
Grid[Table[{Style[cConfigs[[iw, 1]], 12, GrayLevel[0.3]],
  smallChip[grid[[iw]]]}, {iw, 4}],
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.00, 0.50, 0.50], 0.93],
FrameMargins →
12]]];];
(* =====*)
(*PART 7:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And@@resultsAll, "CASE 2: ALL CLAIMS VERIFIED",
  "SOME CLAIMS FAILED"], Bold, White, 16], Background →
  If[And@@resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];

```

Given hypotheses, Case 2 ($u < 0$) :

$$u < 0, \quad k' = \frac{1}{\sqrt{1-u}}, \quad k = k' \sqrt{-u}, \quad k, k' \in (0, 1)$$

For $i = 1, 2, 3, 4$:

$$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0, \quad g_i = \text{dn}(t_i) = \frac{1}{\sqrt{1+\frac{1}{z_i}}}$$

$$p_i, q_i \in \mathbb{R}_{>0} \cup i \mathbb{R}_{>0}, \quad e_0, e', e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$$

$$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$$

$$\mathcal{D}(x_1, t) = -(1 - x_1 t^2) (1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$$

Given formulas:

$$\text{cn}(v + w) = \frac{\text{cn}(v) \text{cn}(w) - \text{sn}(v) \text{sn}(w) \text{dn}(v) \text{dn}(w)}{1 - k^2 \text{sn}(v)^2 \text{sn}(w)^2} \quad \text{the second addition formula}$$

$$\{\text{sn}, \text{cn}, \text{dn}\}(-v) = \{-\text{sn}(v), \text{cn}(v), \text{dn}(v)\} \quad \text{the parity formulas}$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test these formulas ★ :

$$W_2 = \frac{\varepsilon_2 (t \sqrt{u x_1 y_2 (1+z_2) (1+u z_2)} + e_0 e_2 \sqrt{u y_2 z_2 \mathcal{D}(x_1, t)})}{y_2 u (z_2 + x_1 t^2)}$$

$$W_1 = \frac{\varepsilon_1 (t \sqrt{u x_1 y_1 (1+z_1) (1+u z_1)} - e_0 e_1 \sqrt{u y_1 z_1 \mathcal{D}(x_1, t)})}{y_1 u (z_1 + x_1 t^2)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \text{in cases (a) and (b)}$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k \sqrt{|x_1 x_3|} (1 - x_1 t^2)} \quad \text{in case (c)}$$

The parameter normalization for $u < 0$ **Verified****Given hypotheses and formulas, in addition to those above:**

$$\operatorname{sn}(v)^2 + \operatorname{cn}(v)^2 = 1, \quad \operatorname{dn}(v)^2 = (k')^2 + k^2 \operatorname{cn}(v)^2 \quad \text{the Pythagorean relations}$$

$$x_1 p_1^2 = 1, \quad \operatorname{cn}(\varsigma) = \frac{t}{p_1}, \quad \operatorname{cn}(\varsigma)^2 = x_1 t^2$$

Verification:By *the Pythagorean relations* and $k^2 = -u (k')^2$, we have

$$(p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma))^2 = p_1^2 (1 - \operatorname{cn}(\varsigma)^2) ((k')^2 + k^2 \operatorname{cn}(\varsigma)^2) = -\frac{(k')^2 x_1 \mathcal{D}(x_1, t)}{x_1^2} = \left(\frac{i k' R}{|x_1|} \right)^2$$

By *the parity formulas*, replacing ς by $-\varsigma$ preserves $\operatorname{cn}(\varsigma)$ and reverses the sign of $\operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma)$.Thus, we may fix $p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma) = -\frac{e_0 i k' R}{|x_1|}$, so that the two values of e_0 correspond to the two admissible choices of ς .

The middle equality of the displayed chain, that is,

$$p_1^2 (1 - \operatorname{cn}(\varsigma)^2) ((k')^2 + k^2 \operatorname{cn}(\varsigma)^2) = -\frac{(k')^2 x_1 \mathcal{D}(x_1, t)}{x_1^2}, \quad \text{is verified on each sign region:}$$

$$x_1 > 0 \quad \checkmark$$

$$x_1 < 0 \quad \checkmark$$

The derivation: the formulas for W_2 and W_1 for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

For $l = 1, 2$:

$$x_1 p_1^2 = 1, \quad y_l q_l^2 = 1, \quad p_l = p_1$$

$$\operatorname{sn}(t_l) = \frac{i \varepsilon_l \sqrt{(g_l^2 - 1) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{cn}(t_l) = \frac{\varepsilon_l \sqrt{(g_l^2 - (k')^2) p_l^2 q_l^2}}{k p_l q_l}$$

$$\operatorname{cn}(\zeta) = \frac{t}{p_1}, \quad p_1 \operatorname{sn}(\zeta) \operatorname{dn}(\zeta) = -\frac{e_0 i k' R}{|x_1|}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(e' t_l) = \{e' \operatorname{sn}(t_l), \operatorname{cn}(t_l), \operatorname{dn}(t_l)\} \quad \text{by the parity formulas}$$

$$\operatorname{sn}(t_l)^2 = \frac{1 - g_l^2}{k^2} = -\frac{1 - u}{u(1 + z_l)}, \quad 1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 = \frac{z_l + x_1 t^2}{1 + z_l} \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm t_l$ and ζ are not poles and $1 - k^2 \operatorname{sn}(t_l)^2 \operatorname{sn}(\zeta)^2 \neq 0$, the second addition formula, at $v = \zeta$, $w = e' t_l$, holds pointwise and, together with the parity formulas, gives

$$q_l \operatorname{cn}(\zeta + e' t_l) = \frac{(1 + z_l) q_l \operatorname{cn}(t_l) \operatorname{cn}(\zeta) - e' (1 + z_l) q_l \operatorname{sn}(t_l) \operatorname{dn}(t_l) \operatorname{sn}(\zeta) \operatorname{dn}(\zeta)}{z_l + x_1 t^2}$$

Using the given formulas, we get

$$q_l \operatorname{cn}(\zeta + e' t_l) = \frac{\varepsilon_l (t \sqrt{u x_1 y_l (1 + z_l) (1 + u z_l)} - e' e_0 \sqrt{u y_l z_l} \mathcal{D}(x_1, t))}{y_l u (z_l + x_1 t^2)} \quad \star$$

Setting $l = 2$, $e' = -e_2$ in $\star$ gives the first formula of $\star$.

Setting $l = 1$, $e' = e_1$ in $\star$ gives the second formula of $\star$.

The identity $\star$ is verified separately on each sign region:

$z_l > 0, y_l > 0$, hence $x_1 < 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓
$z_l > 0, y_l < 0$, hence $x_1 > 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓
$z_l < -1, y_l > 0$, hence $x_1 > 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓
$z_l < -1, y_l < 0$, hence $x_1 < 0$	for all 8 sign combinations of $e', e_0, \varepsilon_l \in \{\pm 1\}$	✓

The derivation: the formula for U in cases (a) and (b) for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

$$x_3 p_3^2 = 1, \quad y_2 q_2^2 = 1, \quad p_2 = p_1, \quad q_3 = q_2, \quad h_{231}, h_{232}, h_{233} \in \mathbb{R}$$

$$\xi_{23} = \operatorname{sgn}(x_1 x_3), \quad \operatorname{sgn}(u z_2) = \operatorname{sgn}(x_1 y_2)$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}, \quad S_{23}^2 = \xi_{23}$$

$$\{\operatorname{sn}, \operatorname{cn}, \operatorname{dn}\}(e_2 t_2 + e_3 t_3) = \{i S_{23} h_{231}, S_{23} h_{232}, h_{233}\}$$

$$\operatorname{sn}(e_2 t_2 + e_3 t_3) = i S_{23} h_{231}, \quad \operatorname{sn}(e_2 t_2 + e_3 t_3)^2 = -\xi_{23} h_{231}^2,$$

$$\operatorname{sn}(e_2 t_2 + e_3 t_3) \operatorname{dn}(e_2 t_2 + e_3 t_3) = i S_{23} h_{231} h_{233}, \quad \operatorname{cn}(e_2 t_2 + e_3 t_3) = S_{23} h_{232}$$

$$h_{233}^2 - \xi_{23} k^2 h_{231}^2 = 1 \quad (\text{by } \operatorname{dn}(v)^2 + k^2 \operatorname{sn}(v)^2 = 1 \text{ at } v = e_2 t_2 + e_3 t_3)$$

$$x_1 p_1^2 = 1, \quad \operatorname{cn}(\varsigma) = \frac{t}{p_1}, \quad p_1 \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma) = -\frac{e_0 i k' R}{|x_1|}$$

$$\operatorname{sn}(\varsigma)^2 = 1 - \operatorname{cn}(\varsigma)^2 = 1 - x_1 t^2,$$

$$1 - k^2 \operatorname{sn}(\varsigma)^2 \operatorname{sn}(e_2 t_2 + e_3 t_3)^2 = h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2 \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $\pm(e_2 t_2 + e_3 t_3)$ and ς are not poles and $1 - k^2 \operatorname{sn}(\varsigma)^2 \operatorname{sn}(e_2 t_2 + e_3 t_3)^2 \neq 0$,
the [second addition formula](#), at $v = \varsigma$, $w = -e_2 t_2 - e_3 t_3$

, holds pointwise and, together with [the parity formulas](#), gives

$$p_3 \operatorname{cn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{p_3 (\operatorname{cn}(e_2 t_2 + e_3 t_3) \operatorname{cn}(\varsigma) + \operatorname{sn}(e_2 t_2 + e_3 t_3) \operatorname{dn}(e_2 t_2 + e_3 t_3) \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma))}{h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2}$$

Using the given hypotheses and formulas, we get

$$p_3 \operatorname{cn}(\varsigma - e_2 t_2 - e_3 t_3) = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2) (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{|x_1 x_3|} (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)} \quad \blacklozenge$$

which is the third formula of [★](#).

The identity [◆](#) is verified separately on each sign region, with $h_{231}, h_{232}, h_{233}, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

In case (a), both values of ξ_{23} occur, because the sign of $z_2 z_3$

is not constrained; in case (b), $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$,

$g_i^2 - 1 = -\frac{1}{1+z_i}$, and $z_i(1+z_i) > 0$, so $z_2 z_3 > 0$. The eight sign regions cover both cases.

$$x_1 > 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = -1, \quad \operatorname{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 > 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = -1, \quad \operatorname{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 > 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = -1, \quad \operatorname{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 > 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = -1, \quad \operatorname{sgn}(u z_2) = 1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 > 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = -1 \quad \checkmark$$

$$x_1 < 0, \quad x_3 < 0, \quad y_2 < 0, \quad \text{hence } \xi_{23} = 1, \quad \operatorname{sgn}(u z_2) = 1 \quad \checkmark$$

The derivation: the formula for U in case (c) for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

$$\xi_{23} = \text{sgn}(x_1 x_3), \quad \text{sgn}(u z_2) = \text{sgn}(x_1 y_2)$$

$$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}$$

$$t_2 + t_3 = 2 K m + i K', \quad m \in \mathbb{Z}, \quad (-1)^m = S_{23}, \quad e_2 = e_3$$

$$\text{cn}(v + 2 m K) = (-1)^m \text{cn}(v), \quad \text{cn}(v \mp i K') = \frac{\pm i \text{dn}(v)}{k \text{sn}(v)} \quad \text{the shift formulas}$$

$$x_1 p_1^2 = 1, \quad \text{cn}(\varsigma) = \frac{t}{p_1}, \quad p_1 \text{sn}(\varsigma) \text{dn}(\varsigma) = -\frac{e_0 i k' R}{|x_1|}$$

$$\text{sn}(\varsigma)^2 = 1 - \text{cn}(\varsigma)^2 = 1 - x_1 t^2, \quad 1 - x_1 t^2 \neq 0, \quad \text{by the choice of } t$$

Verification:

Since $e_2 = e_3$ and $(-1)^{m e_2} = (-1)^m$, the shift formulas give

$$p_3 \text{cn}(\varsigma - e_2 (t_2 + t_3)) = \frac{(-1)^m e_2 i p_3 \text{dn}(\varsigma)}{k \text{sn}(\varsigma)} = \frac{S_{23} e_2 i p_3 \text{sn}(\varsigma) \text{dn}(\varsigma)}{k \text{sn}(\varsigma)^2}$$

Using the given hypotheses and formulas, we get

$$p_3 \text{cn}(\varsigma - e_2 (t_2 + t_3)) = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k \sqrt{|x_1 x_3|} (1 - x_1 t^2)} \quad \blacktriangle$$

which is the fourth formula of $\star$.

The identity $\blacktriangle$

is verified separately on each of the four sign regions realizable in case (c), with

$e_0, e_2, \varepsilon_2, \varepsilon_3, R$

symbolic, that is, for all their real values at once, which covers all their sign choices:

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$, $g_i^2 - 1 = -\frac{1}{1+z_i}$, and $z_i(1+z_i) > 0$, so $z_2 z_3 > 0$.

$x_1 > 0, x_3 > 0, y_2 > 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = 1$ $\checkmark$

$x_1 > 0, x_3 > 0, y_2 < 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = -1$ $\checkmark$

$x_1 < 0, x_3 < 0, y_2 > 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = -1$ $\checkmark$

$x_1 < 0, x_3 < 0, y_2 < 0$, hence $\xi_{23} = 1, \text{sgn}(u z_2) = 1$ $\checkmark$

CASE 2: ALL CLAIMS VERIFIED

Section 10

Results of Section 10 are used in the proof of Theorem 2

In[2652]:=

```
(* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[UU, XX, X3, Z2, Z3, tt, RR, ee0, ep2, ep3, H1s, H2s, H3s, sX, statusChip,
```

```

smallChip, colGiven, colTest, lblZ2, lblZ3, lblX, zwin, wsign, mkA, mkAB,
mkAP, chkR, bOK, gdict, p6h, p6ok, p7h, p7ok, HAat, HBat, uLemPosat,
uLemNegat, uLemPosCat, uLemNegCat, uThmABat, uThmCat, resultsAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["✗", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
colGiven = RGBColor[0.85, 0.45, 0.00];
colTest = RGBColor[0.10, 0.35, 0.75];
(*Every check below is a rational identity in a fixed set of radical atoms,
decided by Together;
no simplification search is involved.Sign regions:each of z2,
z3 lies in one of the two windows z>0 or z<-1,by z(1+z)>0;
we write x1=sx XX with XX>0,and the sign of x3 is forced by Sign[x1 x3]=Sign[
  z2 z3],the sign of y2 by Sign[y2]=Sign[u x1 z2].On each region the atoms
  sw2=Sqrt[|z2|],sv2=Sqrt[|1+z2|],sb2=Sqrt[1+u z2],etc.are real principal
  roots,because their radicands are positive there by the window facts;
each de-nesting step used below is certified by squaring,
which is a rational identity,and both sides are nonnegative
  by construction.On each locus the parametrization is certified
  by checking the corresponding guard equality rationally.*)
lblZ2[1] = TraditionalForm[HoldForm[Subscript[z, 2] > 0]];
lblZ2[2] = TraditionalForm[HoldForm[Subscript[z, 2] < -1]];
lblZ3[1] = TraditionalForm[HoldForm[Subscript[z, 3] > 0]];
lblZ3[2] = TraditionalForm[HoldForm[Subscript[z, 3] < -1]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, 1] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, 1] < 0]];
zwin[1, Z_] := Z; zwin[2, Z_] := -1 - Z;
wsign[1] = 1; wsign[2] = -1;
sX = Sqrt[XX X3];
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
mkA[w2_, w3_, uv_] :=
  Module[{z2v = zwin[w2, Z2], z3v = zwin[w3, Z3], s2 = wsign[w2], s3 = wsign[w3]}, <|
    "z2" → z2v, "z3" → z3v, "s2" → s2, "s3" → s3, "xi" → s2 s3, "az2" → s2 z2v,
    "az3" → s3 z3v, "a12" → s2 (1 + z2v), "a13" → s3 (1 + z3v), "sw2" → Sqrt[s2 z2v],
    "sw3" → Sqrt[s3 z3v], "sv2" → Sqrt[s2 (1 + z2v)], "sv3" → Sqrt[s3 (1 + z3v)],
    "sb2" → Sqrt[1 + uv z2v], "sb3" → Sqrt[1 + uv z3v] |>];
mkAB[w2_, w3_] := Module[{at = mkA[w2, w3, 0]}, at["sb2"] = at["sv3"] / at["sw3"];
  at["sb3"] = at["sv2"] / at["sw2"]; at];
mkAP[w_] := Module[{at = mkA[w, w, 0], sS},
  sS = Sqrt[at["z2"] + at["z3"] + at["z2"] at["z3"]];
  at["sS"] = sS; at["sb2"] = at["sw3"] at["sv2"] / sS;
  at["sb3"] = at["sw2"] at["sv3"] / sS; at];

```

```

bOK[at_, uv_] :=
  chkR[at["sb2"]^2, 1 + uv at["z2"]] && chkR[at["sb3"]^2, 1 + uv at["z3"]];
gdict[at_, 1] := {at["sv2"] / at["sw2"], at["sv3"] / at["sw3"],
  (1 + at["z2"]) / at["z2"], (1 + at["z3"]) / at["z3"]};
gdict[at_, 2] := {at["sw2"] / at["sv2"], at["sw3"] / at["sv3"],
  at["z2"] / (1 + at["z2"]), at["z3"] / (1 + at["z3"])};
p6h[at_, kk2_, mS_, e2_, e3_, reg_] :=
  Module[{g2, g3, sr1, sr2, sr3, sr4, sr5, d, xi = at["xi"], W},
    {g2, g3} = gdict[at, reg][[1 ;; 2]];
    If[reg == 1, W = at["sw2"] at["sw3"];
      sr1 = at["sb3"] / W; sr2 = at["sb2"] / W;
      sr3 = at["sb2"] at["sb3"] / W; sr4 = 1 / W;
      sr5 = at["sb2"] at["sb3"] / (at["az2"] at["az3"]), W = at["sv2"] at["sv3"];
      sr1 = at["sb3"] / (mS W); sr2 = at["sb2"] / (mS W);
      sr3 = at["sb2"] at["sb3"] / (mS^2 W); sr4 = 1 / W;
      sr5 = at["sb2"] at["sb3"] / (mS^2 at["a12"] at["a13"])]];
    d = kk2 - (g2^2 - 1) (g3^2 - 1);
    {(e2 g3 sr1 + e3 g2 sr2) / d,
      (sr3 + e2 e3 g2 g3 sr4) / d, (kk2 g2 g3 + e2 e3 xi sr5) / d}];
p6ok[at_, kk2_, kp2_, mS_, reg_] :=
  Module[{g2, g3, d2, d3, sr1, sr2, sr3, sr4, sr5, xi = at["xi"], W},
    {g2, g3, d2, d3} = gdict[at, reg];
    If[reg == 1, W = at["sw2"] at["sw3"];
      sr1 = at["sb3"] / W; sr2 = at["sb2"] / W;
      sr3 = at["sb2"] at["sb3"] / W; sr4 = 1 / W;
      sr5 = at["sb2"] at["sb3"] / (at["az2"] at["az3"]), W = at["sv2"] at["sv3"];
      sr1 = at["sb3"] / (mS W); sr2 = at["sb2"] / (mS W);
      sr3 = at["sb2"] at["sb3"] / (mS^2 W); sr4 = 1 / W;
      sr5 = at["sb2"] at["sb3"] / (mS^2 at["a12"] at["a13"])]];
    chkR[g2^2, d2] && chkR[g3^2, d3] && chkR[(g2^2 - 1) (g3^2 - kp2) xi, sr1^2] &&
    chkR[(g3^2 - 1) (g2^2 - kp2) xi, sr2^2] && chkR[(g2^2 - kp2) (g3^2 - kp2) xi,
      sr3^2] && chkR[(g2^2 - 1) (g3^2 - 1) xi, sr4^2] &&
    chkR[(g2^2 - 1) (g2^2 - kp2) (g3^2 - 1) (g3^2 - kp2), sr5^2];
p7h[at_, kkAt_, kp2_, e2_, reg_] :=
  Module[{g2, g3}, {g2, g3} = gdict[at, reg][[1 ;; 2]];
    {e2 (g2^2 - g3^2) / (2 kkAt g2 g3),
      (g2^2 g3^2 - kp2) / (2 kkAt g2 g3), (g2^2 + g3^2) / (2 g2 g3)}];
p7ok[at_, kkAt_, kk2_, reg_] :=
  Module[{g2, g3, d2, d3}, {g2, g3, d2, d3} = gdict[at, reg];
    chkR[g2^2, d2] && chkR[g3^2, d3] && chkR[kkAt^2, kk2];
HAat[at_, uv_, e2_, e3_] :=
  Module[{d = uv at["z2"] at["z3"] - 1, sA2 = at["sw3"] at["sv3"] at["sb3"],
    sA3 = at["sw2"] at["sv2"] at["sb2"], sB = at["sb2"] at["sb3"],
    sC = at["sw2"] at["sv2"] at["sw3"] at["sv3"]},
    {(at["s2"] at["z2"] sA2 + e2 e3 at["s3"] at["z3"] sA3) / d,
      (at["xi"] at["z2"] at["z3"] sB + e2 e3 sC) / d,
      (uv at["xi"] sC + e2 e3 sB) / d}];

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HBat[at_] := Module[{q = at["z2"] at["z3"],
  s = at["z2"] + at["z3"], sC = at["sw2"] at["sv2"] at["sw3"] at["sv3"]},
  {q (at["z3"] - at["z2"]) / (2 sC), q (s + 2) / (2 sC),
  (s + 2 q) at["xi"] / (2 sC)}];
uLemPosat[{h1_, h2_, h3_}, kk2_, xi_, sx_, suz_] :=
  ep2 ep3 suz (XX h2 h3 tt + ee0 h1 RR) / (sX (1 + xi kk2 sx XX h1^2 tt^2));
uLemNegat[{h1_, h2_, h3_}, kk2_, kpAt_, xi_, sx_, suz_] := ep2 ep3 suz
  (XX h2 tt + ee0 kpAt h1 h3 RR) / (sX (h3^2 - xi kk2 sx XX h1^2 tt^2));
uLemPosCat[kkAt_, sx_, suz_] := ep2 ep3 suz sx / (sX kkAt tt);
uLemNegCat[kkAt_, kpAt_, sx_, suz_, e2_] :=
  ep2 ep3 suz ee0 e2 kpAt RR / (kkAt sX (1 - sx XX tt^2));
uThmABat[{H1_, H2_, H3_}, uv_, at_, sx_, e2_] :=
  ep2 ep3 at["s2"] (at["sw2"] at["sw3"] / sX)
  (XX H2 H3 tt + ee0 e2 H1 RR) / (at["z2"] at["z3"] + uv sx XX H1^2 tt^2);
uThmCat[at_, sy2_] := ep2 ep3 (at["sw2"] at["sw3"] / sX) sy2 / tt;
resultsAll = {};
(* =====*)
(*PART 2:THE GIVEN HYPOTHESES AND THE FORMULAS TO TEST*)
(* =====*)
ClearAll[braceGrp];
braceGrp[items_List, sz_ : 13, col_ : GrayLevel[0]] := DisplayForm[
  RowBox[{GridBox[({ToBoxes[Style[#, sz, col]]} & /@ items), GridBoxAlignment ->
    {"Columns" -> {{Left}}}, GridBoxSpacings -> {"Rows" -> {{1.8}}}],
  StyleBox[""], FontColor -> GrayLevel[0.55], SpanMaxSize -> Infinity]]];
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[u ≠ 0]], " ",
    TraditionalForm[HoldForm[u < 1]], " ", TraditionalForm[HoldForm[k]],
    " ", TraditionalForm[HoldForm[k']]], " ∈ (0, 1)"}], 14],
  Style[Row[{"For ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
  Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]],
    " ", TraditionalForm[HoldForm[Subscript[y, i]]], " ",
    TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
    " ", TraditionalForm[HoldForm[Subscript[x, i]]],
    " ", TraditionalForm[HoldForm[Subscript[y, i]]], " ",
    TraditionalForm[HoldForm[Subscript[z, i]]], " ≠ -1, 0"}], 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] = Sign[Subscript[
    z, 2] Subscript[z, 3]] = Sign[Subscript[x, 1] Subscript[x, 3]]],
    " ", TraditionalForm[HoldForm[Sign[Subscript[y, 2]] =
    Sign[u Subscript[x, 1] Subscript[z, 2]]]]], 14],
  Style[Row[{TraditionalForm[HoldForm[u Subscript[x, i]
    Subscript[y, i] Subscript[z, i] > 0]], " ",
    TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
    " ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[e, 0]]], " ",
    TraditionalForm[HoldForm[Subscript[e, i]]], " ",
    TraditionalForm[HoldForm[Subscript[ε, i]]], " ",
    TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"}], 14],

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Style[Row[{TraditionalForm[HoldForm[ $\mathcal{D}[\text{Subscript}[x, 1], t] =$ 
   $-(1 - \text{Subscript}[x, 1] t^2) (1 - u \text{Subscript}[x, 1] t^2)]],$ 
  ", ", TraditionalForm[HoldForm[ $R = \text{Sqrt}[\text{Subscript}[x, 1]$ 
 $\mathcal{D}[\text{Subscript}[x, 1], t]]],$  ", ", TraditionalForm[
  HoldForm[ $\text{Subscript}[x, 1] \mathcal{D}[\text{Subscript}[x, 1], t] \geq 0]]], 14], ""],
Style[Row[{"(i) if ", TraditionalForm[HoldForm[ $u > 0$ ]], " : ",
  TraditionalForm[HoldForm[ $k = \text{Sqrt}[u]$ ]], ", ", TraditionalForm[
  HoldForm[ $k' = \text{Sqrt}[1 - u]$ ]], ", ", TraditionalForm[HoldForm[
   $\text{Subscript}[g, i] = \text{Sqrt}[(1 + \text{Subscript}[z, i]) / \text{Subscript}[z, i]]],$ 
  ", ", TraditionalForm[HoldForm[ $k^2 - (\text{Subsuperscript}[g, 2, 2] - 1)$ 
 $(\text{Subsuperscript}[g, 3, 2] - 1) = (u \text{Subscript}[z, 2] \text{Subscript}[z, 3] -$ 
 $1) / (\text{Subscript}[z, 2] \text{Subscript}[z, 3])]]], 14], ""],
Style[Row[{"(ii) if ", TraditionalForm[HoldForm[ $u < 0$ ]], " : ",
  TraditionalForm[HoldForm[ $k' = 1 / \text{Sqrt}[1 - u]$ ]], ", ", TraditionalForm[
  HoldForm[ $k = k' \text{Sqrt}[-u]$ ]], ", ", TraditionalForm[HoldForm[
   $\text{Subscript}[g, i] = \text{Sqrt}[\text{Subscript}[z, i] / (1 + \text{Subscript}[z, i])]]],$ 
  ", ", TraditionalForm[HoldForm[ $k^2 - (\text{Subsuperscript}[g, 2, 2] - 1)$ 
 $(\text{Subsuperscript}[g, 3, 2] - 1) = -((1 + u (\text{Subscript}[z, 2] +$ 
 $\text{Subscript}[z, 3] + \text{Subscript}[z, 2] \text{Subscript}[z, 3])) /$ 
 $((1 - u) (1 + \text{Subscript}[z, 2]) (1 + \text{Subscript}[z, 3]))]]], 14],
""], Style[Row[{TraditionalForm[HoldForm[ $\text{Subscript}[h, 231]$ ]],
  ", ", TraditionalForm[HoldForm[ $\text{Subscript}[h, 232]$ ]],
  ", ", TraditionalForm[HoldForm[ $\text{Subscript}[h, 233]$ ]],
  ", ", TraditionalForm[HoldForm[ $\text{Subscript}[H, 1]$ ]],
  ", ", TraditionalForm[HoldForm[ $\text{Subscript}[H, 2]$ ]], ", ",
  TraditionalForm[HoldForm[ $\text{Subscript}[H, 3]$ ]], "  $\in \mathbb{R}$ "]], 14],
Style[Row[{TraditionalForm[HoldForm[ $\text{Subscript}[h, 231] = 0$ ]],
  " if and only if ", TraditionalForm[
  HoldForm[ $\text{Subscript}[g, 2] = \text{Subscript}[g, 3]$ ]], " and ",
  TraditionalForm[HoldForm[ $\text{Subscript}[e, 2] = -\text{Subscript}[e, 3]$ ]], " ; ",
  TraditionalForm[HoldForm[ $\text{Subscript}[h, 232] = 0$ ]], " if and only if ",
  TraditionalForm[HoldForm[ $\text{Subscript}[g, 2] \text{Subscript}[g, 3] = k'$ ]],
  " and ", TraditionalForm[
  HoldForm[ $\text{Subscript}[e, 2] = -\text{Subscript}[e, 3]$ ]], " ;"}], 14],
Style[Row[{TraditionalForm[HoldForm[ $\text{Subscript}[h, 233] = 0$ ]],
  " if and only if ", TraditionalForm[
  HoldForm[ $k^2 (\text{Subsuperscript}[g, 2, 2] + \text{Subsuperscript}[g, 3, 2]) =$ 
 $k^2 - (\text{Subsuperscript}[g, 2, 2] - 1) (\text{Subsuperscript}[g, 3, 2] - 1)]],$ 
  " and ", TraditionalForm[HoldForm[ $\text{Subscript}[e, 2] =$ 
 $\text{Subscript}[e, 3]$ ]], " ; in particular, ",
  TraditionalForm[HoldForm[ $\text{Subscript}[h, 233] \neq 0$ ]], " if ",
  TraditionalForm[HoldForm[ $k^2 = (\text{Subsuperscript}[g, 2, 2] - 1)$ 
 $(\text{Subsuperscript}[g, 3, 2] - 1)]], " ."}], 14],
""], Style[Row[{"the equalities ", TraditionalForm[
  HoldForm[ $u \text{Subscript}[z, 2] \text{Subscript}[z, 3] = 1$ ]], " and ",
  TraditionalForm[HoldForm[ $1 + u (\text{Subscript}[z, 2] + \text{Subscript}[$ 
 $z, 3] + \text{Subscript}[z, 2] \text{Subscript}[z, 3]) = 0$ ]], " ",$$$$ 
```

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Style["cannot hold simultaneously", Bold, 14], " ."]], 14],
"", Style["Given formulas and expressions:", Bold, 14],
Style[TraditionalForm[HoldForm[
  U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]] /
    Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] Piecewise[
      {(Abs[Subscript[x, 1]] Subscript[h, 232] Subscript[h, 233] t +
        Subscript[e, 0] Subscript[h, 231] Sqrt[Subscript[x, 1]
          D[Subscript[x, 1], t]) / (1 + Subscript[ξ, 23] k^2
          Subscript[x, 1] Subsuperscript[h, 231, 2] t^2), u > 0},
      {(Abs[Subscript[x, 1]] Subscript[h, 232] t + Subscript[e, 0]
        k' Subscript[h, 231] Subscript[h, 233] Sqrt[Subscript[x, 1]
          D[Subscript[x, 1], t]) / (Subsuperscript[h, 233, 2] -
          Subscript[ξ, 23] k^2 Subscript[x, 1] Subsuperscript[
            h, 231, 2] t^2), u < 0}]]], 15, Bold, colGiven], "",
Grid[{{braceGrp[TraditionalForm[HoldForm[Subscript[h, 231] ==
  (Subscript[e, 2] Subscript[g, 3] Sqrt[(Subsuperscript[g, 2, 2] -
    1) (Subsuperscript[g, 3, 2] - Superscript[k', 2])
    Subscript[ξ, 23]] + Subscript[e, 3] Subscript[g, 2]
    Sqrt[(Subsuperscript[g, 3, 2] - 1) (Subsuperscript[g, 2, 2] -
      Superscript[k', 2]) Subscript[ξ, 23]]) / (k^2 -
      (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]],
  TraditionalForm[HoldForm[Subscript[h, 232] ==
    (Sqrt[(Subsuperscript[g, 2, 2] - Superscript[k', 2])
      (Subsuperscript[g, 3, 2] - Superscript[k', 2]) Subscript[ξ,
        23]] + Subscript[e, 2] Subscript[e, 3] Subscript[g, 2]
      Subscript[g, 3] Sqrt[(Subsuperscript[g, 2, 2] - 1)
        (Subsuperscript[g, 3, 2] - 1) Subscript[ξ, 23]]) / (k^2 -
        (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]],
  TraditionalForm[HoldForm[Subscript[h, 233] == (k^2 Subscript[g, 2]
    Subscript[g, 3] + Subscript[e, 2] Subscript[e, 3] Subscript[ξ,
      23] Sqrt[(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 2,
        2] - Superscript[k', 2]) (Subsuperscript[g, 3, 2] - 1)
        (Subsuperscript[g, 3, 2] - Superscript[k', 2]))] / (k^2 -
        (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]]],
  13, colGiven], Row[{Style["(a) ", Bold, 13], Style[
    TraditionalForm[HoldForm[k^2 ≠ (Subsuperscript[g, 2, 2] - 1)
      (Subsuperscript[g, 3, 2] - 1)]], 13]]}],
Alignment → {Left, Center}, Spacings → {1.5, 0}], "", Grid[
  {{braceGrp[Row[TraditionalForm[HoldForm[Subscript[h, 231] == Subscript[
    e, 2] (Subsuperscript[g, 2, 2] - Subsuperscript[g, 3, 2]) /
    (2 k Subscript[g, 2] Subscript[g, 3])]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232] ==
      (Subsuperscript[g, 2, 2] Subsuperscript[g, 3, 2] - Superscript[
        k', 2]) / (2 k Subscript[g, 2] Subscript[g, 3])]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 233] ==
      (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) /
      (2 Subscript[g, 2] Subscript[g, 3])]]]]], 13, colGiven],

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Row[{Style["(b) ", Bold, 13], Style[TraditionalForm[HoldForm[
  k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
  13], Style[" ", 13], Style[TraditionalForm[
  HoldForm[Subscript[e, 2] == -Subscript[e, 3]], 13]]}],
Alignment -> {Left, Center}, Spacings -> {1.5, 0}], ""],
Grid[{{braceGrp[{Style[TraditionalForm[HoldForm[U == HoldForm[
  Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]] /
  Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] Piecewise[
  {{Sign[Subscript[x, 1]] / (k t), u > 0}, {Subscript[e, 0]
  Subscript[e, 2] k' Sqrt[Subscript[x, 1] D[Subscript[x, 1],
  t]] / (k (1 - Subscript[x, 1] t^2)), u < 0}}]], Bold]],
  14, colGiven], Row[{Style["(c) ", Bold, 13], Style[
  TraditionalForm[HoldForm[
    k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
    13], Style[" ", 13], Style[TraditionalForm[
    HoldForm[Subscript[e, 2] == Subscript[e, 3]], 13]]}],
Alignment -> {Left, Center}, Spacings -> {1.5, 0}], ""],
Style[Row[{"Each ", smallChip[True],
  " below certifies that the statement displayed beside it has been
  verified by a symbolic computation at that point; a ",
  smallChip[False], " would mean that this verification failed."}],
  12, GrayLevel[0.35]], "", Style[
  Row[{"We want to test these formulas ",
    Style["*"], 14, colTest], " :"}], Bold, 14],
Style[Row[{"We show that the formulas for ",
  TraditionalForm[HoldForm[U]],
  " below follow from the given hypotheses, formulas, and
  expressions above,"}], 12, GrayLevel[0.35]],
Style[Row[{"together with the following claim: in each case,
  the denominator of the given formula and the denominator
  of the tested formula vanish for the same values of ",
  TraditionalForm[HoldForm[t]], " , that is, one of them is
  zero if and only if the other is; hence the compared
  formulas are defined for exactly the same ",
  TraditionalForm[HoldForm[t]], " ."}], 12, GrayLevel[0.35]],
"", Style[TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
  Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
  Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  ((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
  Subscript[e, 0] Subscript[e, 2] Subscript[H,
  1] Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]]) /
  (Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
  Subsuperscript[H, 1, 2] t^2))]], 15, Bold, colTest],
"", Grid[{{braceGrp[{TraditionalForm[HoldForm[Subscript[H, 1] ==
  (Abs[Subscript[z, 2]] Sqrt[Subscript[z, 3] (1 + Subscript[z, 3])
  (1 + u Subscript[z, 3])] + Subscript[e, 2] Subscript[e, 3]
  Abs[Subscript[z, 3]] Sqrt[Subscript[z, 2] (1 + Subscript[

```

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z, 2]) (1 + u Subscript[z, 2])) / (u Subscript[z, 2]
Subscript[z, 3] - 1)], TraditionalForm[HoldForm[
Subscript[H, 2] == (Abs[Subscript[z, 2] Subscript[z, 3]] Sqrt[
(1 + u Subscript[z, 2]) (1 + u Subscript[z, 3])] + Subscript[e,
2] Subscript[e, 3] Sqrt[Subscript[z, 2] Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]) /
(u Subscript[z, 2] Subscript[z, 3] - 1)], TraditionalForm[
HoldForm[Subscript[H, 3] == (u Sqrt[Subscript[z, 2] Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3])] Sign[
Subscript[z, 2] Subscript[z, 3] + Subscript[e, 2] Subscript[
e, 3] Sqrt[(1 + u Subscript[z, 2]) (1 + u Subscript[z, 3])]) /
(u Subscript[z, 2] Subscript[z, 3] - 1)]], 13, colTest],
Row[{Style["(a) ", Bold, 13], Style[TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] ≠ 1]], 13]}]},
Alignment → {Left, Center}, Spacings → {1.5, 0}], "",
Grid[{{braceGrp[TraditionalForm[HoldForm[
Subscript[H, 1] == Subscript[z, 2] Subscript[z, 3] (Subscript[z, 3] -
Subscript[z, 2]) / (2 Sqrt[Subscript[z, 2] Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3])])],
TraditionalForm[HoldForm[Subscript[H, 2] == Subscript[z, 2]
Subscript[z, 3] (Subscript[z, 2] + Subscript[z, 3] + 2) /
(2 Sqrt[Subscript[z, 2] Subscript[z, 3] (1 + Subscript[z, 2])
(1 + Subscript[z, 3])])], TraditionalForm[
HoldForm[Subscript[H, 3] == (Subscript[z, 2] + Subscript[z, 3] +
2 Subscript[z, 2] Subscript[z, 3]) Sign[Subscript[z, 2]
Subscript[z, 3] / (2 Sqrt[Subscript[z, 2] Subscript[z, 3]
(1 + Subscript[z, 2]) (1 + Subscript[z, 3])])]],
13, colTest], Row[{Style["(b) ", Bold, 13], Style[
TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]],
13], Style[" ", 13], Style[TraditionalForm[
HoldForm[Subscript[e, 2] == -Subscript[e, 3]], 13]}]},
Alignment → {Left, Center}, Spacings → {1.5, 0}], "",
Grid[{{braceGrp[Style[TraditionalForm[HoldForm[
U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
HoldForm[Sign[Subscript[y, 2] / t]], Bold]], 14, colTest],
Row[{Style["(c) ", Bold, 13], Style[TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], 13],
Style[" ", 13], Style[TraditionalForm[
HoldForm[Subscript[e, 2] == Subscript[e, 3]], 13]}]},
Alignment → {Left, Center}, Spacings → {1.5, 0}],
Spacings → 1.1], Background →
RGBColor[
0.99,
0.96,
0.88],
FrameMargins → 14]];

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(* =====*)
(*PART 3:CASE (a) FOR u>0*)
(* =====*)
ClearAll[regA, aCheckPos, aPreCk];
regA = Flatten[
  Table[{Row[{lblZ2[w2], " ", lblZ3[w3], " ", lblX[sx]}], w2, w3, sx},
    {w2, {1, 2}}, {w3, {1, 2}}, {sx, {1, -1}}, 2];
aCheckPos[{lab_, w2_, w3_, sx_}, {e2_, e3_}] := Module[{at = mkA[w2, w3, UU],
  xi, suz, W, h, H, brOK, numCk, denCk, fullCk}, xi = at["xi"];
  suz = at["s2"];
  W = at["sw2"] at["sw3"];
  h = p6h[at, UU, 1, e2, e3, 1]; H = HAat[at, UU, e2, e3];
  brOK = bOK[at, UU] && p6ok[at, UU, 1 - UU, 1, 1] &&
    chkR[h[[1]] W, xi e2 H[[1]] && chkR[h[[2]] W, xi H[[2]] && chkR[h[[3]] H[[3]]];
  numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
    xi W (XX (xi H2s / W) H3s tt + ee0 (xi e2 H1s / W) RR)];
  denCk = chkR[at["z2"] at["z3"] + UU sx XX H1s^2 tt^2,
    at["z2"] at["z3"] (1 + xi UU sx XX (xi e2 H1s / W)^2 tt^2)];
  fullCk = chkR[uLemPosat[{xi e2 H1s / W, xi H2s / W, H3s}, UU, xi, sx, suz],
    uThmABat[{H1s, H2s, H3s}, UU, at, sx, e2]];
  {brOK, numCk && denCk, fullCk}];
aPreCk[{lab_, w2_, w3_, sx_}] :=
  Module[{at = mkA[w2, w3, UU], xi, W}, xi = at["xi"];
  W = at["sw2"] at["sw3"];
  chkR[at["z2"] at["z3"] / (xi W sX), W / sX]];
Module[{grid, brRegion, idRegion, fullRegion, preAll, ok},
  grid = Table[aCheckPos[w, s], {w, regA}, {s, Tuples[{1, -1}, 2]};
  brRegion = Table[And@@grid[[iw, All, 1]], {iw, Length[regA]}];
  idRegion = Table[And@@grid[[iw, All, 2]], {iw, Length[regA]}];
  fullRegion = Table[And@@grid[[iw, All, 3]], {iw, Length[regA]}];
  preAll = And@@(aPreCk /@ regA);
  ok = (And@@brRegion) && (And@@idRegion) && (And@@fullRegion) && preAll;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
      RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[U]], Bold,
      15, RGBColor[0.10, 0.30, 0.70]], Style[" in case (a) for ", Bold, 15,
      RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u > 0]],
      Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]}],
      Style[Row[{"By ", TraditionalForm[HoldForm[
        Subscript[ξ, 23] = Sign[Subscript[z, 2] Subscript[z, 3]] =
          Sign[Subscript[x, 1] Subscript[x, 3]]], " : ",
        TraditionalForm[HoldForm[Subscript[ξ, 23] Subscript[z, 2]
          Subscript[z, 3] = Abs[Subscript[z, 2] Subscript[z, 3]]], " ",
        TraditionalForm[HoldForm[Subscript[ξ, 23] Subscript[x, 1]
          Subscript[x, 3] = Abs[Subscript[x, 1] Subscript[x, 3]]], " ."}],
        12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],

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Style[Row[{"in this case ", TraditionalForm[HoldForm[
  k^2 ≠ (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]],
  " , that is, ", TraditionalForm[HoldForm[
  u Subscript[z, 2] Subscript[z, 3] ≠ 1]],
  " , and the following identities hold:"], 13], Style[
Row[{TraditionalForm[HoldForm[k = Sqrt[u]]], " , ", TraditionalForm[
  HoldForm[Subsuperscript[g, i, 2] - 1 = 1 / Subscript[z, i]]], " , ",
  TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - Superscript[k', 2] =
    (1 + u Subscript[z, i]) / Subscript[z, i]]],
  " , ", TraditionalForm[HoldForm[Subscript[g, i] =
    Sqrt[Subscript[z, i] (1 + Subscript[z, i]) / Abs[Subscript[z, i]]]],
  " ", Style["♦", 14, RGBColor[0.55, 0.33, 0.10]]], 13],
Style[Row[{"Hence, substituting the identities ", Style["♦", 13,
  RGBColor[0.55, 0.33, 0.10]], " and the equality ", TraditionalForm[
  HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] =
    Abs[Subscript[z, 2] Subscript[z, 3]]], " into ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
  " , ", TraditionalForm[HoldForm[Subscript[h, 232]]], " , ",
  TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
  " , we get, with ", Style[Row[{"the case-(a) ",
    TraditionalForm[HoldForm[Subscript[H, 1]]], " , ",
    TraditionalForm[HoldForm[Subscript[H, 2]]], " , ", TraditionalForm[
      HoldForm[Subscript[H, 3]]}], 13, colTest], " ,"}, 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, 231] =
  Subscript[ξ, 23] Subscript[e, 2] Subscript[H, 1] / Sqrt[Abs[
    Subscript[z, 2] Subscript[z, 3]]]], " , ", TraditionalForm[
  HoldForm[Subscript[h, 232] = Subscript[ξ, 23] Subscript[H, 2] /
    Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]], " , ",
  TraditionalForm[HoldForm[Subscript[h, 233] = Subscript[H, 3]]],
  " ", Style["♦", 14, RGBColor[0.00, 0.45, 0.25]]], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], " , ",
    TraditionalForm[HoldForm[Subscript[e, 3]]]}], 12, GrayLevel[0.3]],
  smallChip[brRegion[[iw]]], {iw, Length[regA]}],
  Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"Thus, substituting the identities ",
  Style["♦", 13, RGBColor[0.00, 0.45, 0.25]],
  " into the numerator and the denominator of ",
  Style["the given U-formula", Italic, 13, colGiven], " , we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
  HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] =
    Subscript[ξ, 23] Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]
    (Abs[Subscript[x, 1] Subscript[h, 232] Subscript[h, 233] t +

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Subscript[e, 0] Subscript[h, 231]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
HoldForm[Subscript[z, 2] Subscript[z, 3] +
u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 =
Subscript[z, 2] Subscript[z, 3] (1 + Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for all 4 sign combinations of ",
TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[e, 3]]]], 12, GrayLevel[0.3]],
smallChip[idRegion[[iw]]], {iw, Length[regA]}],
Alignment → Left, Spacings → {2, 0.7}], ""],
Style["and, finally, the prefactor identity:", 13],
Style[TraditionalForm[
HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
HoldForm[Subscript[z, 2] Subscript[z, 3] / (Subscript[ξ, 23]
Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]])] = Sqrt[Subscript[
z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[
Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] = Abs[
Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] =
Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"]], 13],
Style[Row[{"and is verified separately on each of the 8
sign regions (sign combinations of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
" as above) ", smallChip[preAll]]], 13], ""],
Style[Row[{"Together, the numerator, denominator, and prefactor
identities, along with ", TraditionalForm[
HoldForm[Sign[u Subscript[z, 2]] = Sign[Subscript[z, 2]]],
", , give the formula ", Style["*", 13, colTest],
" in case (a):"]], 13],
Style[Row[{TraditionalForm[HoldForm[U = HoldForm[Subscript[ε, 2]
Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
Subscript[x, 1] D[Subscript[x, 1], t]] / (Subscript[z, 2]
Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
t^2))]], " ", Style["*", 14]]], 14, colTest],

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    "", Style[Row[{"The identity ", Style["*", 13, colTest],
      " is also verified separately on each sign region, with ",
      TraditionalForm[HoldForm[Subscript[H, 1]]],
      " ", TraditionalForm[HoldForm[Subscript[H, 2]]],
      " ", TraditionalForm[HoldForm[Subscript[H, 3]]],
      " ", TraditionalForm[HoldForm[Subscript[e, 0]]],
      " ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
      " ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
      " symbolic, that is, for all their real values at once,
      which covers all their sign choices:"], 13],
    Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]], Style[
      Row[{"for all 4 sign combinations of ",
        TraditionalForm[HoldForm[Subscript[e, 2]]], " ",
        TraditionalForm[HoldForm[Subscript[e, 3]]}], 12, GrayLevel[0.3]],
      smallChip[fullRegion[[iw]]], {iw, Length[regA]}],
      Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
    Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
    FrameMargins →
      12]]];];
(* =====*)
(*PART 4:CASES (b) AND (c) FOR u>0,ON u z2 z3=1*)
(* =====*)
ClearAll[regBpos, bCheckPos, bPreCk, cCheckPos];
regBpos =
  Flatten[Table[{Row[{lblZ2[w], " ", lblZ3[w], " ", lblX[sx]}], w, sx},
    {w, {1, 2}}, {sx, {1, -1}}], 1];
bCheckPos[{lab_, w_, sx_}, e2_] :=
  Module[{at = mkAB[w, w], uv, kkAt, W, d2, d3, h, H, brOK, numCk, denCk, fullCk},
    uv = 1 / (at["z2"] at["z3"]);
    W = at["sw2"] at["sw3"]; kkAt = 1 / W;
    {d2, d3} = gdict[at, 1][[3 ;; 4]];
    h = p7h[at, kkAt, 1 - uv, e2, 1]; H = Hbat[at];
    brOK = bOK[at, uv] && p7ok[at, kkAt, uv, 1] && chkR[kkAt^2, (d2 - 1) (d3 - 1)] &&
      chkR[h[[1]] W, e2 H[[1]]] && chkR[h[[2]] W, H[[2]]] && chkR[h[[3]] W, H[[3]]];
    numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
      W (XX (H2s / W) H3s tt + ee0 (e2 H1s / W) RR)];
    denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
      at["z2"] at["z3"] (1 + sx XX (e2 H1s / W)^2 kkAt^2 tt^2)];
    fullCk = chkR[uLemPosat[{e2 H1s / W, H2s / W, H3s}, uv, 1, sx, at["s2"]],
      uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
    {brOK, numCk && denCk, fullCk}];
bPreCk[{lab_, w_, sx_}] :=
  Module[{at = mkAB[w, w], W}, W = at["sw2"] at["sw3"];
    chkR[at["z2"] at["z3"] / (W sx), W / sx]];
cCheckPos[{lab_, w_, sx_}] :=
  Module[{at = mkAB[w, w], uv, kkAt, d2, d3}, uv = 1 / (at["z2"] at["z3"]);
    kkAt = 1 / (at["sw2"] at["sw3"]);

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{d2, d3} = gdict[at, 1][[3 ;; 4]];
p7ok[at, kkAt, uv, 1] && chkR[kkAt^2, (d2 - 1) (d3 - 1)] &&
  chkR[uLemPosCat[kkAt, sx, at["s2"]], uThmCat[at, at["s2"] sx]]];
Module[{gridB, brRegion, idRegion, fullRegion, preAll, cRegion, ok},
  gridB = Table[bCheckPos[w, s], {w, regBpos}, {s, {1, -1}}];
  brRegion = Table[And@@gridB[[iw, All, 1]], {iw, Length[regBpos]}];
  idRegion = Table[And@@gridB[[iw, All, 2]], {iw, Length[regBpos]}];
  fullRegion = Table[And@@gridB[[iw, All, 3]], {iw, Length[regBpos]}];
  preAll = And@@(bPreCk /@ regBpos);
  cRegion = cCheckPos /@ regBpos;
  ok = (And@@brRegion) && (And@@idRegion) &&
    (And@@fullRegion) && preAll && (And@@cRegion);
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
      RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
      Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
        " in cases (b) and (c) for ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
      Style[TraditionalForm[HoldForm[u > 0]], Bold, 15,
      RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]}],
    Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
      ", because ", TraditionalForm[HoldForm[k^2 ==
        (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
      " and ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
        1 / Subscript[z, i]]], ", so ", TraditionalForm[HoldForm[
        Subscript[z, 2] Subscript[z, 3] > 0]], ", and the signs of ",
      TraditionalForm[HoldForm[Subscript[z, 2]]], " and ",
      TraditionalForm[HoldForm[Subscript[z, 3]]], " coincide."}],
    12, GrayLevel[0.35]], Style[Row[{"By ", TraditionalForm[
      HoldForm[Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
        Sign[Subscript[x, 1] Subscript[x, 3]] == 1]], " : ",
      TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
        Abs[Subscript[z, 2] Subscript[z, 3]]], ", ",
      TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==
        Abs[Subscript[x, 1] Subscript[x, 3]]], " ."}],
    12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
    Style[Row[{"in these cases ", TraditionalForm[HoldForm[
      k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
      ", that is, ", TraditionalForm[HoldForm[
      u Subscript[z, 2] Subscript[z, 3] == 1]],
      ", and the following identities hold:"}], 13],
    Style[Row[TraditionalForm[HoldForm[
      k == 1 / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
      ", ", TraditionalForm[HoldForm[Subscript[g, i] ==
        Sqrt[Subscript[z, i] (1 + Subscript[z, i])] / Abs[Subscript[z, i]]],
      " ", Style["▲", 14, RGBColor[0.55, 0.33, 0.10]]], 13],
    Style[Row[{"Hence, in case (b) ( ", TraditionalForm[

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HoldForm[Subscript[e, 2] == -Subscript[e, 3]],
" ), substituting the identities ",
Style["▲", 13, RGBColor[0.55, 0.33, 0.10]], " and the equality ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
Abs[Subscript[z, 2] Subscript[z, 3]]], " into ", Style[
Row[{"the case-(b) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
" , we get, with ", Style[Row[{"the case-(b) ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[H, 3]]]}], 13, colTest], " ,"], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, 231] == Subscript[e, 2]
Subscript[H, 1] / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] ==
Subscript[H, 2] / Sqrt[Subscript[z, 2] Subscript[z, 3]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233] == Subscript[H, 3]]],
" ", Style["▲", 14, RGBColor[0.00, 0.45, 0.25]]}], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]], Style[
Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
12, GrayLevel[0.3]], smallChip[brRegion[[iw]]]},
{iw, Length[regBpos]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style[Row[{"Thus, substituting the identities ",
Style["▲", 13, RGBColor[0.00, 0.45, 0.25]],
" into the numerator and the denominator of ",
Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] == Sqrt[Subscript[z, 2]
Subscript[z, 3] (Abs[Subscript[x, 1] Subscript[h, 232]
Subscript[h, 233] t + Subscript[e, 0] Subscript[h, 231]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
HoldForm[Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2] Subscript[z, 3]
(1 + k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]], Style[
Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
12, GrayLevel[0.3]], smallChip[idRegion[[iw]]]},
{iw, Length[regBpos]}], Alignment → Left, Spacings → {2, 0.7}],
"",
Style[Row[{"and, finally, ", Style["the prefactor identity", Italic,

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13, RGBColor[0.80, 0.20, 0.50]], " :")], 13], Style[TraditionalForm[
HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
HoldForm[Subscript[z, 2] Subscript[z, 3] /
Sqrt[Subscript[z, 2] Subscript[z, 3]] == Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[Subscript[z, 2]
Subscript[z, 3] == Abs[Subscript[z, 2] Subscript[z, 3]]]],
" and ", TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==
Abs[Subscript[x, 1] Subscript[x, 3]]]], " ,"]], 13],
Style[Row[{"and is verified separately on each of the 4
sign regions (sign combinations of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], " ",
TraditionalForm[HoldForm[Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
" as above) ", smallChip[preAll]]], 13], ""],
Style[Row[{"Together, the numerator, denominator, and prefactor
identities, along with ", TraditionalForm[
HoldForm[Sign[u Subscript[z, 2]] == Sign[Subscript[z, 2]]]],
", , give the formula ", Style["*", 13, colTest],
" in case (b):"]], 13],
Style[Row[{"TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
Subscript[x, 1] D[Subscript[x, 1], t]) / (Subscript[z, 2]
Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
t^2))]], " ", Style["*", 14]]], 14, colTest],
"", Style[Row[{"The identity ", Style["*", 13, colTest],
" is also verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
" symbolic, that is, for all their real values at once,
which covers all their sign choices:"]], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]], Style[
Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
12, GrayLevel[0.3]], smallChip[fullRegion[[iw]]]],
{iw, Length[regBpos]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style[Row[{"In case (c) ( ", TraditionalForm[
HoldForm[Subscript[e, 2] == Subscript[e, 3]]],
" ), the ", Style["given", Italic, 13, colGiven],
" and ", Style["tested", Italic, 13, colTest],
" formulas are their replacements:"]], 13],

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Style[Row[{Style[TraditionalForm[HoldForm[
    U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[
        z, 2]] / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
    HoldForm[Sign[Subscript[x, 1]] / (k t)]], 14, colGiven],
    " and ", Style[TraditionalForm[HoldForm[
    U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sqrt[Subscript[z, 2]
        Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]
    HoldForm[Sign[Subscript[y, 2]] / t]], 14, colTest], " ,"], 14],
Style[Row[{"which coincide because ", TraditionalForm[HoldForm[
    Sign[Subscript[y, 2]] == Sign[u Subscript[x, 1] Subscript[z, 2]]],
    " , ", TraditionalForm[HoldForm[
    k == 1 / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
    " , and ", Style["the prefactor identity", Italic, 13,
    RGBColor[0.80, 0.20, 0.50]], " ; their denominators ",
    TraditionalForm[HoldForm[k t]], " and ",
    TraditionalForm[HoldForm[t]], " vanish for the same t because ",
    TraditionalForm[HoldForm[k ≠ 0]], " ."}], 13],
Style[Row[{"The equality of the ", Style["given", Italic, 13, colGiven],
    " and ", Style["tested", Italic, 13, colTest],
    " formulas is verified separately on each sign region, with ",
    TraditionalForm[HoldForm[Subscript[ε, 2]]],
    " , ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
    " symbolic, that is, for all their real values at once,
    which covers all their sign choices:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]],
    smallChip[cRegion[[iw]]]}, {iw, Length[regBpos]}],
    Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.55, 0.25, 0.70], 0.93],
FrameMargins →
12]]];];

(* =====*)
(*PART 5:CASE (a) FOR u<0*)
(* =====*)
ClearAll[aCheckNeg, aPreCkNeg, h233ConvCk,
    regZ, regMix, bCheckNegII, bPreCkII, cCheckNegII];
regZ = Flatten[Table[
    {Row[{lblZ2[w2], " , ", lblZ3[w3]}], w2, w3}, {w2, {1, 2}}, {w3, {1, 2}}], 1];
regMix = Flatten[Table[{Row[{lblZ2[w2], " , ", lblZ3[3 - w2], " , ", lblX[sx]}],
    w2, 3 - w2, sx}, {w2, {1, 2}}, {sx, {1, -1}}], 1];
h233ConvCk[{lab_, w2_, w3_}] :=
Module[{at = mkA[w2, w3, -UU], kk2 = UU / (1 + UU), d2, d3},
    {d2, d3} = gdict[at, 2][[3 ;; 4]];
    chkR[kk2 (d2 + d3) - (kk2 - (d2 - 1) (d3 - 1)),
    (1 - (-UU) at["z2"] at["z3"]) / ((1 + UU) (1 + at["z2"]) (1 + at["z3"]))];
aCheckNeg[{lab_, w2_, w3_, sx_}, {e2_, e3_}] :=
Module[{at = mkA[w2, w3, -UU], uv = -UU, kk2, mS, kpAt, xi, suz,
    W, h, H, brh, brOK, numCk, denCk, fullCk, kk2 = UU / (1 + UU);

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mS = Sqrt[1 + UU]; kpAt = 1 / mS; xi = at["xi"]; suz = -at["s2"];
W = at["sw2"] at["sw3"];
h = p6h[at, kk2, mS, e2, e3, 2]; H = HAat[at, uv, e2, e3];
brh = {-xi e2 mS H1s / (W H3s), -xi H2s / (W H3s), 1 / H3s};
brOK = bOK[at, uv] && p6ok[at, kk2, 1 / (1 + UU), mS, 2] && chkR[h[[3]] H[[3]], 1] &&
  chkR[h[[1]] W H[[3]], -xi e2 mS H[[1]]] && chkR[h[[2]] W H[[3]], -xi H[[2]]];
numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
  -xi W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)];
denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
  at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - xi kk2 sx XX brh[[1]]^2 tt^2)];
fullCk = chkR[uLemNegat[brh, kk2, kpAt, xi, sx, suz],
  uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
{brOK, numCk && denCk, fullCk}];
aPreCkNeg[{lab_, w2_, w3_, sx_}] :=
Module[{at = mkA[w2, w3, -UU], xi, W}, xi = at["xi"];
W = at["sw2"] at["sw3"];
chkR[at["z2"] at["z3"] / (xi W sX), W / sX]];
bCheckNegII[{lab_, w2_, w3_, sx_}, e2_] :=
Module[{at = mkAB[w2, w3], uv, kk2, mS, kpAt, xi, suz, W, h, H,
  brh, brOK, numCk, denCk, fullCk}, uv = 1 / (at["z2"] at["z3"]);
kk2 = -uv / (1 - uv); mS = Sqrt[1 - uv]; kpAt = 1 / mS; xi = at["xi"];
suz = -at["s2"]; W = at["sw2"] at["sw3"];
h = p6h[at, kk2, mS, e2, -e2, 2]; H = HBat[at];
brh = {-xi e2 mS H1s / (W H3s), -xi H2s / (W H3s), 1 / H3s};
brOK = bOK[at, uv] && p6ok[at, kk2, 1 / (1 - uv), mS, 2] && chkR[h[[3]] H[[3]], 1] &&
  chkR[h[[1]] W H[[3]], -xi e2 mS H[[1]]] && chkR[h[[2]] W H[[3]], -xi H[[2]]];
numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
  -xi W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)];
denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
  at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - xi kk2 sx XX brh[[1]]^2 tt^2)];
fullCk = chkR[uLemNegat[brh, kk2, kpAt, xi, sx, suz],
  uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
{brOK, numCk && denCk, fullCk}];
bPreCkII[{lab_, w2_, w3_, sx_}] :=
Module[{at = mkAB[w2, w3], xi, W}, xi = at["xi"];
W = at["sw2"] at["sw3"];
chkR[at["z2"] at["z3"] / (xi W sX), W / sX]];
cCheckNegII[{lab_, w2_, w3_, sx_}, e2_] :=
Module[{at = mkAB[w2, w3], uv, kk2, mS, kpAt, xi, suz, W, h},
uv = 1 / (at["z2"] at["z3"]);
kk2 = -uv / (1 - uv);
mS = Sqrt[1 - uv];
kpAt = 1 / mS; xi = at["xi"]; suz = -at["s2"];
W = at["sw2"] at["sw3"];
h = p6h[at, kk2, mS, e2, e2, 2];
bOK[at, uv] && p6ok[at, kk2, 1 / (1 - uv), mS, 2] && chkR[h[[3]], 0] &&
  chkR[h[[1]], e2 mS W] && chkR[h[[2]], W] && chkR[kk2 mS^2 W^2, 1] &&

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chkR[h[[3]]^2 - xi kk2 sx XX h[[1]]^2 tt^2, sx XX tt^2] &&
chkR[uLemNegat[h, kk2, kpAt, xi, sx, suz], uThmCat[at, suz sx]]];
Module[{grid, brRegion, idRegion, fullRegion, preAll, convAll, gridB,
  brRegionB, idRegionB, fullRegionB, preAllB, gridC, cRegion, ok},
  grid = Table[aCheckNeg[w, s], {w, regA}, {s, Tuples[{1, -1}, 2]};
  brRegion = Table[And@@grid[[iw, All, 1]], {iw, Length[regA]};
  idRegion = Table[And@@grid[[iw, All, 2]], {iw, Length[regA]};
  fullRegion = Table[And@@grid[[iw, All, 3]], {iw, Length[regA]};
  preAll = And@@ (aPreCkNeg /@ regA);
  convAll = And@@ (h233ConvCk /@ regZ);
  gridB = Table[bCheckNegII[w, s], {w, regMix}, {s, {1, -1}}];
  brRegionB = Table[And@@gridB[[iw, All, 1]], {iw, Length[regMix]};
  idRegionB = Table[And@@gridB[[iw, All, 2]], {iw, Length[regMix]};
  fullRegionB = Table[And@@gridB[[iw, All, 3]], {iw, Length[regMix]};
  preAllB = And@@ (bPreCkII /@ regMix);
  gridC = Table[cCheckNegII[w, s], {w, regMix}, {s, {1, -1}}];
  cRegion = Table[And@@gridC[[iw]], {iw, Length[regMix]};
  ok = (And@@brRegion) && (And@@idRegion) &&
    (And@@fullRegion) && preAll && convAll && (And@@brRegionB) &&
    (And@@idRegionB) && (And@@fullRegionB) && preAllB && (And@@cRegion);
AppendTo[resultsAll, ok];
Print[
  Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
    15, RGBColor[0.00, 0.50, 0.50]], Style[" in case (a) for ", Bold, 15,
    RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u < 0]],
    Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
    Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[h, 233] == 0]],
      " if and only if ", TraditionalForm[
        HoldForm[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) ==
          k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
      " and ", TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[
        e, 3]]], " . When ", TraditionalForm[HoldForm[u < 0]], " ,"}],
    12, GrayLevel[0.35]], Style[TraditionalForm[HoldForm[
      k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) -
        (k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)) ==
        (1 - u Subscript[z, 2] Subscript[z, 3]) /
        ((1 - u) (1 + Subscript[z, 2]) (1 + Subscript[z, 3]))], 12,
      GrayLevel[0.35]], Style[Row[{"which is verified separately on
        each of the 4 sign regions (sign combinations of ",
        TraditionalForm[HoldForm[Subscript[z, 2]]], " , ",
        TraditionalForm[HoldForm[Subscript[z, 3]]], " ) ",
        smallChip[convAll], " , so ", TraditionalForm[
          HoldForm[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) ==
            k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
          " if and only if ", TraditionalForm[HoldForm[u Subscript[z, 2]
            Subscript[z, 3] == 1]], " ."}], 12, GrayLevel[0.35]],

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Style[Row[{"Thus, when ", TraditionalForm[HoldForm[u < 0]],
  " , ", TraditionalForm[HoldForm[Subscript[h, 233] == 0]],
  " if and only if ", TraditionalForm[
    HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
    TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]],
  " . Therefore, in cases (i) and (ii) below, ", TraditionalForm[
    HoldForm[Subscript[h, 233] != 0]], " , and the identity ",
    TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
  " gives ", TraditionalForm[HoldForm[Subscript[H, 3] != 0]],
  " , so the divisions by ", TraditionalForm[
    HoldForm[Subscript[H, 3]]], " are legitimate there."}],
12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
Style[Row[{"in this case ", TraditionalForm[HoldForm[
  k^2 != (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]],
  " , that is, ", TraditionalForm[HoldForm[
    1 + u (Subscript[z, 2] + Subscript[z, 3] +
      Subscript[z, 2] Subscript[z, 3]) != 0]], " ;"}], 13],
Style[Row[{"we consider three possible cases: (i) ",
  TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] != 1]],
  " , (ii) ", TraditionalForm[
    HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
    TraditionalForm[HoldForm[Subscript[e, 2] == -Subscript[e, 3]]],
  " , (iii) ", TraditionalForm[
    HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]], " and ",
    TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]], " ."}],
13], "", Style[Row[{"(i) ", TraditionalForm[
  HoldForm[u Subscript[z, 2] Subscript[z, 3] != 1]], " :"}],
Bold, 13], Style["The following identities hold:", 13],
Style[Row[{TraditionalForm[HoldForm[k == Sqrt[u / (u - 1)]]],
  " , ", TraditionalForm[HoldForm[k' == 1 / Sqrt[1 - u]]],
  " , ", TraditionalForm[HoldForm[
    Subsuperscript[g, i, 2] - 1 == -(1 / (1 + Subscript[z, i]))]], " , ",
    TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - Superscript[k', 2] ==
      -((1 + u Subscript[z, i]) / ((1 - u) (1 + Subscript[z, i])))]],
  " , ", TraditionalForm[HoldForm[Subscript[g, i] == Sqrt[
    Subscript[z, i] (1 + Subscript[z, i]) / Abs[1 + Subscript[z, i]]]],
  " , ", Style["●", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
Style[Row[{TraditionalForm[HoldForm[u^2 Subscript[z, 2]
  Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) -
  (1 + u Subscript[z, 2]) (1 + u Subscript[z, 3]) ==
  (u Subscript[z, 2] Subscript[z, 3] - 1) (1 + u (Subscript[z, 2] +
    Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]))]],
  " , ", Style["●", 14, RGBColor[0.55, 0.05, 0.10]]}], 13],
Style[Row[{"and, by ", TraditionalForm[HoldForm[
  Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
  Sign[Subscript[x, 1] Subscript[x, 3]]], " and ",
  TraditionalForm[HoldForm[Sign[1 + Subscript[z, i]] ==

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Sign[Subscript[z, i]]], " : ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
Abs[Subscript[z, 2] Subscript[z, 3]]], " ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] ==
Abs[Subscript[x, 1] Subscript[x, 3]]], " ", TraditionalForm[
HoldForm[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]], " ."}], 13],
Style[Row[{"Hence, substituting the identities ",
Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
" and the equalities ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] == Abs[
Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
HoldForm[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]], " into ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], " ",
TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
", and using the identity ", Style["●", 13,
RGBColor[0.55, 0.05, 0.10]], " , we get, with ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]]}],
13, colTest], " ,"}, 13], Style[
Row[{TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
", ", TraditionalForm[HoldForm[Subscript[h, 231] ==
-Subscript[ξ, 23] Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
(Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] ==
-Subscript[ξ, 23] Subscript[H, 2] /
(Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
", ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]]}], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
Style[Row[{"for all 4 sign combinations of ",
TraditionalForm[HoldForm[Subscript[e, 2]]], " ",
TraditionalForm[HoldForm[Subscript[e, 3]]]}], 12, GrayLevel[0.3]],
smallChip[brRegion[[iw]]], {iw, Length[regA]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"Thus, substituting the identities ",
Style["●", 13, RGBColor[0.00, 0.45, 0.25]],
" into the numerator and the denominator of ",
Style["the given U-formula", Italic, 13, colGiven], " , we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
Sqrt[Subscript[x, 1] ∅[Subscript[x, 1], t]] ==
-Subscript[ξ, 23] Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]

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Subsuperscript[H, 3, 2] (Abs[Subscript[x, 1]] Subscript[h, 232] t +
  Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
  Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
  HoldForm[Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 ==
    Subscript[z, 2] Subscript[z, 3] Subsuperscript[H, 3, 2]
    (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
    Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
  TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
  TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}, 13],
Grid[Table[{Style[regA[[iw, 1]], 12, GrayLevel[0.3]],
  Style[Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 3]]]}], 12, GrayLevel[0.3]],
  smallChip[idRegion[[iw]]], {iw, Length[regA]}],
  Alignment → Left, Spacings → {2, 0.7}], "",
Style["and, finally, the prefactor identity:", 13],
Style[TraditionalForm[
  HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
  HoldForm[Subscript[z, 2] Subscript[z, 3] / (Subscript[ξ, 23]
    Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]])] == Sqrt[Subscript[
    z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[
  Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] == Abs[
    Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
  HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] ==
    Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"}], 13],
Style[Row[{"and is verified separately on each of the 8
  sign regions (sign combinations of ",
  TraditionalForm[HoldForm[Subscript[z, 2]]], ", ",
  TraditionalForm[HoldForm[Subscript[z, 3]]],
  ", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
  " as above) ", smallChip[preAll]}], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
  identities, along with ", TraditionalForm[
  HoldForm[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
  " and the cancellation of the factor ",
  TraditionalForm[HoldForm[Subsuperscript[H, 3, 2]]],
  " ( ", TraditionalForm[HoldForm[Subscript[H, 3] ≠ 0]],
  " , as noted above), give the formula ",
  Style["*"], 13, colTest], " in case (a):"}], 13],
Style[Row[{TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
  Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
  Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]

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      ((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
        Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
          Subscript[x, 1]  $\mathcal{D}$ [Subscript[x, 1], t]] / (Subscript[z, 2]
          Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
          t^2))]], "      ", Style["*", 14]]], 14, colTest],
"", Style[Row[{"The identity ", Style["*", 13, colTest],
  " is also verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]],
  ", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
  ", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
  ", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
  ", ", TraditionalForm[HoldForm[Subscript[ $\epsilon$ , 2]]],
  ", ", TraditionalForm[HoldForm[Subscript[ $\epsilon$ , 3]]],
  " symbolic, that is, for all their real values at once,
  which covers all their sign choices:"}], 13],
Grid[Table[{Style[regA[[iw], 1], 12, GrayLevel[0.3]], Style[
  Row[{"for all 4 sign combinations of ",
    TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[e, 3]]]}], 12, GrayLevel[0.3]],
  smallChip[fullRegion[[iw]]], {iw, Length[regA]}],
Alignment → Left, Spacings → {2, 0.7}], "",
Style[Row[{"(ii) ", TraditionalForm[
  HoldForm[u Subscript[z, 2] Subscript[z, 3] = 1]], " and ",
  TraditionalForm[HoldForm[Subscript[e, 2] = -Subscript[e, 3]]], " :"}],
Bold, 13], Style[Row[{"Here ", TraditionalForm[
  HoldForm[Subscript[ $\xi$ , 23] = -1]], " because ",
  TraditionalForm[HoldForm[u Subscript[z, 2] Subscript[z, 3] = 1]],
  " and ", TraditionalForm[HoldForm[u < 0]], " so ",
  TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] < 0]],
  " , and the signs of ", TraditionalForm[HoldForm[Subscript[z, 2]]],
  " and ", TraditionalForm[HoldForm[Subscript[z, 3]]],
  " are opposite; the given expressions remain ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
    TraditionalForm[HoldForm[Subscript[h, 233]]]}], 12, colGiven],
  " , and the expressions to test are given by ", Style[
  Row[{"the case-(b) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
    ", ", TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
    TraditionalForm[HoldForm[Subscript[H, 3]]]}], 12, colTest], " ."}],
12, GrayLevel[0.35]], Style[Row[TraditionalForm[HoldForm[
  1 + u Subscript[z, 2] = (1 + Subscript[z, 3]) / Subscript[z, 3]],
  ", ", TraditionalForm[HoldForm[1 + u Subscript[z, 3] =
    (1 + Subscript[z, 2]) / Subscript[z, 2]]],
  "      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]]}], 13],
Style[Row[{"Hence, substituting the identities ",
  Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
  " and ", Style["●", 13, RGBColor[0.40, 0.15, 0.60]],

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" and the equalities ", TraditionalForm[
HoldForm[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] == Abs[
Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
HoldForm[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]], " into ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
", we get, with ", Style[Row[{"the case-(b) ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[H, 3]]]}], 13, colTest], " ,"}, 13], Style[
Row[{TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
", ", TraditionalForm[HoldForm[Subscript[h, 231] ==
-Subscript[ξ, 23] Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
(Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
", ", TraditionalForm[HoldForm[Subscript[h, 232] ==
-Subscript[ξ, 23] Subscript[H, 2] /
(Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
" ", Style["●", 14, RGBColor[0.75, 0.10, 0.40]]}], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regMix[[iw, 1]], 12, GrayLevel[0.3]], Style[
Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
12, GrayLevel[0.3]], smallChip[brRegionB[[iw]]}],
{iw, Length[regMix]}], Alignment → Left, Spacings → {2, 0.7}],
""], Style[Row[{"Thus, substituting the identities ",
Style["●", 13, RGBColor[0.75, 0.10, 0.40]],
" into the numerator and the denominator of ",
Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
HoldForm[Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]] ==
-Subscript[ξ, 23] Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]
Subsuperscript[H, 3, 2] (Abs[Subscript[x, 1] Subscript[h, 232] t +
Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
HoldForm[Subscript[z, 2] Subscript[z, 3] +
u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 ==
Subscript[z, 2] Subscript[z, 3] Subsuperscript[H, 3, 2]
(Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2
Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],

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Grid[Table[{Style[regMix[[iw], 1], 12, GrayLevel[0.3]], Style[
  Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
  12, GrayLevel[0.3]], smallChip[idRegionB[[iw]]]],
{iw, Length[regMix]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style["and, finally, the prefactor identity:", 13],
Style[TraditionalForm[
  HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
  HoldForm[Subscript[z, 2] Subscript[z, 3] / (Subscript[ξ, 23]
  Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]])] == Sqrt[Subscript[
  z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]], 14],
Style[Row[{"which holds by ", TraditionalForm[HoldForm[
  Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] == Abs[
  Subscript[z, 2] Subscript[z, 3]]], " and ", TraditionalForm[
  HoldForm[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] ==
  Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"], 13],
Style[Row[{"and is verified separately on each of the 4
  sign regions (sign combinations of ",
  TraditionalForm[HoldForm[Subscript[z, 2]]], " , ",
  TraditionalForm[HoldForm[Subscript[z, 3]]],
  " , ", TraditionalForm[HoldForm[Subscript[x, 1]]],
  " as above) ", smallChip[preAllB]]], 13], "",
Style[Row[{"Together, the numerator, denominator, and prefactor
  identities, along with ", TraditionalForm[
  HoldForm[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
  " and the cancellation of the factor ",
  TraditionalForm[HoldForm[Subsuperscript[H, 3, 2]]],
  " ( ", TraditionalForm[HoldForm[Subscript[H, 3] ≠ 0]],
  " , as noted above), give the formula ",
  Style["*", 13, colTest], " in case (b):"], 13],
Style[Row[{TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
  Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
  Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  ((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
  Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
  Subscript[x, 1] D[Subscript[x, 1], t]) / (Subscript[z, 2]
  Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2]
  t^2))], " ", Style["*", 14]]], 14, colTest],
"", Style[Row[{"The identity ", Style["*", 13, colTest],
  " is also verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]],
  " , ", TraditionalForm[HoldForm[Subscript[H, 2]]],
  " , ", TraditionalForm[HoldForm[Subscript[H, 3]]],
  " , ", TraditionalForm[HoldForm[Subscript[e, 0]]],
  " , ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
  " , ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
  " symbolic, that is, for all their real values at once,
  which covers all their sign choices:"]], 13],

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Grid[Table[{Style[regMix[[iw], 1]], 12, GrayLevel[0.3]], Style[
  Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " =  $\pm 1$ "},
  12, GrayLevel[0.3]], smallChip[fullRegionB[[iw]]}],
{iw, Length[regMix]}], Alignment  $\rightarrow$  Left, Spacings  $\rightarrow$  {2, 0.7}],
"", Style[Row[{"(iii) ", TraditionalForm[
  HoldForm[u Subscript[z, 2] Subscript[z, 3] = 1]], " and ",
  TraditionalForm[HoldForm[Subscript[e, 2] = Subscript[e, 3]]], " :"}],
Bold, 13], Style[Row[{"As in case (ii), ",
  TraditionalForm[HoldForm[Subscript[ $\xi$ , 23] = -1]],
" ; the given expressions remain ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 233]]]}],
  12, colGiven], " , and the expression to test is ",
  Style["the case-(c) U-formula", Italic, 12, colTest],
" ."}], 12, GrayLevel[0.35]],
Style[Row[{"Hence, substituting the identities ",
  Style["●", 13, RGBColor[0.55, 0.33, 0.10]], " and ",
  Style["●", 13, RGBColor[0.40, 0.15, 0.60]], " into ", Style[
  Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
    ", ", TraditionalForm[HoldForm[Subscript[h, 233]]]}],
  13, colGiven], " , we get"}], 13], Style[
  Row[{TraditionalForm[HoldForm[Subscript[h, 233] = 0]], " , ",
    TraditionalForm[HoldForm[k Subscript[h, 231] = Subscript[e, 2]]],
    " , ", TraditionalForm[HoldForm[
      Subscript[h, 232] = Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]],
    " , ", TraditionalForm[HoldForm[Subsuperscript[h, 233, 2] -
      Subscript[ $\xi$ , 23] k^2 Subscript[x, 1]
      Subsuperscript[h, 231, 2] t^2 = Subscript[x, 1] t^2]],
    " ", Style["●", 14, RGBColor[0.00, 0.55, 0.65]]}], 13],
Style[Row[{"Thus, by the identities ", Style["●", 13,
  RGBColor[0.00, 0.55, 0.65]], " , the ",
  TraditionalForm[HoldForm[Subscript[e, 0]]], "-term of ",
  Style["the given U-formula", Italic, 13, colGiven],
" vanishes and it reduces to"}], 13], Style[TraditionalForm[
  HoldForm[U = HoldForm[Subscript[ $\epsilon$ , 2] Subscript[ $\epsilon$ , 3] Sqrt[Subscript[
    z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  HoldForm[Sign[Subscript[y, 2] / t]]], 14, colTest],
Style[Row[{"which is ", Style["the tested case-(c) formula",
  Italic, 13, colTest], " ; the given denominator reduces to ",
  TraditionalForm[HoldForm[Subscript[x, 1] t^2]],
" , which vanishes for the same t as the tested denominator ",
  TraditionalForm[HoldForm[t]], " , namely ",
  TraditionalForm[HoldForm[t = 0]], " . The identities ",
  Style["●", 13, RGBColor[0.00, 0.55, 0.65]],
" and the equality of the ", Style["given", Italic, 13, colGiven],

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" and ", Style["tested", Italic, 13, colTest],
" formulas are verified separately on each sign region:"]], 13],
Grid[Table[{Style[regMix[[iw, 1]], 12, GrayLevel[0.3]], Style[
  Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
  12, GrayLevel[0.3]], smallChip[cRegion[[iw]]]}, {iw, Length[regMix]}],
  Alignment → Left, Spacings → {2, 0.7}], Spacings → 1],
Background → Lighter[RGBColor[0.00, 0.50, 0.50],
  0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 6:CASES (b) AND (c) FOR u<0,ON 1+u(z2+z3+z2 z3)=0*)
(* =====*)
ClearAll[pbCheck, pbPreCk, pcCheck];
pbCheck[{lab_, w_, sx_}, e2_] :=
Module[{at = mkAP[w], uv, kk2, mS, kpAt, kkAt, suz, W, d2, d3, h, H, brh, brOK,
  numCk, denCk, fullCk}, uv = -1 / (at["z2"] + at["z3"] + at["z2"] at["z3"]);
kk2 = -uv / (1 - uv); mS = at["sv2"] at["sv3"] / at["sS"];
kpAt = 1 / mS; kkAt = 1 / (at["sv2"] at["sv3"]); suz = -at["s2"];
W = at["sw2"] at["sw3"];
{d2, d3} = gdict[at, 2][[3 ;; 4]];
h = p7h[at, kkAt, 1 / (1 - uv), e2, 2]; H = HAat[at, uv, e2, -e2];
brh = {-e2 mS H1s / (W H3s), -H2s / (W H3s), 1 / H3s};
brOK = bOK[at, uv] && p7ok[at, kkAt, kk2, 2] && chkR[kk2, (d2 - 1) (d3 - 1)] &&
  chkR[mS^2, 1 - uv] && chkR[kpAt, kkAt at["sS"]] && chkR[h[[3]] H[[3]], 1] &&
  chkR[h[[1]] W H[[3]], -e2 mS H[[1]]] && chkR[h[[2]] W H[[3]], -H[[2]]];
numCk = chkR[XX H2s H3s tt + ee0 e2 H1s RR,
  -W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)];
denCk = chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
  at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - kk2 sx XX brh[[1]]^2 tt^2)];
fullCk = chkR[uLemNegat[brh, kk2, kpAt, 1, sx, suz],
  uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]];
{brOK, numCk && denCk, fullCk}];
pbPreCk[{lab_, w_, sx_}] :=
Module[{at = mkAP[w], W}, W = at["sw2"] at["sw3"];
  chkR[at["z2"] at["z3"] / (W sx), W / sx]];
pcCheck[{lab_, w_, sx_}, e2_] :=
Module[{at = mkAP[w], uv, kk2, mS, kpAt, kkAt, suz, W, d2, d3, H, H1v, H2v},
  uv = -1 / (at["z2"] + at["z3"] + at["z2"] at["z3"]);
kk2 = -uv / (1 - uv); mS = at["sv2"] at["sv3"] / at["sS"];
kpAt = 1 / mS; kkAt = 1 / (at["sv2"] at["sv3"]); suz = -at["s2"];
W = at["sw2"] at["sw3"];
{d2, d3} = gdict[at, 2][[3 ;; 4]];
H = HAat[at, uv, e2, e2];
H1v = -W at["sS"];
H2v = -at["sw2"] at["sv2"] at["sw3"] at["sv3"];
{bOK[at, uv] && chkR[kkAt^2, kk2] && chkR[kk2, (d2 - 1) (d3 - 1)] &&
  chkR[mS^2, 1 - uv] && chkR[kpAt, kkAt at["sS"]] &&
  chkR[H[[3]], 0] && chkR[H[[1]], H1v] && chkR[H[[2]], H2v],

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chkR[ee0 e2 kpAt RR, -(kkAt/W) ee0 e2 H1v RR] &&
chkR[kkAt (1 - sx XX tt^2), (kkAt/(at["z2"] at["z3"]))
(at["z2"] at["z3"] + uv sx XX H1v^2 tt^2)],
chkR[uThmABat[{H1v, H2v, 0}], uv, at, sx, e2],
uLemNegCat[kkAt, kpAt, sx, suz, e2]]];
Module[{gridB, brRegion, idRegion, fullRegion,
preAll, gridC, cBrRegion, cIdRegion, cRegion, ok},
gridB = Table[pbCheck[w, s], {w, regBpos}, {s, {1, -1}}];
brRegion = Table[And@@gridB[[iw, All, 1]], {iw, Length[regBpos]}];
idRegion = Table[And@@gridB[[iw, All, 2]], {iw, Length[regBpos]}];
fullRegion = Table[And@@gridB[[iw, All, 3]], {iw, Length[regBpos]}];
preAll = And@@(pbPreCk/@regBpos);
gridC = Table[pcCheck[w, s], {w, regBpos}, {s, {1, -1}}];
cBrRegion = Table[And@@gridC[[iw, All, 1]], {iw, Length[regBpos]}];
cIdRegion = Table[And@@gridC[[iw, All, 2]], {iw, Length[regBpos]}];
cRegion = Table[And@@gridC[[iw, All, 3]], {iw, Length[regBpos]}];
ok = (And@@brRegion) && (And@@idRegion) && (And@@fullRegion) &&
preAll && (And@@cBrRegion) && (And@@cIdRegion) && (And@@cRegion);
AppendTo[resultsAll, ok];
Print[
Panel[Column[{Row[{Style["The derivation: the formula for ", Bold, 15,
RGBColor[0.75, 0.35, 0.05]], Style[TraditionalForm[HoldForm[U]],
Bold, 15, RGBColor[0.75, 0.35, 0.05]], Style[
" in cases (b) and (c) for ", Bold, 15, RGBColor[0.75, 0.35, 0.05]],
Style[TraditionalForm[HoldForm[u < 0]], Bold, 15,
RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]]},
Style[Row[{"Here ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
", because ", TraditionalForm[HoldForm[k^2 ==
(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
" and ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
-(1/(1 + Subscript[z, i]))]], ", so ", TraditionalForm[
HoldForm[(1 + Subscript[z, 2]) (1 + Subscript[z, 3]) > 0]],
" , and, by ", TraditionalForm[HoldForm[Sign[1 + Subscript[z, i]] ==
Sign[Subscript[z, i]]], ", the signs of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], " and ",
TraditionalForm[HoldForm[Subscript[z, 3]]], " coincide."}],
12, GrayLevel[0.35]], Style[Row[{"By ", TraditionalForm[
HoldForm[Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
Sign[Subscript[x, 1] Subscript[x, 3]] == 1]], " : ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
Abs[Subscript[z, 2] Subscript[z, 3]]], ", ",
TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==
Abs[Subscript[x, 1] Subscript[x, 3]]], ", ",
TraditionalForm[HoldForm[(1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]], " ."}], 12,
GrayLevel[0.35]], Style[Row[{"Moreover, ", TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] ≠ 1]],

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", because the equalities ", TraditionalForm[
HoldForm[u Subscript[z, 2] Subscript[z, 3] == 1]],
" and ", TraditionalForm[HoldForm[1 + u (Subscript[z, 2] +
Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]) == 0]],
" cannot hold simultaneously; hence the expressions
to test are given by ", Style[
Row[{"the case-(a) ", TraditionalForm[HoldForm[Subscript[H, 1]]],
", ", TraditionalForm[HoldForm[Subscript[H, 2]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 3]]]}], 12, colTest],
" . Also ", TraditionalForm[HoldForm[Subscript[h, 233] != 0]],
" (as stated in the hypotheses), and,
in case (b), the identity ",
TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
" gives ", TraditionalForm[HoldForm[Subscript[H, 3] != 0]],
", so the divisions by ", TraditionalForm[
HoldForm[Subscript[H, 3]]], " are legitimate there."}],
12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
Style[Row[{"in these cases ", TraditionalForm[HoldForm[
k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)]],
", that is, ", TraditionalForm[HoldForm[1 + u (Subscript[z, 2] +
Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]) == 0]],
", and the following identities hold:"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[g, i] == Sqrt[
Subscript[z, i] (1 + Subscript[z, i])] / Abs[1 + Subscript[z, i]]],
", ", TraditionalForm[HoldForm[1 + u Subscript[z, 2] ==
Subscript[z, 3] (1 + Subscript[z, 2]) / (Subscript[z, 2] +
Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3])]], ", ",
TraditionalForm[HoldForm[1 + u Subscript[z, 3] == Subscript[z, 2]
(1 + Subscript[z, 3]) / (Subscript[z, 2] + Subscript[z, 3] +
Subscript[z, 2] Subscript[z, 3])]], ", ", TraditionalForm[
HoldForm[k == 1 / Sqrt[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]],
", ", TraditionalForm[HoldForm[k' == k Sqrt[Subscript[z,
2] + Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]]]],
" ", Style["■", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
Style[Row[{"Hence, in case (b) ( ", TraditionalForm[
HoldForm[Subscript[e, 2] == -Subscript[e, 3]]],
" ), substituting the identities ",
Style["■", 13, RGBColor[0.55, 0.33, 0.10]], " and the equality ",
TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] ==
Abs[Subscript[z, 2] Subscript[z, 3]]], " into ", Style[
Row[{"the case-(b) ", TraditionalForm[HoldForm[Subscript[h, 231]]],
", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
TraditionalForm[HoldForm[Subscript[h, 233]]]}], 13, colGiven],
", we get, with ", Style[Row[{"the case-(a) ",
TraditionalForm[HoldForm[Subscript[H, 1]]], ", ",
TraditionalForm[HoldForm[Subscript[H, 2]]], ", ", TraditionalForm[
HoldForm[Subscript[H, 3]]]}], 13, colTest], ", "}], 13], Style[

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Row[{TraditionalForm[HoldForm[Subscript[h, 233] Subscript[H, 3] == 1]],
  ", ", TraditionalForm[HoldForm[
    Subscript[h, 231] == -Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
      (Sqrt[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
  ", ", TraditionalForm[HoldForm[Subscript[h, 232] == -Subscript[H, 2] /
    (Sqrt[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])]],
  " ", Style["■", 14, RGBColor[0.00, 0.45, 0.25]]}], 13],
Style["which are verified separately on each sign region:", 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]], Style[
  Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
  12, GrayLevel[0.3]], smallChip[brRegion[[iw]]}],
{iw, Length[regBpos]}], Alignment → Left, Spacings → {2, 0.7}],
"", Style[Row[{"Thus, substituting the identities ",
  Style["■", 13, RGBColor[0.00, 0.45, 0.25]],
  " into the numerator and the denominator of ",
  Style["the given U-formula", Italic, 13, colGiven], " , we get"}], 13],
Style["the numerator identity:", 13], Style[TraditionalForm[
  HoldForm[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]] ==
    -Sqrt[Subscript[z, 2] Subscript[z, 3]] Subsuperscript[H, 3, 2]
    (Abs[Subscript[x, 1]] Subscript[h, 232] t +
    Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
    Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]])]], 14],
Style["the denominator identity:", 13], Style[TraditionalForm[
  HoldForm[Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
    Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2] Subscript[z, 3]
    Subsuperscript[H, 3, 2] (Subsuperscript[h, 233, 2] -
    k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
Style[Row[{"which are verified separately on each sign region, with ",
  TraditionalForm[HoldForm[Subscript[H, 1]]], " , ",
  TraditionalForm[HoldForm[Subscript[H, 2]]], " , ",
  TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
Grid[Table[{Style[regBpos[[iw, 1]], 12, GrayLevel[0.3]], Style[
  Row[{"for ", TraditionalForm[HoldForm[Subscript[e, 2]]], " = ±1"}],
  12, GrayLevel[0.3]], smallChip[idRegion[[iw]]}],
{iw, Length[regBpos]}], Alignment → Left, Spacings → {2, 0.7}],
Style[Row[{"and, finally, ", Style["the prefactor identity", Italic,
  13, RGBColor[0.80, 0.20, 0.50]], " :"}], 13], Style[TraditionalForm[
  HoldForm[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]
    HoldForm[Subscript[z, 2] Subscript[z, 3] /
      Sqrt[Subscript[z, 2] Subscript[z, 3]] == Sqrt[Subscript[z, 2]
      Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]], 14],
Style[Row[{"which holds by ", TraditionalForm[
  HoldForm[Subscript[z, 2] Subscript[z, 3] ==
    Abs[Subscript[z, 2] Subscript[z, 3]]], " and ",
  TraditionalForm[HoldForm[Subscript[x, 1] Subscript[x, 3] ==

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Abs[Subscript[x, 1] Subscript[x, 3]]], " ,"]], 13],
Style[Row[{"and is verified separately on each of the 4
sign regions (sign combinations of ",
TraditionalForm[HoldForm[Subscript[z, 2]]], " ,",
TraditionalForm[HoldForm[Subscript[z, 3]]],
", ", TraditionalForm[HoldForm[Subscript[x, 1]]],
" as above) ", smallChip[preAll]]], 13], ""],
Style[Row[{"Together, the numerator, denominator, and prefactor
identities, along with ", TraditionalForm[
HoldForm[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]]],
" and the cancellation of the factor ",
TraditionalForm[HoldForm[Subsuperscript[H, 3, 2]]],
" ( ", TraditionalForm[HoldForm[Subscript[H, 3] != 0]],
" , as noted above), give the formula ",
Style["*", 13, colTest], " in case (a):"]], 13],
Style[Row[TraditionalForm[HoldForm[U == HoldForm[Subscript[ε, 2]
Subscript[ε, 3] Sign[Subscript[z, 2]] Sqrt[Subscript[z, 2]
Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
((Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
Sqrt[Subscript[x, 1] D[Subscript[x, 1], t]]) /
(Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
Subsuperscript[H, 1, 2] t^2))]],
" ", Style["*", 14]]], 14, colTest], "", Style[
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" is also verified separately on each sign region, with ",
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", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]],
", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
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which covers all their sign choices:"]], 13],
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", ", TraditionalForm[HoldForm[Subscript[H, 2]]],
", ", TraditionalForm[HoldForm[Subscript[H, 3]]]]],
12, colTest], " , and the given expression is ",
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12, GrayLevel[0.35]], Style[TraditionalForm[HoldForm[

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U = HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]] /
  Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] ·
  HoldForm[Subscript[e, 0] Subscript[e, 2] k' Sqrt[Subscript[x, 1]
    ∂[Subscript[x, 1], t]] / (k (1 - Subscript[x, 1] t^2))]]],
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    "", TraditionalForm[HoldForm[Subscript[H, 3]]]}],
  13, colTest], " , we get"}], 13],
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      Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3])]]], "",
  TraditionalForm[HoldForm[Subscript[H, 2] = -Sqrt[Subscript[z, 2]
    Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])]]],
  " ", Style["■", 14, RGBColor[0.40, 0.15, 0.60]]}], 13],
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  Style["■", 13, RGBColor[0.40, 0.15, 0.60]], " and the equality ",
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    Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]]],
  " into the numerator and the denominator of ", Style[
  "the given case-(c) U-formula", Italic, 13, colGiven], " , we get"}],
13], Style["the numerator identity:", 13],
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  k' Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]] =
  - (k / Sqrt[Subscript[z, 2] Subscript[z, 3]])
  (Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
    Sqrt[Subscript[x, 1] ∂[Subscript[x, 1], t]])]], 14],
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Style[Row[{"which are verified separately on each sign region, with ",
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  TraditionalForm[HoldForm[Subscript[H, 2]]], "",
  TraditionalForm[HoldForm[Subscript[H, 3]]], " symbolic:"}], 13],
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"", Style[Row[{"Together, the numerator, denominator, and ",
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  Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
  ((Abs[Subscript[x, 1] Subscript[H, 2] Subscript[H, 3] t +
  Subscript[e, 0] Subscript[e, 2] Subscript[H, 1] Sqrt[
    Subscript[x, 1] D[Subscript[x, 1], t]] / (Subscript[z, 2]
  Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H,
    1, 2] t^2))]], " ", Style["*", 14]]], 14, colTest],
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    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2]],
  " vanish for the same t, because ", TraditionalForm[
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  " ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
  " symbolic, that is, for all their real values at once,
  which covers all their sign choices:"]], 13],
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(* =====*)
(*PART 7:SUMMARY*)
(* =====*)
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Given hypotheses:

$$u \neq 0, \quad u < 1, \quad k, k' \in (0, 1)$$

For $i = 1, 2, 3, 4$:

$$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0$$

$$\xi_{\alpha\beta} = \sin(z_\alpha z_\beta) - \sin(y_\alpha y_\beta) - \sin(y_\alpha) - \sin(y_\beta z_\alpha)$$

$$\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3), \quad \text{sgn}(y_2) = \text{sgn}(u x_1 z_2)$$

$$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$$

$$e_0, e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$$

$$\mathcal{D}(x_1, t) = -(1 - x_1 t^2)(1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$$

$$(i) \text{ if } u > 0: \quad k = \sqrt{u}, \quad k' = \sqrt{1-u}, \quad g_i = \sqrt{\frac{1+z_i}{z_i}}, \quad k^2 - (g_2^2 - 1)(g_3^2 - 1) = \frac{u z_2 z_3 - 1}{z_2 z_3}$$

$$(ii) \text{ if } u < 0: \quad k' = \frac{1}{\sqrt{1-u}}, \quad k = k' \sqrt{-u},$$

$$g_i = \sqrt{\frac{z_i}{1+z_i}}, \quad k^2 - (g_2^2 - 1)(g_3^2 - 1) = -\frac{1+u(z_2+z_3+z_2 z_3)}{(1-u)(1+z_2)(1+z_3)}$$

$$h_{231}, h_{232}, h_{233}, H_1, H_2, H_3 \in \mathbb{R}$$

$$h_{231} = 0 \text{ if and only if } g_2 = g_3 \text{ and } e_2 = -e_3$$

$$; \quad h_{232} = 0 \text{ if and only if } g_2 g_3 = k' \text{ and } e_2 = -e_3 ;$$

$$h_{233} = 0 \text{ if and only if } k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$$

$$\text{and } e_2 = e_3 ; \text{ in particular, } h_{233} \neq 0 \text{ if } k^2 = (g_2^2 - 1)(g_3^2 - 1) .$$

the equalities $u z_2 z_3 = 1$ and

$1 + u(z_2 + z_3 + z_2 z_3) = 0$ **cannot hold simultaneously** .

Given formulas and expressions:

$$U = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \left(\left\{ \begin{array}{ll} \frac{|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)}}{1 + \xi_{23} k^2 x_1 h_{231}^2 t^2} & u > 0 \\ \frac{|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)}}{h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2} & u < 0 \end{array} \right\} \right)$$

$$\left. \begin{aligned} h_{231} &= \frac{e_2 g_3 \sqrt{(g_2^2 - 1)(g_3^2 - k'^2)} \xi_{23} + e_3 g_2 \sqrt{(g_3^2 - 1)(g_2^2 - k'^2)} \xi_{23}}{k^2 - (g_2^2 - 1)(g_3^2 - 1)} \\ h_{232} &= \frac{\sqrt{(g_2^2 - k'^2)(g_3^2 - k'^2)} \xi_{23} + e_2 e_3 g_2 g_3 \sqrt{(g_2^2 - 1)(g_3^2 - 1)} \xi_{23}}{k^2 - (g_2^2 - 1)(g_3^2 - 1)} \\ h_{233} &= \frac{k^2 g_2 g_3 + e_2 e_3 \xi_{23} \sqrt{(g_2^2 - 1)(g_2^2 - k'^2)(g_3^2 - 1)(g_3^2 - k'^2)}}{k^2 - (g_2^2 - 1)(g_3^2 - 1)} \end{aligned} \right\} \quad \text{(a) } k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$$

$$h_{231} = \frac{e_2 (g_2^2 - g_3^2)}{2 k g_2 g_3}, \quad h_{232} = \frac{g_2^2 g_3^2 - k'^2}{2 k g_2 g_3}, \quad h_{233} = \frac{g_2^2 + g_3^2}{2 g_2 g_3} \quad \text{(b) } k^2 = (g_2^2 - 1)(g_3^2 - 1), \quad e_2 = -e_3$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \left(\left\{ \begin{array}{ll} \frac{\text{sgn}(x_1)}{k t} & u > 0 \\ \frac{e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k (1 - x_1 t^2)} & u < 0 \end{array} \right\} \right) \quad \text{(c) } k^2 = (g_2^2 - 1)(g_3^2 - 1), \quad e_2 = e_3$$

Each  below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a  would mean that this verification failed.

We want to test these formulas ★ :

We show that the formulas for U

below follow from the given hypotheses, formulas, and expressions above, together with the following claim: in each case, the denominator of the given formula and the denominator of the tested formula vanish for the same values of t , that is, one of them is zero if and only if the other is; hence the compared formulas are defined for exactly the same t .

$$U = \frac{\left(\varepsilon_2 \varepsilon_3 \operatorname{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \left(|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} \right)}{z_2 z_3 + u x_1 H_1^2 t^2}$$

$$\left. \begin{aligned} H_1 &= \frac{|z_2| \sqrt{z_3(1+z_3)(1+u z_3)} + e_2 e_3 |z_3| \sqrt{z_2(1+z_2)(1+u z_2)}}{u z_2 z_3 - 1} \\ H_2 &= \frac{|z_2 z_3| \sqrt{(1+u z_2)(1+u z_3)} + e_2 e_3 \sqrt{z_2 z_3(1+z_2)(1+z_3)}}{u z_2 z_3 - 1} \\ H_3 &= \frac{u \sqrt{z_2 z_3(1+z_2)(1+z_3)} \operatorname{sgn}(z_2 z_3) + e_2 e_3 \sqrt{(1+u z_2)(1+u z_3)}}{u z_2 z_3 - 1} \end{aligned} \right\} \quad \text{(a) } u z_2 z_3 \neq 1$$

$$\left. \begin{aligned} H_1 &= \frac{z_2 z_3 (z_3 - z_2)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \\ H_2 &= \frac{z_2 z_3 (z_2 + z_3 + 2)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \\ H_3 &= \frac{(z_2 + z_3 + 2 z_2 z_3) \operatorname{sgn}(z_2 z_3)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \end{aligned} \right\} \quad \text{(b) } u z_2 z_3 = 1, \quad e_2 = -e_3$$

$$U = \left(\varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \frac{\operatorname{sgn}(y_2)}{t} \quad \text{(c) } u z_2 z_3 = 1, \quad e_2 = e_3$$

The derivation: the formula for U in case (a) for $u > 0$

Verified

By $\xi_{23} = \operatorname{sgn}(z_2 z_3) = \operatorname{sgn}(x_1 x_3)$: $\xi_{23} z_2 z_3 = |z_2 z_3|$, $\xi_{23} x_1 x_3 = |x_1 x_3|$.

Verification:

in this case $k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$, that is, $u z_2 z_3 \neq 1$, and the following identities hold:

$$k = \sqrt{u}, \quad g_i^2 - 1 = \frac{1}{z_i}, \quad g_i^2 - k^2 = \frac{1+u z_i}{z_i}, \quad g_i = \frac{\sqrt{z_i(1+z_i)}}{|z_i|} \quad \blacklozenge$$

Hence, substituting the identities $\blacklozenge$ and the equality $\xi_{23} z_2 z_3 = |z_2 z_3|$ into the case-(a) $h_{231}, h_{232}, h_{233}$, we get, with the case-(a) H_1, H_2, H_3 ,

$$h_{231} = \frac{\xi_{23} e_2 H_1}{\sqrt{|z_2 z_3|}}, \quad h_{232} = \frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|}}, \quad h_{233} = H_3 \quad \blacklozenge$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

Thus, substituting the identities ♦

into the numerator and the denominator of *the given U-formula*, we get
the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = \xi_{23} \sqrt{|z_2 z_3|} (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 (1 + \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\xi_{23} \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 8 sign regions (sign combinations of z_2, z_3, x_1 as above) ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = \text{sgn}(z_2)$, give the formula ★ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

The derivation: the formula for U in cases (b) and (c) for $u > 0$

Verified

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and

$g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$, and the signs of z_2 and z_3 coincide.

By $\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3) = 1$: $z_2 z_3 = |z_2 z_3|$, $x_1 x_3 = |x_1 x_3|$.

Verification:

in these cases $k^2 = (g_2^2 - 1)(g_3^2 - 1)$, that is, $u z_2 z_3 = 1$, and the following identities hold:

$$k = \frac{1}{\sqrt{z_2 z_3}}, \quad g_i = \frac{\sqrt{z_i(1+z_i)}}{|z_i|} \quad \blacktriangle$$

Hence, in case (b) ($e_2 = -e_3$), substituting the identities $\blacktriangle$ and the equality

$z_2 z_3 = |z_2 z_3|$ into the case-(b) $h_{231}, h_{232}, h_{233}$, we get, with the case-(b) H_1, H_2, H_3 ,

$$h_{231} = \frac{e_2 H_1}{\sqrt{z_2 z_3}}, \quad h_{232} = \frac{H_2}{\sqrt{z_2 z_3}}, \quad h_{233} = H_3 \quad \blacktriangle$$

which are verified separately on each sign region:

$$z_2 > 0, \quad z_3 > 0, \quad x_1 > 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

$$z_2 > 0, \quad z_3 > 0, \quad x_1 < 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

$$z_2 < -1, \quad z_3 < -1, \quad x_1 > 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

$$z_2 < -1, \quad z_3 < -1, \quad x_1 < 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

Thus, substituting the identities $\blacktriangle$

into the numerator and the denominator of the given U -formula, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = \sqrt{z_2 z_3} (|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 (1 + k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$$z_2 > 0, \quad z_3 > 0, \quad x_1 > 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

$$z_2 > 0, \quad z_3 > 0, \quad x_1 < 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

$$z_2 < -1, \quad z_3 < -1, \quad x_1 > 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

$$z_2 < -1, \quad z_3 < -1, \quad x_1 < 0 \quad \text{for } e_2 = \pm 1 \quad \checkmark$$

and, finally, the prefactor identity :

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\sqrt{z_2 z_3}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $z_2 z_3 = |z_2 z_3|$ and $x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3, x_1 as above) $\checkmark$

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = \text{sgn}(z_2)$, give the formula $\star$ in case (b):

$$U = \frac{(e_2 e_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, e_2, e_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1$	✓

In case (c) ($e_2 = e_3$), the **given** and **tested** formulas are their replacements:

$$U = \frac{e_2 e_3 \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \frac{\operatorname{sgn}(x_1)}{k t} \quad \text{and} \quad U = \left(e_2 e_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \frac{\operatorname{sgn}(y_2)}{t},$$

which coincide because $\operatorname{sgn}(y_2) = \operatorname{sgn}(u x_1 z_2)$, $k = \frac{1}{\sqrt{z_2 z_3}}$, and **the prefactor identity**; their denominators $k t$ and t vanish for the same t because $k \neq 0$.

The equality of the **given** and **tested**

formulas is verified separately on each sign region, with e_2, e_3

symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	✓

The derivation: the formula for U in case (a) for $u < 0$

Verified

Here $h_{233} = 0$ if and only if $k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$ and $e_2 = e_3$. When $u < 0$,

$$k^2 (g_2^2 + g_3^2) - (k^2 - (g_2^2 - 1)(g_3^2 - 1)) = \frac{1 - u z_2 z_3}{(1 - u)(1 + z_2)(1 + z_3)}$$

which is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3) ✓, so $k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$ if and only if $u z_2 z_3 = 1$.

Thus, when $u < 0$, $h_{233} = 0$ if and only if $u z_2 z_3 = 1$ and $e_2 = e_3$. Therefore, in cases (i) and (ii) below, $h_{233} \neq 0$, and the identity $h_{233} H_3 = 1$ gives $H_3 \neq 0$, so the divisions by H_3 are legitimate there.

Verification:

in this case $k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$, that is, $1 + u(z_2 + z_3 + z_2 z_3) \neq 0$;

we consider three possible cases: (i) $u z_2 z_3 \neq 1$

, (ii) $u z_2 z_3 = 1$ and $e_2 = -e_3$, (iii) $u z_2 z_3 = 1$ and $e_2 = e_3$.

(i) $u z_2 z_3 \neq 1$:

The following identities hold:

$$k = \sqrt{\frac{u}{u-1}}, \quad k' = \frac{1}{\sqrt{1-u}}, \quad g_i^2 - 1 = -\frac{1}{1+z_i}, \quad g_i^2 - k'^2 = -\frac{1+u z_i}{(1-u)(1+z_i)}, \quad g_i = \frac{\sqrt{z_i(1+z_i)}}{|1+z_i|} \quad \bullet$$

$$u^2 z_2 z_3 (1 + z_2)(1 + z_3) - (1 + u z_2)(1 + u z_3) = (u z_2 z_3 - 1)(1 + u(z_2 + z_3 + z_2 z_3)) \quad \bullet$$

and, by $\xi_{23} = \operatorname{sgn}(z_2 z_3) = \operatorname{sgn}(x_1 x_3)$ and $\operatorname{sgn}(1 + z_i) = \operatorname{sgn}(z_i)$:

$$\xi_{23} z_2 z_3 = |z_2 z_3|, \quad \xi_{23} x_1 x_3 = |x_1 x_3|, \quad \xi_{23} (1 + z_2)(1 + z_3) = |(1 + z_2)(1 + z_3)|.$$

Hence, substituting the identities $\bullet$ and the equalities $\xi_{23} z_2 z_3 = |z_2 z_3|$

and $\xi_{23} (1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$ into the case-(a) $h_{231}, h_{232}, h_{233}$, and using the identity $\bullet$, we get, with the case-(a) H_1, H_2, H_3 ,

$$h_{233} H_3 = 1, \quad h_{231} = -\frac{\xi_{23} e_2 \sqrt{1-u} H_1}{\sqrt{|z_2 z_3|} H_3}, \quad h_{232} = -\frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|} H_3} \quad \bullet$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

Thus, substituting the identities $\bullet$

into the numerator and the denominator of the given U -formula, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = -\xi_{23} \sqrt{|z_2 z_3|} H_3^2 (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\xi_{23} \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 8 sign regions (sign combinations of z_2, z_3, x_1 as above) ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = -\text{sgn}(z_2)$ and the cancellation of the factor H_3^2 ($H_3 \neq 0$, as noted above), give the formula $\star$ in case (a):

$$U = \frac{(e_2 e_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity $\star$ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, e_2, e_3$

symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for all 4 sign combinations of e_2, e_3	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for all 4 sign combinations of e_2, e_3	✓

(ii) $u z_2 z_3 = 1$ and $e_2 = -e_3$:

Here $\xi_{23} = -1$, because $u z_2 z_3 = 1$ and $u < 0$, so $z_2 z_3 < 0$, and the signs of z_2 and z_3 are opposite; the given expressions remain **the case-(a) $h_{231}, h_{232}, h_{233}$** , and the expressions to test are given by **the case-(b) H_1, H_2, H_3** .

$$1 + u z_2 = \frac{1+z_3}{z_3}, \quad 1 + u z_3 = \frac{1+z_2}{z_2} \quad \bullet$$

Hence, substituting the identities $\bullet$ and $\bullet$ and the equalities

$\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} (1 + z_2)(1 + z_3) = |(1 + z_2)(1 + z_3)|$ into **the case-(a) $h_{231}, h_{232}, h_{233}$** , we get, with **the case-(b) H_1, H_2, H_3** ,

$$h_{233} H_3 = 1, \quad h_{231} = -\frac{\xi_{23} e_2 \sqrt{1-u} H_1}{\sqrt{|z_2 z_3|} H_3}, \quad h_{232} = -\frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|} H_3} \quad \bullet$$

which are verified separately on each sign region:

$z_2 > 0, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1$	✓

Thus, substituting the identities $\bullet$

into the numerator and the denominator of **the given U-formula**, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = -\xi_{23} \sqrt{|z_2 z_3|} H_3^2 (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 > 0, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1$	✓

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\xi_{23} \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23} x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3, x_1 as above) ✓

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u z_2) = -\text{sgn}(z_2)$ and the cancellation of the factor H_3^2 ($H_3 \neq 0$, as noted above), give the formula ★ in case (b):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 < -1, x_1 > 0$ for $e_2 = \pm 1$ ✓
 $z_2 > 0, z_3 < -1, x_1 < 0$ for $e_2 = \pm 1$ ✓
 $z_2 < -1, z_3 > 0, x_1 > 0$ for $e_2 = \pm 1$ ✓
 $z_2 < -1, z_3 > 0, x_1 < 0$ for $e_2 = \pm 1$ ✓

(iii) $u z_2 z_3 = 1$ and $e_2 = e_3$:

As in case (ii), $\xi_{23} = -1$; the given expressions remain

the case-(a) $h_{231}, h_{232}, h_{233}$, and the expression to test is the case-(c) U -formula .

Hence, substituting the identities ● and ● into the case-(a) $h_{231}, h_{232}, h_{233}$, we get

$$h_{233} = 0, \quad k h_{231} = e_2, \quad h_{232} = \sqrt{|z_2 z_3|}, \quad h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2 = x_1 t^2 \quad \bullet$$

Thus, by the identities ●, the e_0 -term of the given U -formula vanishes and it reduces to

$$U = \left(\varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}} \right) \frac{\text{sgn}(y_2)}{t}$$

which is the tested case-(c) formula; the given denominator reduces to $x_1 t^2$, which vanishes for the same t as the tested denominator

t , namely $t = 0$. The identities ● and the equality of the given

and tested formulas are verified separately on each sign region:

$z_2 > 0, z_3 < -1, x_1 > 0$ for $e_2 = \pm 1$ ✓
 $z_2 > 0, z_3 < -1, x_1 < 0$ for $e_2 = \pm 1$ ✓
 $z_2 < -1, z_3 > 0, x_1 > 0$ for $e_2 = \pm 1$ ✓
 $z_2 < -1, z_3 > 0, x_1 < 0$ for $e_2 = \pm 1$ ✓

The derivation: the formula for U

in cases (b) and (c) for $u < 0$

Verified

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = -\frac{1}{1+z_i}$, so

$(1+z_2)(1+z_3) > 0$, and, by $\text{sgn}(1+z_i) = \text{sgn}(z_i)$, the signs of z_2 and z_3 coincide.

By $\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3) = 1$: $z_2 z_3 = |z_2 z_3|$,
 $x_1 x_3 = |x_1 x_3|$, $(1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$.

Moreover, $u z_2 z_3 \neq 1$, because the equalities $u z_2 z_3 = 1$ and $1 + u(z_2 + z_3 + z_2 z_3) = 0$ cannot hold simultaneously; hence the expressions to test are given by

the case-(a) H_1, H_2, H_3 . Also $h_{233} \neq 0$

(as stated in the hypotheses), and, in case (b), the identity $h_{233} H_3 = 1$ gives $H_3 \neq 0$, so the divisions by H_3 are legitimate there.

Verification:

in these cases $k^2 = (g_2^2 - 1)(g_3^2 - 1)$, that is,

$1 + u(z_2 + z_3 + z_2 z_3) = 0$, and the following identities hold:

$$g_i = \frac{\sqrt{z_i(1+z_i)}}{|1+z_i|}, \quad 1 + u z_2 = \frac{z_3(1+z_2)}{z_2+z_3+z_2 z_3}, \quad 1 + u z_3 = \frac{z_2(1+z_3)}{z_2+z_3+z_2 z_3}$$

$$, \quad k = \frac{1}{\sqrt{(1+z_2)(1+z_3)}}, \quad k' = k \sqrt{z_2 + z_3 + z_2 z_3} \quad \blacksquare$$

Hence, in case (b) ($e_2 = -e_3$), substituting the identities $\blacksquare$ and the equality

$z_2 z_3 = |z_2 z_3|$ into the case-(b) $h_{231}, h_{232}, h_{233}$, we get, with the case-(a) H_1, H_2, H_3 ,

$$h_{233} H_3 = 1, \quad h_{231} = -\frac{e_2 \sqrt{1-u} H_1}{\sqrt{z_2 z_3} H_3}, \quad h_{232} = -\frac{H_2}{\sqrt{z_2 z_3} H_3} \quad \blacksquare$$

which are verified separately on each sign region:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1$	✓

Thus, substituting the identities $\blacksquare$

into the numerator and the denominator of the given U-formula, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} =$$

$$- \sqrt{z_2 z_3} H_3^2 (|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)})$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 > 0, z_3 > 0, x_1 < 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 < -1, x_1 > 0$	for $e_2 = \pm 1$	✓
$z_2 < -1, z_3 < -1, x_1 < 0$	for $e_2 = \pm 1$	✓

and, finally, the prefactor identity :

$$\frac{1}{\sqrt{|x_1 x_3|}} \frac{z_2 z_3}{\sqrt{z_2 z_3}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $z_2 z_3 = |z_2 z_3|$ and $x_1 x_3 = |x_1 x_3|$,

and is verified separately on each of the 4 sign regions (sign combinations of z_2, z_3, x_1 as above) ✓

Together, the numerator, denominator, and prefactor identities, along with

$\text{sgn}(u z_2) = -\text{sgn}(z_2)$ and the cancellation of the factor H_3^2

($H_3 \neq 0$, as noted above), give the formula $\star$ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \text{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

The identity $\star$ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0, z_3 > 0, x_1 > 0$	for $e_2 = \pm 1$	✓
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$$\begin{array}{lll}
z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark
\end{array}$$

In case (c) ($e_2 = e_3$), the expressions to test remain

the case-(a) H_1, H_2, H_3 , and the given expression is the case-(c) U -formula :

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_3|}} \cdot \frac{e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)}}{k(1 - x_1 t^2)}$$

Substituting the identities ■ into the case-(a) H_1, H_2, H_3 , we get

$$H_3 = 0, H_1 = -\sqrt{z_2 z_3 (z_2 + z_3 + z_2 z_3)}, H_2 = -\sqrt{z_2 z_3 (1 + z_2)(1 + z_3)} \quad \blacksquare$$

which are verified separately on each sign region:

$$\begin{array}{lll}
z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark
\end{array}$$

Thus, substituting the identities ■ and the equality $k' = k \sqrt{z_2 + z_3 + z_2 z_3}$

into the numerator and the denominator of the given case-(c) U -formula, we get the numerator identity:

$$e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1, t)} = -\frac{k(|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{\sqrt{z_2 z_3}}$$

the denominator identity:

$$k(1 - x_1 t^2) = \frac{k(z_2 z_3 + u x_1 H_1^2 t^2)}{z_2 z_3}$$

which are verified separately on each sign region, with H_1, H_2, H_3 symbolic:

$$\begin{array}{lll}
z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark
\end{array}$$

Together, the numerator, denominator, and prefactor

identities, along with $\operatorname{sgn}(u z_2) = -\operatorname{sgn}(z_2)$, give the formula ★ in case (a):

$$U = \frac{(\varepsilon_2 \varepsilon_3 \operatorname{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}) (|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)})}{z_2 z_3 + u x_1 H_1^2 t^2} \quad \star$$

By the denominator identity, the denominators $k(1 - x_1 t^2)$

and $z_2 z_3 + u x_1 H_1^2 t^2$ vanish for the same t , because $k, z_2 z_3 \neq 0$.

The identity ★ is also verified separately on each sign region, with $e_0, \varepsilon_2, \varepsilon_3$

symbolic, that is, for all their real values at once, which covers all their sign choices:

$$\begin{array}{lll}
z_2 > 0, z_3 > 0, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 > 0, z_3 > 0, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 > 0 & \text{for } e_2 = \pm 1 & \checkmark \\
z_2 < -1, z_3 < -1, x_1 < 0 & \text{for } e_2 = \pm 1 & \checkmark
\end{array}$$

ALL CLAIMS VERIFIED